General Diagnostic Requirement Specification

CN200 统一诊断规范要求

CN200 Project

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Requirement wording conventions/用词约定要求

Within this document the following conventions are applied when describing specific compliance requirements:
本规范中当描述指定的需要遵从的要求时，将使用以下约定：

The word “Should” denotes a preference or desired conformance.
词语“应该”表明了一种倾向性的或期望的性能要求。

The word “Shall” is used in the following ways:
词语“应”用在以下情况：

1. To state a binding requirement on the special network, or the ECUs which comprise the network, which is verifiable by external manipulation and/or observation of an ECU or the network;
   用于陈述对指定网络或组成网络的 ECU 的约束性要求，该约束性要求可以通过对网络或组成网络的 ECU 进行外部操作和/或检查得到验证。
   2. To state a binding requirement upon an ECU’s requirements document that is verifiable through a review of the document.
   用于陈述 ECU 要求文档的约束性要求，该约束性要求可以通过对该文档的审核得到验证。

The word “Must” is used to state a binding requirement upon ECUs on the network which have a corresponding ECU requirements document. These requirements will be verified as part of the component verification.
词语“必须”用于陈述网络中有 ECU 要求文档的 ECU 的约束性要求，该约束性要求须作为零部件验证工作的一部分。

NOTE: In the event of a conflict between the text of this specification and the documents cited herein, the text of this specification takes precedence.
注：当本规范内容与本规范所引用的文档内容产生冲突时，以本规范内容为准。

NOTE: Nothing in this specification supersedes applicable laws and regulations unless a specific exemption has been obtained.
注：本规范内容不能替代任何法律与法规要求，除非获得特定的豁免。

NOTE: If there has any difference between Chinese version and English version, the English version takes precedence.
注：本规范中若有中英文描述不一致的地方，以英文描述为准。

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1 Introduction/引言

1.1 Scope/范围

Based on the international standard ISO 14229-1, ISO 15765-2, ISO 15765-3 and ISO 15031-6, this specification further clarify and refine the requirements of data transmission and protocol implementation, also presented SGMW about diagnosis communication enterprise custom requirements. This document defines the diagnostic application layer protocol requirements for ECU's and tester used within SGMW specified CAN bus networks. This specification contains the protocol services, timings and behavior to provide uniformity and guidance for incorporation diagnostic protocol functionality.
在国际标准ISO 14229-1、ISO 15765-2、ISO 15765-3及ISO 15031-6的基础上，本规范进一步明确并细化了数据传输及协议执行方面的要求，同时还提出了SGMW关于诊断通信的企业自定义要求。对于SGMW CAN网络中的ECU及诊断仪，本规范对其提出了诊断应用层协议方面的要求。本规范定义了传输介质、定时参数、协议服务以及软件刷新流程及相关要求。

This specification shall be used by SGMW to communicate diagnostic requirements for CAN bus system. It also serves as the main input for design of the CAN bus system diagnostics and should preferably be read by diagnostics system design engineers and CAN-bus systems design engineer to design the vehicle network system and diagnostic system, ECU module design engineers and ECU module suppliers to Design and development of ECU module functions, Diagnostic test engineers to verify the vehicle diagnostic system, The production line process engineers and After-sales diagnostic management engineers to development of off-line diagnostic equipment.
本规范应（shall）作为SGMW CAN网络诊断设计开发的主要输入资料。在诊断系统设计工程师与总线系统设计工程师设计整车网络系统及诊断系统、ECU模块设计工程师及相关的ECU模块供应商设计开发ECU模块功能、诊断测试工程师验证整车诊断系统、生产线工艺工程师与售后服务诊断管理工程师开发离线诊断设备时，均应该（should）仔细阅读此规范。

1.2 Organization of document/文档结构

This specification covers the general diagnostic requirements for all ECUs of SGMW CN200 project. Section 1 is the part of Introduction. Section 2 lists the normative references used by this document. Section 3 gives the terms, definitions and abbreviations appear in this specification. The general requirements for diagnostic communication are specified in section 4. The requirements of physical layer are specified in section 5, data link layer in section 6, network layer in section 7, and application layer in section 8. In section 9, diagnostic service documentation structure, are specified. In sections 10 to 15, the diagnostic services in Diagnostic and communication management functional unit, Data transmission functional unit, Stored data transmission functional unit, InputOutput control functional unit, Remote activation of routine functional unit, Upload download functional unit, which is based on ISO 14229-1 (UDS), are defined respectively and specially by SGMW. The requirements for programming ECUs on an SGMW CAN network, as well as the programming process, are defined in chapter 16 .
本规范提出了SGMW CN200车型CAN网络中所有ECU节点的统一诊断要求。第1 章为引言部分；第2 章列出了本规范引用的标准；第3 章列出了本规范使用的术语、定义及缩写；第4 章规定了诊断通信的一般要求；第5 章规定了物理层要求；第6 章规定了数据链路层要求；第7 章规定了网络层要求；第8 章规定了应用层要求；第9 章规定了诊断服务结构；第10 到15 章规定了诊断和通信管理功能单元、数据传输功能单元、存储数据传输功能单元、输入输出控制功能单元、程序远程激活功能单元以及上传下载功能单元中实现的服务以及SGMW自身定义的服务。第16 章规定了SGMW CAN网络中ECU软件刷新

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的相关技术要求以及软件刷新流程。

The annexes A to G give the definitions about the parameter used in the Data transmission functional unit, Stored data transmission functional unit, InputOutput control functional unit, Remote activation of routine functional unit, Upload download functional unit, Address and Length Format Identifier parameter values, Data transmission functional unit data parameter definitions respectively.
附录A到G给出了上述相应功能单元使用的参数、地址、长度及格式定义。

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2 Normative references/引用标准

This document references ISO, GM and other publications as listed in the following sub sections.
本章列出了本规范所引用的ISO、GM以及其它正式发布的标准文档。

2.1 ISO Publications/ISO 标准

[ISO 11898-2]	Road Vehicles - Controller Area Networks (CAN) - Part 2: High-speed medium access unit ISO 2003
[ISO 15765-1]	Road Vehicles - Diagnostics on Controller Area Networks (CAN) - Part 1: General information ISO 2004
[ISO 15765-2]	Road Vehicles - Diagnostics on Controller Area Networks (CAN) - Part 2: Network layer services ISO 2004
[ISO 15765-3]	Road Vehicles - Diagnostics on Controller Area Networks (CAN) - Part 3: Implementation of diagnostic services ISO 2004
[ISO 14229]	Road Vehicles - Unified diagnostic services (UDS) - Specification and requirements ISO 2006
[ISO 15031-5]	Road Vehicles - Communication between vehicle and external equipment for emission-related diagnostic - Part 5: Emission-related diagnostic services ISO 2005
[ISO 15031-6]	Road Vehicles - Communication between vehicle and external equipment for emission-related diagnostic - Part 6: Diagnostic trouble code definitions ISO 2005

2.2 GM Publications/GM 标准

GMW 3107	GMLAN Functional Handler Specification
GMW 3104	GMLAN Communication Strategy Specification
GMW 3110	GMLAN Enhanced Diagnostic Test Mode Specification
GMW 3122	GM High Speed CAN Physical Layer Specification

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3 Terms, definitions and abbreviations/术语、定义及缩写

3.1 ISO Terms/ISO 术语

This document makes use of terms defined in the ISO 15765-2, ISO 15765-3 and ISO 14229-1.
本规范沿用ISO 15765-2、ISO 15765-3与ISO 14229-1中定义的术语。

3.2 Specific Terms Defined By This Document/本规范定义的专用术语

Table 3-1Specific Terms Defined By This Document/本规范定义的专用术语

Abbreviation/缩写	Description/描述
msec	**mili**l**sec**ond
CANId	**CAN Id**entifier
ABS	**A**nti-lock **B**rake **S**ystem
BCM	**B**ody **C**ontrol **M**odule
ECM	**E**ngine **C**ontrol **M**odule
EPS	**E**lectronic **P**ower **S**teering
IC	**I**nstrument **C**luster
SDM	Restraint **S**ensing and **D**iagnostic **M**odule
TCM	**T**ransmission **C**ontrol **M**odule

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4 General requirement/一般要求

4.1 Communication layer structure/通信层结构

The interaction between the diagnostic communication layers in the tester and the ECU(s) is based on the Open Systems Interconnection (O.S.I.) Basic Reference Model in accordance with ISO 7498 and ISO/IEC 10731, which structures communication systems into seven layers. Each layer provides a set of service primitives, which transports a certain type of parameters and data.
诊断仪与ECU的诊断通信层交互是以开放系统互联模型为基础的，基本参考模型符合ISO 7498与ISO/IEC 10731标准要求，它将通信系统分为七层。每一层都提供了一组用于传输特定类型参数与数据的服务原语。

To achieve this, when mapped on this model, the services of the SGMW diagnostic specification defined by this document are broken into:

- enhanced diagnostic services (layer 7), which is based on ISO 14229-1 and ISO 15765-3,
- session services (layers 5), which is based on ISO 15765-3,
- network layer services(layer 3), which is based on ISO 15765-2,
- data link layer services(layer 2), which is based on ISO 11898-1,
- physical layer services(layer 1), CN200 high speed CAN physical layer requirement, which is based on ISO 11898-2.

为了实现以上要求，映射到开放系统互联模型，本规范定义的SGMW诊断规范服务可分为：

- 基于ISO 14229-1和ISO 15765-3的增强型诊断服务（第七层），
- 基于ISO 15765-3的会话层服务（第五层），
- 基于ISO 15765-2的网络层服务（第三层），
- 基于ISO 11898-1的数据链路层服务（第二层），
- 基于ISO 11898-2的CN200高速CAN物理层要求与物理层服务（第一层）。

in accordance with the following Table 4-1.
如Table 4-1所示。

Table 4-1 Communication layer structure/通信层结构

OSI layer OSI 层		SGMW Enhanced diagnostics SGMW 增强型诊断	OBD diagnostics OBD 诊断
7	Application layer 应用层	This specification, which is based on ISO14229-1 and ISO 15765-3 基于 ISO14229-1 和 ISO 15765-3 的本规范	ISO 15031-5
6	Presentation layer 表示层	N/A	N/A
5	Session layer 会话层	This specification, which is based on ISO15765-3 基于 ISO15765-3 的本规范	N/A
4	Transport layer 传输层	N/A	N/A
3	Network layer 网络层	This specification, which is based on ISO15765-2 基于 ISO15765-2 的本规范	ISO 15765-4

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2	Data link layer 数据链路层	ISO11898-1	ISO 15765-4
1	Physical layer 物理层	CN200 high speed CAN physical layer requirement, which is based on ISO 11898-2 基于 ISO 11898-2 的 CN200 高速 CAN 物理层要求	ISO 15765-4

4.2 Diagnostic network topology/诊断网络拓扑

This section gives the diagnostic network topology used in the SGMW CN200 project. Similar to the on-board communication network, the diagnostic network consists of DLC connector and seven in-vehicle ECUs, which are ECM, TCM, ABS, IC, EPS, SDM and BCM.
本章节给出了SGMW CN200项目所采用的诊断网络拓扑，与在线通信网络类似，诊断网络包含DLC诊断接头以及7个车载ECU，分别是ECM、TCM、ABS、IC、EPS、SDM与BCM，如Figure 4-1所示。

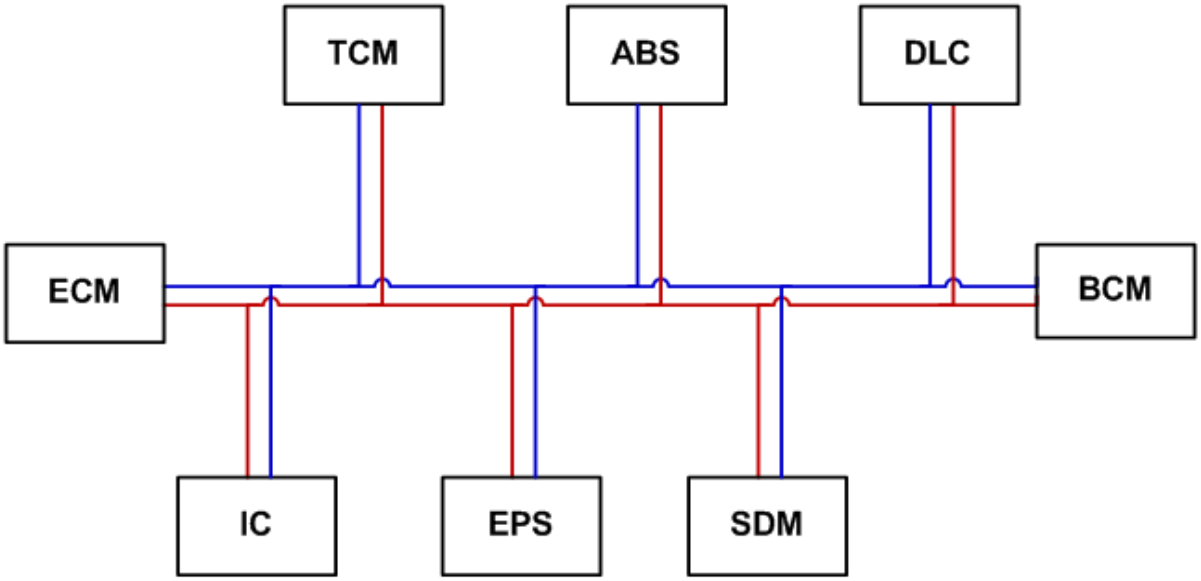

Figure 4-1 Diagnostic network topology/诊断网络拓扑

NOTE: Both ECM and BCM have bus termination (120Ω) internally.
注：ECM与BCM内含有总线终端电阻（120Ω）。

4.3 CAN identifiers definition /CANId 定义

The following 错误！未找到引用源。 provides an overview about the definition of CAN identifiers.
错误！未找到引用源。定义了 CANId：

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5 Physical layer requirement/物理层要求

The tester and ECU shall comply with the key rules as follow:

1. The normal transmission rate of high speed bus: 500kbits/sec,
2. The tester shall NOT have bus termination internally, because that two onboard ECUs are specifically assigned to perform bus termination.

诊断仪和 ECU 应（shall）遵从以下关键规则：

1. 高速 CAN 总线正常通信波特率为 500Kbits/s，
2. 诊断仪内部不能有终端电阻，因为两个在线 ECU 已经被明确分配作为总线终端。

Other more detailed requirements, with which the tester and the ECUs shall comply, can be found in “CN200 high speed CAN physical layer requirement”.
诊断仪与 ECU 应（shall）遵从的其它更多详细要求请参考“CN200 高速 CAN 物理层要求”。

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6 Data link layer requirement/数据链路层要求

The tester and ECU shall communicate with each other with high speed CAN bus.
The tester and ECU shall comply with the key rules as follow:

1. Only the standard (11bit CAN identifier) CAN data frame shall be used for diagnostic communication.

诊断仪与 ECU 应（shall）通过高速 CAN 总线进行通信。
诊断仪和 ECU（shall）遵从以下关键规则：

1. 诊断通信中只允许使用标准（11 位 CAN 标识符）CAN 数据帧。

Other more detailed requirements, with which the tester and the ECUs shall comply, can be found in ISO 11898-1.
诊断仪与 ECU 应（shall）遵从的其它更多详细要求请参考 ISO 11898-1。

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7 Network layer requirement/网络层要求

The following subsections describe the requirement of network layer, which is based on ISO 15765-2, used by the tester and the ECUs for diagnostic communication.
Other than the requirement defined in the following subsections, more detailed requirements of network layer protocol, with which the tester and the ECUs shall comply, can be found in ISO 15765-2.
本章节所描述的网络层要求是基于 ISO 15765-2 制定的，用于诊断仪和 ECU 之间的诊断通信。
除了以下章节定义的要求外，更多针对诊断仪与 ECU 应（shall）遵从的网络层的详细要求请参照 ISO 15765-2。

NOTE: In this document, the exchange of network layer data is supported by only one addressing format: normal addressing, which is defined in ISO 15765-2.
注：本规范网络层的数据交互仅支持一种编址格式：正常编址，此编址格式在 ISO 15765-2 内做了定义。

7.1 Network layer protocol/网络层协议

7.1.1 Protocol functions/协议功能

The network layer protocol performs the following functions:
a) transmission/reception of messages up to 4095 data bytes;
b) reporting of transmission/reception completion (or failure).
网络层协议具有以下功能：
a）一次可发送或接收最多4095个数据字节；
b）报告发送或接收完成（或失败）。

7.1.2 Single frame transmission/单帧传输

Transmission of messages up to seven (normal addressing) data bytes is performed via transmission of a unique N_PDU, called SF.
Figure 7-1provides an overview about Single frame transmission.
发送消息多达7个数据字节时（正常编址），可通过发送一个网络层协议数据单元，称之为单帧（SF）。
Figure 7-1 为单帧传输示意图。

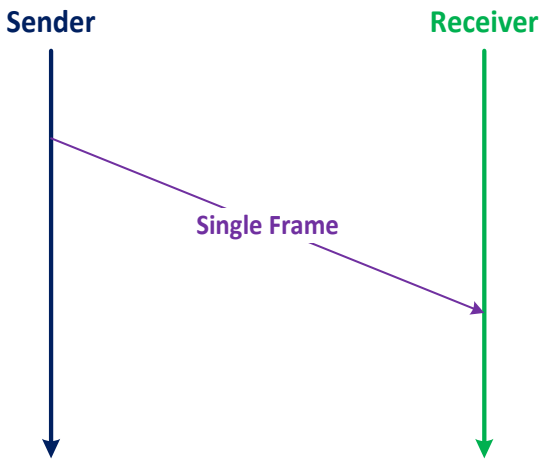

Figure 7-1 Single frame transmission/单帧传输

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7.1.3 Multiple frame transmission/多帧传输

Transmission of longer messages is performed via segmentation of the message and transmission of multiple N_PDUs. Reception of longer messages is performed via reception of multiple N_PDUs and reassembly of the received data bytes (concatenation). The multiple N_PDUs are called FirstFrame (for the first N_PDU of the message) and ConsecutiveFrame (for all the following N_PDUs).
通过报文分割并发送多个网络层协议数据单元（N_PDU），来完成超过7个字节的消息的发送；通过接收多个N_PDU并对接收的一系列数据字节进行重组，来完成超过7个字节的消息的接收。多个N_PDU分为首帧（报文的第一个协议数据单元）与后续帧（接下来的所有协议数据单元）。

The receiver of a multiple N_PDU message has the possibility of adapting the transmission throughput to its capability by means of the flow control mechanism using the FlowControl protocol data units (FC N_PDU).
多N_PDU报文接收器能够调整传输吞吐量以适应自身接收能力，这种调整是依靠流控机制采用流控协议数据单元来实现的（FC N_PDU）。

Messages longer than seven data bytes are segmented into
-a FirstFrame protocol data unit (FF N_PDU), containing the first six (normal addressing) data bytes, and one or more ConsecutiveFrame protocol data units (CF N_PDU), containing each seven data bytes. The CF N_PDU contains only the remaining data bytes, and may therefore be less than seven data bytes long.
超过7个数据字节的消息被分为首帧与后续帧，其中首帧包含前6个数据字节的网络层协议数据单元（FF N_PDU），后续帧包含剩余数据字节的网络层协议数据单元（CF N_PDU）。每一个CF N_PDU最多包含7个数据字节，由于CF N_PDU包含了FF N_PDU以外的其余字节，所以数据长度有可能小于7。

Figure 7-2 provides an overview about multiple frame transmission.
Figure 7-2 为多帧传输示意图。

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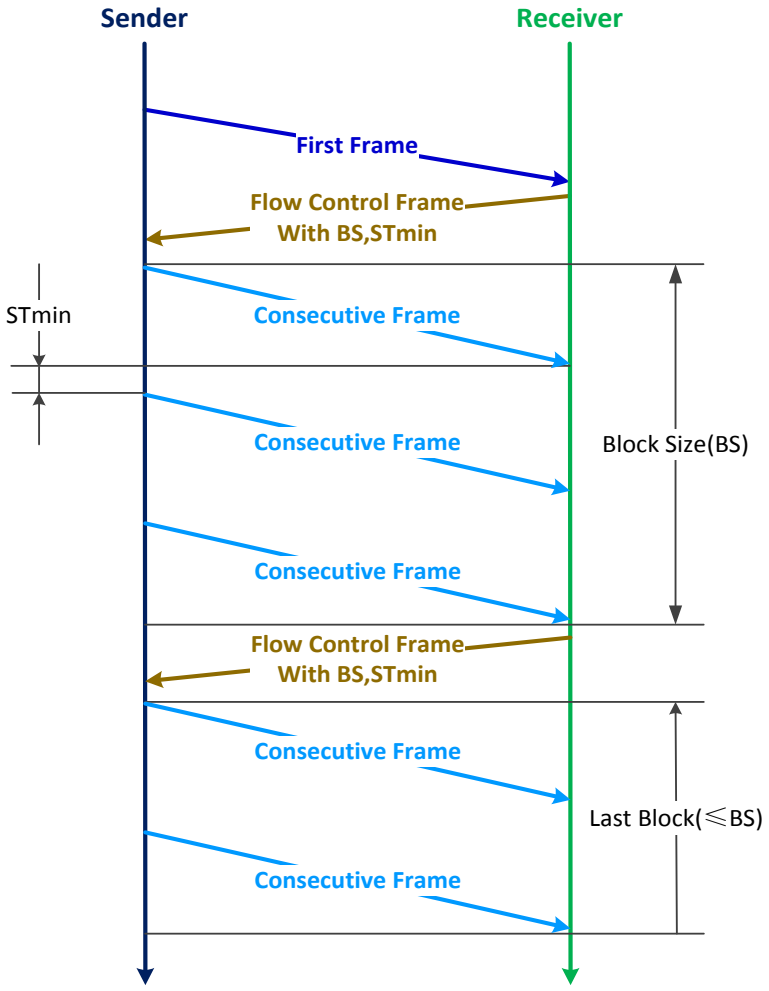

Figure 7-2 multiple frame transmission/多帧传输

The message length is transmitted in the FF N_PDU. All CF N_PDUs are numbered by the sender to help the receiver re-assemble them in the same order.
消息长度信息包含在FF N_PDU。发送器对所有的CF N_PDU进行编号，这有助于接收器按照相应的编号顺序对消息进行重组。

The flow control mechanism (see Figure 7-2) allows the receiver to inform the sender about the receiver's capabilities. Since different ECUs may have different capabilities, the flow control sent by the receiver informs the sender about its capabilities. The sender shall conform to the receiver's capabilities.
流控机制（Figure 7-2）允许接收器把自身的接收能力通知给发送器。由于不同节点的接收能力可能不同，所以需要接收器把自身的接收能力通知给发送器，发送器应（shall）调整自身的发送策略以适应接收器的接收能力。

These capabilities are defined as follows.
-BlockSize (BS): the maximum number of N_PDUs the receiver allows the sender to send, before waiting for an authorization to continue transmission of the following N_PDUs.
-SeparationTimeMin (ST_{min}): the minimum time the sender is to wait between the transmissions of two

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CF N_PDUs.
接收器的接收能力定义如下。
-数据块大小：在允许发送器继续传输后续帧之前，接收器允许发送器发送数据帧的最大数量。
-最小间隔时间：发送器传输的两个连续后续帧之间最小时间间隔。

All blocks, except the last one, will consist of BS N_PDUs. The last one will contain the remaining N_PDUs (1 up to BS).
除最后一个数据块以外，其余数据块包含协议数据单元的数量为BS。最后一个数据块包含剩余的协议数据单元数量为1到BS。

Each time the sender/receiver waits for an N_PDU from the receiver/sender, a timeout mechanism allows detection of a transmission failure.
在发送器（接收器）等待接收接收器（发送器）发送的协议数据单元时，允许采用一种超时机制来检测传输的成败。

By means of FC N_PDUs, the receiver has the possibility of authorizing transmission of the following CF N_PDUs, to delay transmission of that authorization or to deny the reception of a segmented message in case the number of bytes to be transferred exceeds the number of bytes that can be stored in the receiver buffer.
通过流控帧，接收器具有控制发送器发送后续帧的能力，它可以控制延时传输，并且当发送器所发送的字节数超出了接收器缓存的存储能力时拒绝接收。

7.1.4 Network layer protocol data units/网络层协议数据单元

7.1.4.1 Protocol data unit types/协议数据单元

The communication between the peer protocol entities of the network layer in different nodes is done by means of exchanging N_PDUs.
不同节点网络层对等实体之间的通信是通过网络层协议数据单元的交互完成的。

This part specifies four different types of network layer protocol data units — single-frame (SF N_PDU), first-frame (FF N_PDU), consecutive-frame (CF N_PDU) and flow-control (FC N_PDU) — which are used to establish a communication path between the peer network layer entities, to exchange communication parameters, to transmit user data and to release communication resources.
本部分定义了四种不同类型的网络层协议数据单元：单帧（SF N_PDU）、首帧（FF N_PDU）、后续帧（CF N_PDU）及流控帧（FC N_PDU）。这些网络层协议数据单元用来在网络层对等实体之间建立通信路径、交换通信参数、传输用户数据与释放通信资源。

7.1.4.2 SF N_PDU/单帧网络层协议数据单元

The SF N_PDU is identified by the single-frame protocol control information (SF N_PCI). The SF N_PDU shall be sent out by the sending network entity and can be received by one or multiple receiving network entities. It shall be sent out to transfer a service data unit that can be transferred via a single service request to the data link layer.
单帧网络层协议数据单元通过单帧协议控制信息（SF N_PCI）来识别。单帧网络层协议数据单元应（shall）被发送端网络实体发送出去，可以由一个或多个接收端网络实体接收。单帧网络层协议数据单元应（shall）被发送出去用来传输服务数据单元，该服务数据单元可以通过对数据链路层的单一服务请求发送出去。

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7.1.4.3 FF N_PDU/首帧网络层协议数据单元

The FF N_PDU is identified by the first-frame protocol control information (FF N_PCI). The FF N_PDU shall be sent out by the sending network entity and received by a unique receiving network entity for the duration of the segmented message transmission. It identifies the first N_PDU of a segmented message transmitted by a network sending entity and received by a receiving network entity. The receiving network layer entity shall start assembling the segmented message on receipt of an FF N_PDU.
首帧协议控制信息（FF N_PCI）是首帧网络层协议数据单元（FF N_PDU）的标识。在分段信息传输过程中，发送端网络实体应（shall）发送FF N_PDU，对应地，接收端网络实体应（shall）接收FF N_PDU。上述发送接收机制可用来识别分段消息的FF N_PDU。当接收到一个FF N_PDU时，接收端网络层实体开始重组分段信息。

7.1.4.4 CF N_PDU/后续帧网络层协议数据单元

The CF N_PDU is identified by the consecutive-frame protocol control information (CF N_PCI). The CF N_PDU transfers segments (N_Data) of the service data unit message data (<MessageData>). All N_PDUs transmitted by the sending entity after the FF N_PDU shall be encoded as CF N_PDUs. The receiving entity shall pass the assembled message to the service user of the network receiving entity after the last CF N_PDU has been received. The CF N_PDU shall be sent out by the sending network entity and received by a unique receiving network entity for the duration of the segmented message transmission.
后续帧协议控制信息（CF N_PCI）是后续帧网络层协议数据单元（CF N_PDU）的标识。CF N_PDU用来传输分段的服务数据单元消息数据。在FF N_PDU之后被发送实体发送的所有N_PDU应（shall）与CF N_PDU的编码方式一致。在接收到最后一个CF N_PDU之后，接收端网络实体应（shall）把已重组的消息传递给接收端网络实体的服务使用者。在分段消息传输过程中，发送端网络实体应（shall）发送CF N_PDU，对应地，接收端网络实体应（shall）接收CF N_PDU。

7.1.4.5 FC N_PDU/流控网络层协议数据单元

The FC N_PDU is identified by the flow-control protocol control information (FC N_PCI). The FC N_PDU instructs a sending network entity to start, stop or resume transmission of CF N_PDUs. It shall be sent by the receiving network layer entity to the sending network layer entity, when ready to receive more data, after correct reception of
a) an FF N_PDU, or
b) the last CF N_PDU of a block of consecutive frames, if further consecutive frames need to be sent.
The FC N_PDU can also inform a sending network entity to pause transmission of CF N_PDUs during a segmented message transmission or to abort the transmission of a segmented message if the length information (FF_DL) in the FF N_PDU transmitted by the sending entity exceeds the buffer size of the receiving entity.
流控协议控制信息（FC N_PCI）是流控网络层协议数据单元（FC N_PDU）的标识。FC N_PDU控制发送端网络实体开始、停止或者恢复CF N_PDU的传输。当接收端网络实体正确接收到：
a).FF N_PDU，或
b).前一个后续帧数据块中的最后一个CF N_PDU（如果需要发送更多的后续帧），
并且已经准备好接收后续数据时，它应发送一个FC N_PDU给发送端网络实体。
另外，在分段信息传输过程中，FC N_PDU可以要求发送端网络实体暂停CF N_PDU的传输。如果发送端实体发送的FF N_PDU内的数据长度信息（FF_DL）超过接收端实体的缓存空间，FC N_PDU还可以要求发送端网络实体终止整个分段信息的传输。

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7.1.4.6 Protocol data unit field description/协议数据单元场域描述

7.1.4.6.1 N_PDU format/协议数据单元格式

The protocol data unit (N_PDU) enables the transfer of data between the network layer in one node and the network layer in one or more other nodes (peer protocol entities). All N_PDUs consist of three (3) fields (Table 7-1).
N_PDU能够使能一个节点的网络层与另一个或多个节点的网络层（对等协议实体）之间传输数据。所有的N_PDU都包含3个场域（Table 7-1）。

Table 7-1 N_PDU format/N_PDU 格式

Address information 地址信息	Protocol control information 协议控制信息	Data field 数据场
N_AI	N_PCI	N_Data

7.1.4.6.2 Address information (N_AI)/地址信息（N_AI）

The N_AI is used to identify the communicating peer entities of the network layer. The N_AI information received in the N_SDU shall be copied and included in the N_PDU. If the message data (<MessageData> and <Length>) received in the N_SDU is so long that segmentation is needed for the network layer to transmit the complete message, the N_AI shall be copied and included (repeated) in every N_PDU that is transmitted.
N_AI标识了网络层之间进行信息交互的对等实体。N_SDU中接收的N_AI应（shall）复制到N_PDU。如果N_SDU中接收的消息数据（<消息数据>与<长度>）太长，以至于需要网络层分段才能传输整个消息，那么N_AI应（shall）复制（重复）到传输的每一个N_PDU中。

This field contains address information that identifies the type of message exchanged, and the recipient(s) and sender between whom data exchange takes place. The address information consists of message addresses.
N_AI的场域中包含了一些地址信息，这些地址信息标识了交互消息的类型以及发生数据交换的接收端与发送端。N_AI的地址信息由消息地址组成。

7.1.4.6.3 Protocol control information (N_PCI)/协议控制信息（N_PCI）

This field identifies the type of N_PDUs exchanged. It is also used to exchange other control parameters between the communicating network layer entities.
N_PCI标识了相互交换的N_PDU类型。同时，它还可以用于通信网络层实体之间其它控制参数的交换。

7.1.4.6.4 Data Field (N_Data)/数据场（N_Data）

The N_Data in the N_PDU is used to transmit the service user data received in the <MessageData> parameter in the N_USData.request service call. The <MessageData>, if needed, is segmented into smaller parts that each fit into the N_PDU data field before they are transmitted over the network.
The size of N_Data depends on the N_PDU type and the address format chosen.
N_PDU中的N_Data用来传输服务用户数据，该服务用户数据包含在接收的N_USData.request服务原语的<消息数据>参数中。如果需要的话，在通过网络传输之前，<消息数据>参数被划分成可填充到N_PDU数据场中的多个小块。
N_Data的大小取决于N_PDU类型及选定的地址格式。

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7.1.5 Protocol control information specification/协议控制信息说明

7.1.5.1 N_PCI/网络层协议控制信息

The N_PCI field identifies the type of N_PDUs exchanged. It is also used to exchange other control parameters between the communicating network layer entities. Each N_PDU is identified by means of an N_PCI as following table(Table 7-2).
N_PCI场标识了相互交换的N_PDU类型，它还可以用于通信网络层实体之间其它控制参数的交换。
N_PCI标识的N_PDU类型如Table 7-2所示：

Table 7-2 N_PDU type and N_PCI bytes/N_PDU 类型与 N_PCI 字节

N_PDU type	N_PCI bytes			
	Byte #1		Byte #2	Byte #3
	Bits 7-4	Bits 3-0		
Single Frame (SF)	N_PCItype=0	SF_DL	N/A	N/A
First Frame (FF)	N_PCItype=1	FF_DL		N/A
Consecutive Frame (CF)	N_PCItype=2	SN	N/A	N/A
Flow Control (FC)	N_PCItype=3	FS	BS	ST_{min}

7.1.5.2 SingleFrame N_PCI parameter definition/SFN_PCI 参数定义

7.1.5.2.1 SF N_PCI byte/SFN_PCI 字节

The following table (Table 7-3) provides an overview of the SF N_PCI byte.
Table 7-3对SF N_PCI字节进行了描述.

Table 7-3 SF N_PCI byte/ SF N_PCI 字节

N_PDU name	SF N_PCI byte							
	Byte#1							
	7	6	5	4	3	2	1	0
SingleFrame/单帧	0	0	0	0	SF_DL			

The parameter SingleFrame DataLength (SF_DL) is used in the SF N_PDU to specify the number of service user data bytes.
SF N_PDU中的单帧数据长度（SF_DL）参数表示服务用户数据字节的数量。

NOTE: SF_DL = 7 is only allowed with normal addressing.
注：只有在正常编址模式下SF_DL的值才可能为7.

7.1.5.2.2 SF_DL error handling/ SF_DL 错误处理

If the network layer receives an SF with an SF_DL equal to zero (0), then the network layer shall ignore the received SF N_PDU.
If the network layer receives an SF with an SF_DL greater than 7, then the network layer shall ignore the received SF N_PDU.
如果接收到一个SF_DL为0的单帧（SF），那么网络层应（shall）忽略所接收的SF N_PDU。
如果接收到一个SF_DL大于7的单帧（SF），那么网络层应（shall）忽略所接收的SF N_PDU。

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7.1.5.3 FirstFrame N_PCI parameter definition/ FF N_PCI 参数定义

7.1.5.3.1 FF N_PCI byte/ FF N_PCI 字节

The following table (Table 7-4) provides an overview of the FF N_PCI byte.
Table 7-4对FF N_PCI字节进行了描述.

Table 7-4 FF N_PCI byte/ FF N_PCI 字节

N_PDU name	FF N_PCI byte								
	Byte#1								Byte#2
	7	6	5	4	3	2	1	0	7 - 0
FirstFrame	0	0	0	1	FF_DL				

The parameter FF_DL is used in the FF N_PDU to specify the number of service user data bytes. The encoding of the segmented message length results in a twelve (12) bit length value (FF_DL) where the least significant bit (LSB) is specified to be bit “0” of the N_PCI byte #2 and the most significant bit (MSB) is bit three (3) of the N_PCI byte #1. The maximum segmented message length supported is equal to 4 095 bytes of user data. It shall be assigned the value of the service parameter <Length>.
FF N_PDU中的首帧数据长度（FF_DL）参数表示服务用户数据字节的数量。分段消息长度（FF_DL）可解码成一个长度占12个位的值，该值的最低位为N_PCI第2个字节中的第0位，该值的最高位为N_PCI第1个字节中的第3位。所支持分段消息的最大长度为4095个用户数据字节，服务参数<长度>值应（shall）分配给FF_DL。

NOTE: The range of FF_DL is 8-FFF (Hex) with normal addressing.
注：在正常编址模式下FF_DL的范围为8-FFF（Hex）。

7.1.5.3.2 FirstFrame DataLength (FF_DL) parameter definition/ FF_DL 参数定义

The parameter FF_DL is used in the FF N_PDU to specify the number of service user data bytes. The following table (Table 7-5) provides the definition of FF_DL.
FF N_PDU中的FF_DL参数表示服务用户数据字节的数量。
Table 7-5对FF_DL进行了定义。

Table 7-5 FF_DL parameter definition/ FF_DL 参数定义

Hex value	Description/描述
0-6	**Invalid/无效** This range of values is invalid. 该范围值无效。
7	**FirstFrame DataLength (FF_DL) with extended addressing or mixed addressing/扩展编址与混合编制下的首帧数据长度（FF_DL）** FF_DL = 7 is only allowed with extended or mixed addressing. NOTE：In this specification , only normal addressing format is supported. 仅在扩展编址或混合编址模式下FF_DL值才能为7. NOTE:本规范只支持正常编址模式。
8-FFF	**FirstFrame DataLength (FF_DL)/首帧数据长度（FF_DL）** The encoding of the segmented message length results in a twelve (12) bit length value (FF_DL) where the least significant bit (LSB) is specified to be bit “0” of the N_PCI byte #2 and the most significant bit (MSB) is bit three (3) of the N_PCI byte #1. The maximum segmented message length supported is equal to 4 095 bytes of user data. It shall be assigned the value of the service parameter <Length>.

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	分段消息长度（FF_DL）可解码成一个长度占12个位的值，该值的最低位为N_PCI第2个字节中的第0位，该值的最高位为N_PCI第1个字节中的第3位。所支持分段消息的最大长度为4095个用户数据字节，服务参数<长度>值应（shall）分配给FF_DL。

7.1.5.3.3 FF_DL error handling/ FF_DL 错误处理

If the network layer receives an FF with an FF_DL that is greater than the available receiver buffer size, then this shall be considered as an error condition. The network layer shall abort the message reception and send an FC N_PDU with the parameter FlowStatus = Overflow.
如果接收到一个FF_DL大于接收端可用缓存大小（这种情况被当作一个错误条件）的首帧（FF）。那么网络层应（shall）终止消息接收，并发送一个流状态（FlowStatus）参数为溢出（Overflow）的FC N_PDU。

If the network layer receives an FF with an FF_DL that is less than eight, then the network layer shall ignore the received FF N_PDU and not transmit an FC N_PDU.
如果接收到一个FF_DL小于8的首帧（FF），那么网络层应（shall）忽略接收的FF N_PDU，并且不会发送FC N_PDU。

7.1.5.4 ConsecutiveFrame N_PCI parameter definition/ CF N_PCI 参数定义

7.1.5.4.1 CF N_PCI byte/ CF N_PCI 字节

The following table (Table 7-6) provides an overview of the CF N_PCI byte.
Table 7-6对CF N_PCI字节进行了描述.

Table 7-6 CF N_PCI byte/ CF N_PCI 字节

N_PDU name	CF N_PCI byte							
	Byte#1							
	7	6	5	4	3	2	1	0
ConsecutiveFrame	0	0	1	0	SN			

7.1.5.4.2 SequenceNumber (SN) parameter definition/序列号参数定义

The parameter SN is used in the CF N_PDU to specify the order of the consecutive frames.
-The SN shall start with zero (0) for all segmented messages. The FF shall be assigned the value zero (0).It does not include an explicit SequenceNumber in the N_PCI field but shall be treated as the segment number zero (0).
-The SN of the first CF that immediately follows the FF shall be set to one (1).
-The SN shall be incremented by one (1) for each new CF that is transmitted during a segmented message transmission.
-The SN value shall not be affected by any FC frame.
-When the SN reaches the value of fifteen (15), it shall wraparound and be set to zero (0) for the next CF.
The following table (Table 7-7) provides the definition of SN.
CF N_PDU中的参数SN表示后续帧的顺序。
对所有分段消息来说，SN应（shall）从0开始。其中，首帧（FF）中的SN为0。虽然首帧（FF）中的N_PCI 场没有包含一个明确的序列号，但是首帧（FF）的SN仍被当作0.
紧跟在首帧（FF）之后的第一个后续帧（CF）的SN为1.
在分段消息传输过程中，每发送一个新的CF，SN的值应（shall）加1。
任何FC帧不应（shall not）影响SN的值。

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当达到15时，SN的值应（shall）在发送下一个后续帧时重新设为0.
Table 7-7对SN进行了定义。

Table 7-7 Summary of SN definition/SN 定义简介

N_PDU	FF	CF	CF	CF	CF	CF	CF	CF
SN(Hex)	0	1	…	E	F	0	1	…

Hex value	Description/描述
0-F	**SequenceNumber (SN)/序列号** The SequenceNumber (SN) shall be encoded in the lower nibble bits of N_PCI byte #1. The SN shall be set to a value within the range of zero (0) to fifteen (15). SN占据N_PCI中第一个字节的低4位，其值的范围为0~F（Hex）。

7.1.5.4.3 SN error handling/SN 错误处理

If a CF N_PDU message is received with an incorrect sequence number, then proper error handling shall take place in the network layer. The message reception shall be aborted, and the network layer shall make an N_USData.indication service call with the parameter <N_Result> = N_WRONG_SN to the adjacent upper layer.
如果接收的CF N_PDU消息中SN不正确，那么网络层应（shall）对错误进行合适的处理。在这种情况下，网络层应（shall）停止消息接收，并通过调用<N_Result> = N_WRONG_SN参数，使用N_USData.indication服务原语给邻近的上一层以指示。

7.1.5.5 FlowControl N_PCI parameter definition/ FC N_PCI 参数定义

7.1.5.5.1 FlowControl N_PCI bytes/ FC N_PCI 字节

The following table (Table 7-8) provides an overview of the FC N_PCI byte.
Table 7-8对FC N_PCI字节进行了描述。

Table 7-8 FC N_PCI bytes/ FC N_PCI 字节

N_PDU name	FC N_PCI byte									
	Byte#1								Byte#2	Byte#3
	7	6	5	4	3	2	1	0		
FlowControl	0	0	1	1	FS				BS	ST_{min}

7.1.5.5.2 FlowStatus (FS) parameter definition/流状态参数定义

The parameter FlowStatus (FS) indicates whether the sending network entity can proceed with the message transmission.
The definition of FS as the following table (Table 7-9) .
参数FS表示接收端告知发送端网络实体能否继续执行消息的传输。
Table 7-9定义了FS。

Table 7-9 FS parameter definition/FS 参数定义

Hex value Bit(3-0)	Description of FS
0	**ContinueToSend (CTS)/继续发送** The FlowControl ContinueToSend parameter shall be encoded by setting the lower nibble of

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	the N_PCI byte #1 to “0”. It shall cause the sender to resume the sending of Consecutive frames. The meaning of this value is that the receiver is ready to receive a maximum of BS number of Consecutive frames. N_PCI中第一个字节的低4位设为0表示FS状态为CTS，此时发送端可继续发送后续帧。同时，还表示接收端已准备好接收最大数量为BS的后续帧。
1	**Wait (WT)/等待** The FlowControl Wait parameter shall be encoded by setting the lower nibble of the N_PCI byte #1 to “1”. It shall cause the sender to continue to wait for a new FlowControl N_PDU and to restart its N_BS timer. N_PCI中第一个字节的低4位设为1表示FS状态为WT，此时发送端应（shall）继续等待新的FC N_PDU，并重启N_BS定时器。
2	**Overflow (OVFLW)/溢出** The FlowControl Overflow parameter shall be encoded by setting the lower nibble of the N_PCI byte #1 to “2”. It shall cause the sender to abort the transmission of a segmented message and make an N_USData.confirm service call with the parameter <N_Result>=N_BUFFER_OVFLW. This N_PCI FlowStatus parameter value is only allowed to be transmitted in the FlowControl N_PDU that follows the FirstFrame N_PDU and shall only be used in case the message length FF_DL of the received FirstFrame N_PDU exceeds the buffer size of the receiving entity. N_PCI中第一个字节的低4位设为2表示FS状态为OVFLW，此时发送端应（shall）终止发送分段消息，并通过调用<N_Result>=N_BUFFER_OVFLW参数，使用N_USData.confirm服务原语给邻近的上一层以确认。OVFLW只允许在紧跟FF N_PDU之后的FC N_PDU中进行传输，并且，仅当在接收的FF N_PDU中的FF_DL超过了接收端实体缓存大小的情况下才能使用。

7.1.5.5.3 FS error handling/FS 错误处理

If an FC N_PDU message is received with an invalid (reserved) FS parameter value, then proper error handling shall take place in the network layer. The message transmission shall be aborted, and the network layer shall make an N_USData.confirm service call with the parameter <N_Result> = N_INVALID_FS to the adjacent upper layer.
如果接收的FC N_PDU消息中FS参数值无效（预留），那么网络层应（shall）对错误进行合适的处理。在这种情况下，网络层应（shall）终止消息发送，并通过调用<N_Result> =N_INVALID_FS参数，使用N_USData.confirm服务原语给邻近的上一层以确认。

7.1.5.5.4 BlockSize (BS) parameter definition/数据块参数定义

The BS parameter shall be encoded in byte #2 of the FC N_PCI.
The units of BS are the absolute number of CF N_PDUs per block.
The definition of BS as the following table (Table 7-10) .
BS参数占据FC N_PCI中的第二个字节。
BS的大小表示每个数据块中CF N_PDU的数目。
Table 7-10定义BS。

Table 7-10 BS parameter definition/BS 参数定义

Hex value	Description of BS
00	**BlockSize (BS)/数据块大小** The BS parameter value zero (0) shall be used to indicate to the sender that no more FC frames shall be sent during the transmission of the segmented message. The sending network layer entity shall send all remaining consecutive frames without any stop for further FC frames from the receiving network layer entity. BS值为0表示，在发送端发送分段消息期间接收端不会发送后续的FC，发送端网络层实体应（shall）一次性把所有后续帧发送出去，而不用停下来接收接收端网络实体发送的FC。

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01-FF	**BlockSize (BS) /数据块大小** This range of BS parameter values shall be used to indicate to the sender the maximum number of consecutive frames that can be received without an intermediate FC frame from the receiving network entity. 该范围的BS参数值表示，发送端不用接收接收端网络实体发送的FC而能发送的后续帧的最大数目。

NOTE: In this specification the value of BS is defined to be 0 for tester and ECU.
注：本规范中，诊断仪与ECU流控消息中的BS值为0.

7.1.5.5.5 Definition of SeparationTime (ST_{min}) parameter/ ST_{min} 参数定义

The ST_{min} parameter value specifies the minimum time gap allowed between the transmission of consecutive frame network protocol data units.
The definition of ST_{min} as the following Table 7-11.
ST_{min}参数表示发送相邻的后续帧网络协议数据单元所允许的最小时间间隔。
Table 7-11定义了ST_{min}。

Table 7-11 ST_{min} parameter definition/ ST_{min} 参数定义

Hex value	Description of ST_{min}
00-7F	**SeparationTime (ST_{min}) range:0ms–127ms** The units of ST_{min} in the range 00 hex – 7F hex are absolute milliseconds (ms). 该范围的ST_{min}值表示发送相邻的CF N_PDU所允许的最小时间间隔为该值对应的毫秒数。
80-F0	**Reserved**
F1-F9	**SeparationTime (ST_{min}) range:100μs–900μs** The units of ST_{min} in the range F1 hex – F9 hex are even 100 microseconds (μs), where parameter value F1 hex represents 100 μs and parameter value F9 hex represents 900 μs. 该范围的ST_{min}值发送相邻的CF N_PDU所允许的最小时间间隔为：F1（Hex）表示100μs，F2（Hex）表示200μs，. . . . . ，F9（Hex）表示900μs。
FA-FF	**Reserved**

The measurement of the ST_{min} starts after completion of transmission of a ConsecutiveFrame (CF) and ends at the request for the transmission of the next CF.
发送端在完成发送一CF时开始ST_{min}计时，在请求发送下一个CF时停止ST_{min}计时。

7.1.5.5.6 ST error handling/ST 错误处理

If an FC N_PDU message is received with a reserved ST parameter value, then the sending network entity shall use the longest ST value specified by this part of ISO 15765 (7F hex , i.e. 127ms) instead of the value received from the receiving network entity for the duration of the ongoing segmented message transmission.
如果接收的FC N_PDU消息中ST参数值为预留值，那么发送端网络实体应（shall）使用ISO 15765（7F（hex）即：127ms）中定义的最大值来替代从接收端网络实体接收的值，并在后续分段消息传输中使用该时间间隔。

NOTE: In this specification the value of ST_{min} is defined to be 0 for tester, 0-10 for ECU.
注：本规范中，诊断仪发送的流控消息中ST_{min}要求为0，ECU发送的流控消息中ST_{min}要求为：0~10之间的任意值.

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7.1.6 Maximum number of FC.Wait frame transmissions (N_WFTmax)/发送流控等待帧的最大次数(N_WFTmax)

The purpose of this variable is to avoid communication sender nodes being potentially hooked-up in case of a fault condition whereby the latter could be waiting continuously.
-The N_WFTmax parameter shall indicate how many FC N_PDU WTs can be transmitted by the receiver in a row.
-The N_WFTmax parameter upper limit shall be SGMW-Specified.
-The N_WFTmax parameter shall only be used on the receiving network entity during message reception.
-If N_WFTmax parameter value is set to zero (0), then flow control shall rely upon flow control continue to send FC N_PDU CTS only. Flow control wait (FC N_PDU WT) shall not be used by that network entity.
使用该参数的目的是避免发送流控消息的节点陷入一种让其余节点持续等待的错误状态。
-N_WFTmax参数表示接收端能够发送的FC N_PDU WT最大次数。
-在消息接收过程中，N_WFTmax参数应（shall）只用于接收端网络实体。
-如果N_WFTmax参数值设为0，那么该网络实体的流控应（shall）使用FC N_PDU CTS，而不能使用FC N_PDU WT。

NOTE: In this specification the value of N_WFTmax is defined to be 0 for tester and ECU.
注：本规范中，测试仪与ECU的N_WFTmax值定义为0.

For the N_PCI field, other more detailed parameter definitions and requirements, with which the tester and the ECUs shall comply, can be found in ISO 15765-2.
对于N_PCI域来说，诊断仪与ECU应（shall）遵从的其它详细参数定义与要求见ISO 15765-2。

7.1.7 Network layer timing/网络层定时

7.1.7.1 Timing parameters/定时参数

Figure 7-3 shows the network layer timing parameters, while the following table (Table 7-12) defines the network layer timing parameter values and their corresponding start and end positions based on the data link layer services.
Figure 7-3中标明了网络层的定时参数，同时，Table 7-12定义了网络层定时参数值，并定义了基于数据链路层服务的这些参数值的起始时刻与停止时刻。

Performance requirement values are the binding communication requirements to be met by each communication peer in order to be compliant with the specification. A certain application may define specific performance requirements within the ranges defined in the following table (Table 7-12).
对每个满足本规范要求的通信实体来说，性能要求参数值是一种必须满足的具有约束力的通信要求。某一特定的应用可能会在Table 7-12中所定义的范围内定义明确的性能请求。

Timeout values are defined to be higher than the values for the performance requirements in order to ensure a working system and to overcome communication conditions where the performance requirement can absolutely not be met (e.g. high bus load). Specified timeout values shall be treated as the lower limit for any given implementation. The real timeout shall occur no later than at the specified timeout value + 50 %. The network layer shall issue an appropriate service primitive to the network layer service user upon detection of an error condition.

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定义超时值大于性能要求参数值的目的是确保系统正常工作并克服性能要求参数值肯定不能满足的通信状态（例如：高总线负载）。对于任何给定的执行步骤来说，定义的超时值应（shall）作为对应定时参数的下限。真正的超时时间应（shall）不迟于指定超时值的50％。当检测到错误条件时，网络层应（shall）向网络层服务用户提供相应的服务原语。

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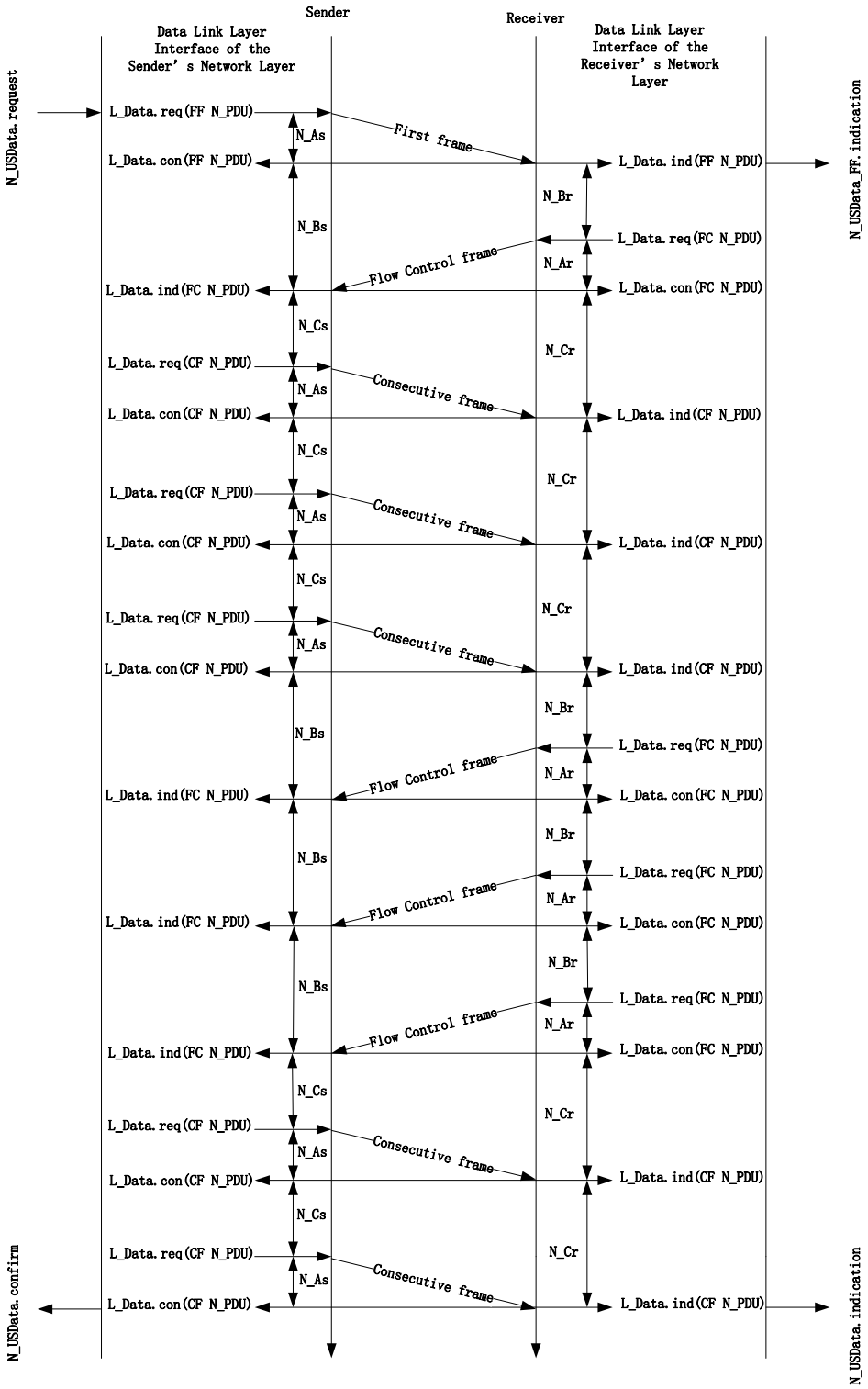

Figure 7-3 Placement of network layer timing parameters/网络层定时参数

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The following table defines network layer timing parameter, which shall be used by the tester and the ECUs.
下表定义了诊断仪与ECU应（shall）使用的网络层定时参数。

Table 7-12 Network layer timing parameters/网络层定时参数

Parameter 参数	Description 描述	Timeout value 超时值	Performance requirement value 性能要求值
N_As / N_Ar	Time from transmit request until a CAN frame transmit confirmation is received. CAN 数据帧经过数据链路层发送的时间。	250ms[1]	-
N_Bs	Time that the transmitter of a multi-frame (MF) message shall wait to receive a flow control (FC) frame before timing out with a network layer error. 发送端接收流控制帧的等待时间。	250ms[1]	-
N_Br	Maximum time for the receiver of a multi-frame message to transmit a flow control frame after receiving the first frame (FF). 接收端发送流控制帧的间隔时间。	-	<100ms[1]
N_Cs	Maximum time from either receiving a flow control continue to send (FC.CTS) frame or receiving the transmit confirmation of last consecutive frame (CF) until transmitting the next CF. 发送端发送连续帧的间隔时间。	-	<100ms[1]
N_Cr	Time that the receiver of a multi-frame (MF) message shall wait to receive a consecutive frame (CF) after sending a flow control (FC) frame or receiving the previous consecutive frame (CF) before timing out with a network layer error. 接收端接收连续帧的等待时间。	250ms[1]	-

1. For OBD ECUs, whose network layer only supports a single set of timing parameters shall support the timing values specified ISO15765-4 (which references ISO15765-2) for both OBD and enhanced diagnostic communication. Non OBD ECUs, and OBD ECUs which can support different timing values for OBD versus enhanced diagnostics, shall support the values in this table for enhanced diagnostic communication.
对于仅支持一组网络层定时参数的OBD ECU来说，它应（shall）支持ISO15765-4（参考ISO15765-2）中对OBD与增强型诊断通信规定的定时参数。非OBD ECU应（shall）支持本表格定义的增强型诊断通信参数值；同样地，如果OBD ECU对于OBD诊断通信与增强型诊断通信使用不同的网络层定时参数，那么它也应（shall）支持本表格定义的增强型诊断通信参数值。

For the network layer timing parameters, other more detailed requirements, with which the tester and the ECUs shall comply, can be found in ISO 15765-2.
对诊断仪与ECU应（shall）遵从的网络层定时参数的其它更详细要求见ISO 15765-2。

7.1.7.2 Network layer timeouts/网络层超时设定

The following table (Table 7-13) defines the cause and action in a network layer timeout.
Table 7-13定义了网络层超时的原因与超时后执行的动作。

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Table 7-13 Network layer timeouts/网络层超时参数

Timeout 超时	Cause 原因	Action 执行动作
N_As	Any N_PDU not transmitted in time on the sender side. 发送端未按时发送任何一个N_PDU。	Abort message transmission and issue N_USData.confirm with <N_Result> = N_TIMEOUT_A. 发送端停止消息发送，并且向相邻的上层发出<N_Result> = N_TIMEOUT_A的N_USData.confirm。
N_Ar	Any N_PDU not transmitted in time on the receiver side. 接收端未按时发送任何一个N_PDU。	Abort message reception and issue N_USData.indication with <N_Result> = N_TIMEOUT_A. 接收端停止消息接收，并且向相邻的上层发出<N_Result> = N_TIMEOUT_A的N_USData.indication
N_Bs	FlowControl N_PDU not received (lost, overwritten) on the sender side or preceding FirstFrame N_PDU or ConsecutiveFrame N_PDU not received (lost, overwritten) on the receiver side. 发送端未接收到（丢失或被覆盖）FC N_PDU或者接收端未接收到（丢失或被覆盖）之前的FF N_PDU或CF N_PDU。	Abort message transmission and issue N_USData.confirm with <N_Result> = N_TIMEOUT_Bs. 发送端停止消息发送，并且向相邻的上层发出<N_Result> = N_TIMEOUT_Bs的N_USData.confirm
N_Cr	ConsecutiveFrame N_PDU not received (lost, overwritten) on the receiver side or preceding FC N_PDU not received (lost, overwritten) on the sender side. 接收端未接收到（丢失或被覆盖）CF N_PDU或者发送端未接收到（丢失或被覆盖）之前的FC N_PDU。	Abort message reception and issue N_USData.indication with <N_Result> = N_TIMEOUT_Cr. 接收端停止消息接收，并且向相邻的上层发出<N_Result> = N_TIMEOUT_Cr的N_USData.indication

7.1.7.3 Unexpected arrival of N_PDU/接收非期望 N_PDU 的处理方式

An unexpected N_PDU is defined as one that has been received by a node outside the expected order of N_PDUs. It could be an N_PDU defined by this part of ISO 15765 (SF N_PDU, FF N_PDU, CF N_PDU or FC N_PDU) that is received out of the normal expected order, or else it could be an unknown N_PDU that cannot be interpreted by the definitions given in this document.
接收端节点未按照期望顺序接收到的N_PDU定义为非期望N_PDU，它可能是ISO 15765（SF N_PDU、FF N_PDU、CF N_PDU或FC N_PDU）定义的没有按照正常期望顺序接收的N_PDU，或者是一种不能被本规范中的定义所解释的无法识别的N_PDU。

Depending on the network layer design decision to support full- or half-duplex communication, the interpretation of “unexpected” differs :
a) with half-duplex, point-to-point communication between two nodes is only possible in one direction at a time;
b) with full-duplex, point-to-point communication between two nodes is possible in both directions at one time.
依据网络层设计策略支持全双工通信还是支持半双工通信的不同，对“非期望”的解释也不同：
a) 在半双工通信模式下，同一时刻两节点之间的点对点通信只能为单向；
b) 在全双工通信模式下，同一时刻两节点之间的点对点通信可能为双向。

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In addition to the network layer design decision, the possibility has to be considered that a reception or transmission from/to a node with the same address information (N_AI), as contained in the received unexpected N_PDU, is in progress.
除了网络层设计策略之外，可能的原因为从某一节点接收或向某一节点发送带有相同地址信息（N_AI）的动作正在执行，并且该N_AI与接收到的非期望N_PDU中包含的一致。

As a general rule, arrival of an unexpected N_PDU from any node shall be ignored, with the exception of SF N_PDUs and physically addressed FF N_PDUs; functionally addressed first frames shall be ignored. When the specified action is to ignore an unexpected N_PDU, this means that the network layer shall not notify the upper layers of its arrival.
一般来说，除了接收SF N_PDU与物理寻址的FF N_PDU以外，接收端应（shall）忽略从任何节点接收到的非期望N_PDU；功能寻址的首帧应（shall）忽略。如果接收端的处理方式为忽略非期望的N_PDU，那么这意味着网络层不应（shall not）把非期望N_PDU的到来通知上一层。

The following table (Table 7-14) defines the network layer behavior in the case of the reception of an unexpected N_PDU, in consideration of the actual network layer internal status (NWL status) and the design decision to support half- or full-duplex communication. It has to be understood that the received N_PDU will contain the same N_AI as the reception or transmission that could be in progress at the time the N_PDU is received.
考虑到实际的网络层内部状态（NWL status）以及支持半双工或全双工通信的设计策略，Table 7-14定义了当接收到一非期望的N_PDU时网络层的行为。

Table 7-14 network layer behavior/网络层行为

NWL status	Reception of				
	SF N_PDU	FF N_PDU	CF N_PDU	FC N_PDU	Unknown N_PDU
Segmented Transmit in Progress 正在进行分段消息发送	Full-duplex: If a reception is in progress, see corresponding cell below in this table, otherwise process the SF N_PDU as the start of a new reception. 全双工： 如果接收正在进行，那么处理方式见本表格内下面的相应单元格，否则，接收端把SF N_PDU作为新接收的开始。	Full-duplex: If a reception is in progress, see corresponding cell below in this table, otherwise process the FF N_PDU as the start of a new reception. 全双工： 如果接收正在进行，那么处理方式见本表格内下面的相应单元格，否则，接收端把FF N_PDU作为新接收的开始。	Full-duplex: If a reception is in progress, see corresponding cell below in this table. 全双工： 如果接收正在进行，那么处理方式见本表格内下面的相应单元格。	If awaited, process the FC N_PDU, otherwise ignore it. 如果发送端在等待FC N_PDU，那么接收端就发送FC N_PDU，否则，接收端将忽略FC N_PDU。	Ignore 忽略
	Half-duplex: Ignore 半双工： 忽略	Half-duplex: Ignore 半双工： 忽略	Half-duplex: Ignore 半双工： 忽略		
Segmented Receive in Progress 正在进行分段消息接收	Terminate the current reception, report an N_USData.indication, with <N_Result> set to N_UNEXP_PDU, to the upper layer, and process the SF	Terminate the current reception, report an N_USData.indication, with <N_Result> set to N_UNEXP_PDU, to the upper layer, and process the FF N_PDU	If awaited, process the CF N_PDU in the on-going reception and perform the required	Full-duplex: If a transmission is in progress, see corresponding cell above in	Ignore 忽略

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	N_PDU as the start of a new reception. 停止当前的接收，向上一层报告一个<N_Result>= N_UNEXP_PDU，的N_USData.indication，并且，接收端把SF N_PDU作为新接收的开始。	as the start of a new reception. 停止当前的接收，向上一层报告一个<N_Result>= N_UNEXP_PDU，的N_USData.indication，并且，接收端把FF N_PDU作为新接收的开始。	checks (e.g. SN in right order), otherwise ignore it. 如果接收端在等待接收，那么在正在进行的接收中接收CF N_PDU，并执行所需的检查，否则，接收端将忽略CF N_PDU。	this table. 全双工： 如果发送正在进行，那么处理方式见本表格内上面的相应单元格。 Half-duplex: Ignore 半双工： 忽略	
Idle[a] 空闲	Process the SF N_PDU as the start of a new reception.	Process the FF N_PDU as the start of a new reception.	Ignore 忽略	Ignore 忽略	Ignore 忽略
a. Neither a segmented transmission nor segmented reception is in progress. 未进行分段消息的发送与接收。					

7.1.7.4 Wait frame error handling/等待帧错误处理

When the receiver has transmitted N_WFTmax flow control wait network protocol data units (FC N_PDU WT) in a row and following this it cannot meet the performance requirement for the transmission of a flow control continue to send network protocol data unit (FC NPDU CTS), then the receiver side shall abort the message reception and issue an N_USData.indication with <N_Result> set to N_WFT_OVRN to the higher layer.
当接收端已经连续发送了N_WFTmax “流控等待网络协议数据单元（FC N_PDU WT）”，并且当前状态与发送“流控继续发送网络协议数据单元（FC NPDU CTS）”的执行条件不匹配，那么接收端应（shall）该停止消息的接收并且使用<N_Result> = N_WFT_OVRN的N_USData.indication服务原语向上层指示。

The sender of the message is informed about the aborted message reception via an N_USData.confirm with <N_Result> set to N_TIMEOUT_Bs (because of the missing FlowControl N_PDU from the receiver an N_Bs timeout occurs in the sender).
接收端通过<N_Result> =N_TIMEOUT_Bs的N_USData.confirm服务，把消息接收的停止告诉消息的发送端（由于缺少来自消息接收端的FC N_PDU，导致发送端N_Bs计时器超时）。

7.1.7.5 Interleaving of messages/消息的穿插

The network layer protocol shall be capable of carrying out parallel transmission of different messages that are not mapped onto the same N_AI. This is necessary to ensure that the receiving peer is able to reassemble in a consistent manner the received network protocol data units. This scheme for example enables gateway operation that needs to handle different message transmissions concurrently across distinct sub-networks.
网络层协议应（shall）具有一种能力，通过这种能力，能够执行映射到不同N_AI的不同消息的并行传输。这对于确保接收实体按照固定的方法重组接收到的网络协议数据单元来说是很有必要的。例如，该方案使能的需要处理同时跨越不同子网的不同消息传输的网关操作。

7.2 Data Link Layer usage/数据链路层的使用

This section defines the requirements for the usage of the data link layer services of the diagnostic

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communication system.
本节定义了诊断通信系统数据链路层服务使用要求。

7.2.1 Mapping of the N_PDU fields/ N_PDU 场的映射

This section specifies the mapping mechanisms for addressing format, based on the data link layer services and service parameters defined in section 6.
基于数据链路层服务与第6章定义的服务参数，本节定义了指定编址格式下N_PDU场的映射机制。

NOTE: In this document, the exchange of network layer data is supported by only one addressing format: normal addressing, which is defined in ISO 15765-2.
注: 本规范网络层的数据交互仅支持一种编址格式：正常编址，此编址格式在 ISO 15765-2 内做了定义。

The mapping of N_PDU parameters into CAN frame at the case of normal addressing as following table (Table 7-15).
在正常编址情况下，N_PDU 与 CAN 数据帧的映射如 Table 7-15 所示。

Table 7-15 Mapping of the N_PDU fields/ N_PDU 场域的映射

N_PDU type	CAN Identifier	CAN frame data field			
		Byte1	Byte2	Byte3	Byte4-8
Single Frame (SF)	N_AI	N_PCI	N_Data		
First Frame (FF)	N_AI	N_PCI		N_Data	
Consecutive Frame (CF)	N_AI	N_PCI	N_Data		
Flow Control (FC)	N_AI	N_PCI			N/A

For the mapping of the N_PDU fields, other more detailed requirements, with which the tester and the ECUs shall comply, can be found in ISO 15765-2.
对于N_PDU域的映射来说，诊断仪与ECU需遵从的其它更详细要求见ISO 15765-2。

7.2.2 Mapping between N_AI and diagnostic CAN identifiers (CANIds)/ N_AI 与诊断 CANIds 的映射

Diagnostic CAN Identifiers (CANIds) are required in order to separate diagnostic messages from Normal Communication messages.
使用诊断CAN标识符（CANId）的目的是把诊断消息从正常通信消息中分离出来。

The diagnostic CANId in a request message determines the type of addressing used for that message, either physically addressed (targeted to only a single ECU) or functionally addressed (targeted to one or multiple ECUs).
请求信息中的诊断CANId规定了该消息所使用的寻址类型：物理寻址（指向单一的ECU）或者功能寻址（指向一个或多个ECU）。

Correspondingly, the diagnostic CANId in a response message determines the addressed ECU used for the request message, such as: either OBD ECU (emission-related ECU) or other ECU (non emission-related ECU) etc.
相应地，响应消息中的诊断CANId规定了请求消息中被寻址的ECU，例如：OBD相关的ECU（排放相关的ECU）或者其余ECU（非排放相关的ECU）等。

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7.2.3 CAN frame Data Length Code (DLC)/CAN 帧 DLC

7.2.3.1 DLC parameter/DLC 参数

The DLC parameter specifies the number of data bytes transmitted in a CAN frame. This part of ISO 15765 does not specify any requirements concerning the length of the data field in a CAN frame other than those implied by the size of the network layer protocol data units.
DLC参数表示传输的CAN帧中包含的数据字节个数。ISO 15765中除了规定网络层协议数据单元大小外，没有对一个CAN数据帧中数据场的长度做任何要求。

7.2.3.2 CAN frame data padding/CAN 帧数据填充

The DLC is always set to 8. If the N_PDU to be transmitted is shorter than 8 bytes, then the sender has to set the DLC to the maximum value 8 (padding of unused data bytes). In particular this can be the case for an SF, FC frame or last CF of a segmented message.
DLC的值总为8，如果发送的N_PDU中的数据长度小于8，那么发送端需要把DLC设为最大值8（填充未使用数据字节）。尤其是对于SF、FC或分段消息的最后一个CF。

The DLC parameter of the CAN frame is set by the sender and read by the receiver to determine the number of data bytes per CAN frame to be processed by the network layer. The DLC parameter cannot be used to determine the message length; this information shall be extracted from the N_PCI information in the beginning of a message.
发送端设置CAN数据帧中的DLC参数，接收端通过读取该DLC参数来确定网络层要处理的每个CAN数据帧中的数据字节数，DLC参数不能用来确定消息长度；消息长度应（shall）从消息开始时的N_PCI信息中提取。

NOTE: In this document, the tester and the ECUs shall comply the following definitions and requirements:
a. The CAN DLC contained in every diagnostic CAN frame shall always be set to eight (8). Any diagnostic CAN frame with a DLC less than eight (8) shall be ignored by the received entity of both tester and ECUs.
b. Tester shall always use values of 0x55 in its request message, and ECU shall always use values of 0xAA in its response message to pad the unused data bytes
注：本规范中，诊断仪与ECU应（shall）遵从以下定义与要求：
a. 每个诊断CAN数据帧中的DLC应（shall）设为8，诊断仪与ECU的接收实体应（shall）忽略任何DLC小于8的诊断CAN数据帧。
b. 诊断仪的请求消息中总是使用0x55（Hex）来填充未使用字节，ECU的响应消息中总是使用0xAA（Hex）来填充未使用字节。

7.2.3.3 Data Length Code error handling/DLC 错误处理

Depending on the N_PCI value, the network layer can calculate the smallest expected value for the CAN DLC parameter in a received CAN frame.
The reception of a CAN frame with a DLC value smaller than expected (less than 8 for applications which pad the CAN frames) shall be ignored by the network layer without any further action.
依据N_PCI的值，网络层能够计算出接收的CAN数据帧中DLC参数的最小期望值。
接收的CAN数据帧中DLC长度小于期望值（使用填充规则时小于8）时，网络层应（shall）忽略该CAN数据帧。

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8 Application and session layers requirement/应用层与会话层要求

This section defines the requirements for the Application and Session layer services of the diagnostic communication system.
This section is based on ISO 15765-3 and ISO 14229-1, which defines the implementation of unified diagnostic services (UDS) on CAN bus. For this, the on-vehicle ECUs shall comply with the requirement of server in ISO 15765-3 and ISO 14229-1; correspondingly, the tester shall comply with the requirement of the client in ISO 15765-3 and ISO 14229-1.
本章定义了诊断通信系统的应用层和会话层的要求。
本章的相关技术要求是基于ISO 15765-3与ISO 14229-1标准（这两个标准规定了统一诊断服务（UDS）在CAN总线实现的相关要求）而制定的。因此，车载ECU应（shall）遵从ISO 15765-3中对服务器的要求，同时，诊断仪应（shall）遵从ISO 15765-3和ISO 14229-1中对客户端的要求。

8.1 Network layer interface/网络层接口

This section defines the requirements for the interface between the network layer and the application layer of the diagnostic communication system.
Requirements of network layer interface, with which the tester and the ECUs shall comply, can be found in ISO 15765-3.
本章节定义了诊断通信系统的网络层与应用层之间的接口要求。
ISO 15765-3中规定了诊断仪与ECU应（shall）遵从的网络层接口要求。

8.2 Application and session layer/应用层与会话层

This section defines the requirements for the application and session layer protocol of the diagnostic communication system.
Other than the requirement defined in the following subsections, more detailed requirements of application and session layer, with which the tester and the ECUs shall comply, can be found in ISO 15765-3 and ISO 14229-1.
本章节定义了诊断通信系统的应用层与会话层协议要求。
除下述章节定义的要求之外，诊断仪与ECU应（shall）遵从的应用层与会话层详细要求请参考ISO 15765-3和ISO 14229-1.

8.2.1 Application layer protocol/应用层协议

8.2.1.1 Service identifier (SID) overview/服务标识符（SID）概述

Each service shall be identified by its service Identifier (SID). The SID must be included in each diagnostic request message.
The table below (Table 8-1) provides an overview about Service Identifier (SID) used by different protocols in relation to SGMW SID assignments.
每个服务的标识符（SID）标识了该服务本身。每个诊断请求消息必须（must）包含一个SID。
在参考多个不同协议的基础上，SGMW定义了自己的SID配置，Table 8-1概述了所有的服务标识符（SID）的配置情况。

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Table 8-1 Service identifier overview/服务标识符概述

Service Identifier 服务标识符 (Hex Value)	Service type 服务类型 (bit 6)	Where defined 参考标准
00 - 0F	OBD service requests (bit 6 = 0)	ISO 15031-5
10 - 3E	ISO 14229 service requests (bit 6 = 0)	ISO 14229,this document
3F	Not applicable	Reserved by document
40 - 4F	OBD service Positive Response (bit 6 = 1)	ISO 15031-5
50 - 7E	ISO 14229 Positive Response (bit 6 = 1)	ISO 14229, this document
7F	Negative Response service identifier	ISO 14229, this document
80	Not applicable	Reserved by ISO 14229
81 - 82	Not applicable	Reserved by ISO 14229
83 - 88	ISO 14229 service requests (bit 6 = 0)	ISO 14229, this document
89 - 9F	Service requests (bit 6 = 0)	Reserved for future expansion as needed
A0 - B9	Service requests (bit 6 = 0)	Defined by vehicle manufacturer
BA - BE	Service requests (bit 6 = 0)	Defined by system supplier
BF	Not applicable	Reserved by document
C0	Not applicable	Reserved by ISO 14229
C1 - C2	Not applicable	Reserved by ISO 14229
C3 - C8	ISO 14229 Positive Response (bit 6 = 1)	ISO 14229, this document
C9 - DF	Positive Response (bit 6 = 1)	Reserved for future expansion as needed
E0 - F9	Positive Response (bit 6 = 1)	Defined by vehicle manufacturer
FA - FE	Positive Response (bit 6 = 1)	Defined by system supplier
FF	Not applicable	Reserved by document

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8.2.1.2 Diagnostic Service Table Overview/诊断服务概述

This chapter gives a fundamental overview of the diagnostic services.
There are three sessions defined for this specification, which are Default Session, programming Session and Extended Session.
本章节列出了诊断服务的基本概述。
本规范定义了3种会话模式：默认会话模式、编程会话模式与扩展会话模式。

The following table (Table 8-2) indicates the addressed formats which shall be supported for each service. The table followed shows the services which are allowed during the defaultSession, programmingSession and extendedDiagnosticSession. An ECU is allowed to support more other diagnostic services per session after confirmed by SGMW.
Table 8-2明确了每个服务应（shall）支持的寻址模式，并列出了在3种会话模式下所支持的服务。如要增加某个ECU在每个会话模式下支持的服务，需经过SGMW确认。

Table 8-2 Diagnostic Service Table Overview/诊断服务列表概述

Service Name 服务名称	SID (Hex)	Sub-function/子功能 (Hex)	Session/会话			Security Access[1] /安全访问			Addressing method/寻址方式	
			Default 默认	Programming 编程	Extended 扩展	Level 1 等级 1	Level 2 等级 2	Level 3 等级 3	Physical 物理	Functional 功能
Diagnostic and communication management functional unit/诊断与通信管理功能单元										
DiagnosticSessionControl	10	01	Y[2]	Y	Y	N[2]	N	N	Y	Y
		02	N	Y	Y	N	N	N	Y	N
		03	Y	N	Y	N	N	N	Y	Y
ECUReset	11	01	Y	Y	Y	N	N	N	Y	Y
		03	Y	Y	Y	N	N	N	Y	Y
SecurityAccess	27	01/02	N	N	Y	N	N	N	Y	N
		03/04	N	N	Y	N	N	N	Y	N

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Service Name 服务名称	SID (Hex)	Sub-function/子功能 (Hex)	Session/会话			Security Access[1] /安全访问			Addressing method/寻址方式	
			Default 默认	Programming 编程	Extended 扩展	Level 1 等级 1	Level 2 等级 2	Level 3 等级 3	Physical 物理	Functional 功能
		05/06	N	Y	N	N	N	N	Y	N
CommunicationControl	28	00/03	N	N	Y	N	N	N	Y	Y
TesterPresent	3E	00/80	Y	Y	Y	N	N	N	Y	Y
AccessTimingParameter	83	-	-	-	-	-	-	-	-	-
SecuredDataTransmission	84	-	-	-	-	-	-	-	-	-
ControlDTCSetting	85	01/02	N	N	Y	N	N	N	Y	Y
ResponseOnEvent	86	-	-	-	-	-	-	-	-	-
LinkControl	87	01/83	N	N	Y	N	N	N	Y	Y
ProgrammingMode	A5	01/03	N	N	Y	N	N	N	Y	Y
Data transmission functional unit/数据传输功能单元										
ReadDataByIdentifier	22	-	Y	Y	Y	N	N	N	Y	Y[3]
ReadMemoryByAddress	23	-	-	-	-	-	-	-	-	-
ReadScalingDataByIdentifier	24	-	-	-	-	-	-	-	-	-
ReadDataByPeriodicIdentifier	2A	-	-	-	-	-	-	-	-	-
DynamicallyDefineDataIdentifier	2C	01/03	Y	N	Y	N	N	N	Y	N
WriteDataByIdentifier	2E	-	N	Y	Y	Y[4]	Y[4]	Y[4]	Y	N
WriteMemoryByAddress	3D	-	-	-	-	-	-	-	-	-
Stored data transmission functional unit/存储数据传输功能单元										
ClearDiagnosticInformation	14	-	Y	N	Y	N	N	N	Y	Y
ReadDTCInformation	19	01/02/03/04/06/0A	Y	N	Y	N	N	N	Y	Y

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Service Name 服务名称	SID (Hex)	Sub-function/子功能 (Hex)	Session/会话			Security Access[1] /安全访问			Addressing method/寻址方式	
			Default 默认	Programming 编程	Extended 扩展	Level 1 等级 1	Level 2 等级 2	Level 3 等级 3	Physical 物理	Functional 功能
InputOutput control functional unit/输入输出控制功能单元										
InputOutputControlByIdentifier	2F	-	N	N	Y	Y/N[5]	N	N	Y	N
Remote activation of routine functional unit/程序远程激活功能单元										
RoutineControl	31	01/02/03	N	Y	Y	Y/N[6]	Y/N[6]	Y/N[6]	Y	N
Upload download functional unit/上传下载功能单元										
RequestDownload	34	-	N	Y	N	N	N	Y	Y	N
RequestUpload	35	-	-	-	-	-	-	-	-	-
TransferData	36	-	N	Y	N	N	N	N	Y	N
RequestTransferExit	37	-	N	Y	N	N	N	N	Y	N

注：

1. The definition of the security access level can be found in chapter 10.3.2.2 of this specification.
   有关安全访问等级的定义可参见本规范中的第 10.3.2.2 章节。
2. 'Y': YES; 'N': NO.
   "Y"：支持；"N"：不支持。
3. If the ReadDataByIdentifier (22 hex) service request message is a single frame, the functional addressing method shall be supported; otherwise, the functional addressing method is not required to be supported.
   如果通过标识符读数据（22 hex）服务请求消息为单帧，则应（shall）支持功能寻址，否则，不需要支持功能寻址。
4. All the WriteDataByIdentifier (2E hex) service in all ECUs shall need the security access, and the security level needed by this service depends on each DID, which is defined in the individual ECU diagnostic specification. For those ECU/vehicle DIDs belonged to vehicle-manufacturer-specific DID, their security level used to be written is defined in annex C.2 of this specification.
   所有 ECU 的通过标识符写数据（2E hex）服务均应（shall）需要安全访问，该服务所需的安全等级取决于（ECU 独立诊断规范中定义的）具体 DID。对于那些属于企业级 DID 的 ECU/车辆标识符选项 DID，它们所使用的"写"安全等级在本规范中的附录 C.2 进行了定义。
5. For some ECUs, the InputOutputControlByIdentifier (2F hex) service may need the security access, the definition whether this service needs the security or not and (if need) the security level needed by the service depends on each DID, which is defined in the individual ECU diagnostic specification.
   对于某些 ECU 而言，通过标识符进行输入输出控制（2F hex）服务可能（may）需要安全访问，该服务是否需要安全及其所需的安全等级取决于

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Service Name 服务名称	SID (Hex)	Sub-function/子功能 (Hex)	Session/会话			Security Access[1]/安全访问			Addressing method/寻址方式	
			Default 默认	Programming 编程	Extended 扩展	Level 1 等级 1	Level 2 等级 2	Level 3 等级 3	Physical 物理	Functional 功能
（ECU 独立诊断规范中定义的）具体 DID。 6. For some ECUs, the RoutineControl (31 hex) service may need the security access, the definition whether this service needs the security or not and (if need) the security level needed by the service depends on each RID, which is defined in the individual ECU diagnostic specification. 对于某些 ECU 而言，例程控制（31 hex）服务可能（may）需要安全访问，该服务是否需要安全及其所需的安全等级取决于（ECU 独立诊断规范中定义的）具体 RID。										

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8.2.1.3 Diagnostic communication implementation rules for tester/诊断仪的诊断通信执行规则

Pre-definitions:
In applying the rules which follow, a diagnostic request is considered in process, until the ECU has fulfilled one of the following (depending on the specific service requested):
- The ECU sent the last consecutive frame of an USDT multi-frame positive response;
- The ECU sent an USDT negative response message indicating there is no positive response forthcoming (response code other than $78);
- The ECU sent an USDT single frame positive response message.

预定义：
应用下列规则时，在ECU执行下列任一动作之前（取决于具体的服务请求），假定ECU一直在处理诊断请求：
ECU发送未确认分段数据传输（USDT）多帧肯定响应的最后一个后续帧；
ECU发送一帧否定响应码不为$78的USDT否定响应消息以表明后续无肯定响应消息；
ECU发送一个USDT的单帧肯定响应消息。

Diagnostic communication implementation rules for tester as following:
A tester is not allowed to send a physically addressed request any time a previous physically addressed request is in process.
1. A tester is not allowed to send a physical request any time a functionally request has been received that will result in a response, and the response to the functionally request is in process.
2. A tester is not allowed to send a functionally request that results in a response if a previous physically addressed request is in process.
3. A tester is not allowed to send a functionally request that results in a response if a previous functionally addressed request is in process.
4. After sending a functionally addressed request which will result in a response, a tester is not allowed to send a second functionally addressed request that does not result in a response (e.g., TesterPresent request) until at least 30 msec has elapsed after sending the first request.
5. A tester shall wait a minimum of 30 msec after sending a functionally addressed request that does not result in a response before sending another functionally addressed request message.
6. A tester can send a functionally addressed request message that does not result in a response (e.g. functional tester present) at any point in time during which the ECU is in the process of receiving or responding to a physically addressed diagnostic request (In other words the functionally addressed tester present request may even interleave the transmission of frames of a multi-frame physical request message or a multi-frame physical response message).
7. A tester may also send a diagnostic physical request immediately following a diagnostic functionally request as long as the functionally request does not require a response.

诊断仪的诊断通信执行规则如下：
在先前的物理寻址请求正在执行期间，不允许诊断仪发送针对同一 ECU 的物理寻址请求。
要求肯定响应的功能寻址请求已经接收到，并且功能寻址请求的响应正在处理期间，不允许诊断仪发送任何物理寻址请求。
在先前的物理寻址请求处理期间，不允许诊断仪发送要求肯定响应的功能寻址请求。
在先前的功能寻址请求处理期间，不允许诊断仪发送要求肯定响应的功能寻址请求。
在发送一个要求肯定响应的功能寻址请求之后 30ms 内，不允许诊断仪发送第二个不要求肯定响应的功能寻址请求（例如：诊断仪在线服务请求（3E hex））。
在发送一个不要求肯定响应的功能寻址请求之后 30ms 内，不允许诊断仪再发送另一个功能寻址请求。

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在 ECU 处于接收或响应物理寻址诊断请求期间，诊断仪可以在任意时间点发送不要求肯定响应的功能寻址请求（例如：诊断仪在线服务请求（3E hex））。（换而言之，功能寻址的诊断仪在线服务请求可以穿插在多帧物理寻址请求或多帧物理寻址响应之间）。
在发送不要求肯定响应的功能寻址请求之后，诊断仪可立即发送一个物理寻址请求。

The following table (Table 8-3) provides an additional overview about the required handling in the tester also taking into consideration the ECU specific rules given in section 8.2.1.3.The “Previous Request” column refers to the last diagnostic message sent by the tester and “New Request” column refers to the diagnostic message that the tester is attempting to send next.
在8.2.1.3章节给出的ECU具体规则基础上，Table 8-3提供了诊断仪关于寻址请求处理的附加描述。“先前的请求”列是诊断仪之前发送的诊断报文，“新的请求”列是诊断仪将要发送的诊断报文。

Table 8-3 Tester communication implementation rules

New Request addressed type	Previous Request addressed type	Handling in the tester
physically	physically	Wait until the completion of the previous physically request (pos. response or negative response other than $78) before transmitting a new physically request.
physically	functionally with response	Wait until the completion of the previous functionally request that requires a response (pos. response or negative response other than $78) before transmitting the physically request.
physically	functionally without response	The tester is allowed to transmit the physically request immediately (e.g. functionally TesterPresent followed by any physically request).
functionally without response	physically	The tester is allowed to transmit the functionally request that does not require a response immediately (e.g. any physically request followed by a functionally TesterPresent).
functionally without response	functionally with response	Wait 30ms after the transmission of the functionally request that requires a response before transmitting the functionally request that does not require a response.
functionally without response	functionally without response	Wait 30ms after the transmission of the functionally request that does not require a response before transmitting the next functionally request that does not require a response.
functionally with response	physically	Wait until the completion of the previous physically request (pos. response or negative response other than $78) before transmitting the functionally request that requires a response.
functionally with response	functionally with response	Wait until the completion of the previous functionally request that requires a response (pos. response or negative response other than $78) before transmitting the next functionally request that requires a response.
functionally with response	functionally without response	Wait 30ms after the transmission of the functionally request that does not require a response before transmitting the functionally request that requires a response.

Table 8-3 诊断仪通信执行规则

新的寻址请求类型	先前的寻址请求类型	诊断仪处理方式
物理寻址请求	物理寻址请求	在完成先前的物理寻址请求（发出肯定响应或除$78 之外的否定响应）之后，才能发送新的物理寻址请求。

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新的寻址请求类型	先前的寻址请求类型	诊断仪处理方式
物理寻址请求	要求肯定响应的功能寻址请求	在完成先前的要求肯定响应的功能寻址请求（发出肯定响应或除$78 之外的否定响应）之后，才能发送新的物理寻址请求。
物理寻址请求	不要求肯定响应的功能寻址请求	允许诊断仪立刻发送物理寻址请求（例如：功能寻址的“诊断仪在线”请求后，紧跟着一个物理寻址请求）。
不要求肯定响应的功能寻址请求	物理寻址请求	允许诊断仪立刻发送不要求肯定响应的功能寻址请求（例如：任何一个物理寻址请求后，可以紧跟一个功能寻址的“诊断仪在线”请求）。
不要求肯定响应的功能寻址请求	要求肯定响应的功能寻址请求	在发送一个要求肯定响应的功能寻址请求后，需要等待30ms，才能发送不要求肯定响应的功能寻址请求。
不要求肯定响应的功能寻址请求	不要求肯定响应的功能寻址请求	在发送先前的不要求肯定响应的功能寻址请求后，需要等待 30ms，才能发送下一个不要求肯定响应的功能寻址请求。
要求肯定响应的功能寻址请求	物理寻址请求	在完成先前的物理寻址请求（发出肯定响应或除$78 之外的否定响应）之后，才能发送要求肯定响应的功能请求。
要求肯定响应的功能寻址请求	要求肯定响应的功能寻址请求	在完成先前的要求肯定响应的功能寻址请求（发出肯定响应或除$78 之外的否定响应）之后，才能发送下一个要求肯定响应的功能寻址请求。
要求肯定响应的功能寻址请求	不要求肯定响应的功能寻址请求	在发送不要求肯定响应的功能寻址请求后，需要等待30ms，才能发送要求肯定响应的功能寻址请求。

8.2.1.4 Diagnostic communication implementation rules for ECU/ECU 的诊断通信执行规则

1. The ECU shall be able to receive and process all tester requests which comply with the above rules.
2. If a polling rate is used by an ECU to service CAN controller message object(s), then the polling rate for diagnostic message object(s) shall be < 30msec.

ECU应（shall）能接收并处理来自诊断仪的遵从上述规则的所有请求。
如果ECU采用轮询方式来检测CAN控制器消息，那么诊断消息的轮询周期应（shall）小于30ms。

Note: The above rule provides the tester a fixed maximum wait time between requests to avoid overwriting diagnostic requests in the ECUs receive message object(s). This wait time does not apply to the case where a physically request is followed by a functionally request that does not require a response (e.g. physically request followed by a functionally TesterPresent) or the case where a functionally request that does not require a response is followed by a physically request (e.g. functionally TesterPresent followed by a physically request) – see section 8.2.1.3.
注：上述规则规定了诊断仪相邻两次请求之间的最大等待时间，这样就避免了在ECU接收消息时，新的诊断请求消息覆盖先前的诊断请求消息。此等待时间不用于物理寻址请求之后跟着一个不要求肯定响应的功能寻址请求（例如：物理寻址请求之后跟着一个“诊断仪在线”的功能寻址请求）或一个不要求肯定响应的功能寻址请求之后跟着一个物理寻址请求的情况（例如：“诊断仪在线”功能寻址请求之后跟着一个物理寻址请求），见章节8.2.1.3。

8.2.1.5 ECU response implementation rules/ECU 响应执行规则

For ECU response implementation rules, with which the tester and the ECUs shall comply, can be found in ISO 14229-1.

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ISO 14229-1中规定了ECU应（shall）遵从的有关ECU响应执行方面的规则。

8.2.2 Application layer and diagnostic session management timing/应用层和诊断会话管理时间参数

This section defines the requirements for the application and session layer protocol of the diagnostic communication system.
Other than the requirement defined in the following subsections, more detailed requirements of application and session layer, with which the tester and the ECUs shall comply, can be found in ISO 15765-3.
本章节定义了诊断通信系统的应用层与会话层协议要求。
除了下述章节所定义的要求之外，诊断仪与ECU应（shall）遵从的其他详细的应用层与会话层要求见ISO 15765-3。

8.2.2.1 Application layer timing parameter definitions/应用层时间参数定义

The following table (Table 8-4) defines the timing parameter of application layer, which shall be used by the tester and the ECUs.
诊断仪和ECU应（shall）遵从Table 8-4定义的应用层时间参数。

Table 8-4 Application layer timing parameter definitions

Timing Parameter	Description	Type	Min.	Max.
$P2_{CAN_Tester}$	Timeout for the tester to wait after the successful transmission of a request message (indicated via N_USData.con) for the start of related incoming response messages (N_USDataFF.ind for a multi-frame message or N_USData.ind for a Single Frame message). Name of Parameter in ISO 15765-3: $P2_{CAN_Client}$	Timer reload value	$P2_{CAN_ECU_max}$ +$P2_{CAN}$[1]	N/A[1]
$P2^*_{CAN_Tester}$	Enhanced timeout for the tester to wait after the reception of a negative response message with response code 78 hex (indicated via N_USData.ind) for the start of incoming response messages (N_USDataFF.ind for a multi-frame message or N_USData.ind for a Single Frame message). Name of Parameter in ISO 15765-3: $P2^*_{CAN_Client}$	Timer reload value	$P2^*_{CAN_ECU_max}$ +$P2_{CAN_rsp}$[1]	N/A[1]
$P2_{CAN_ECU}$	Performance requirement for the ECU to start with the response message after the reception of a request message (indicated via N_USData.ind). The ECU has to make sure that a single frame response message, or the first frame of a multi-frame response message is completed (successfully transmitted onto the CAN network) within $P2_{CAN_ECU}$. Name of Parameter in ISO 15765-3: $P2_{CAN_Server}$	Performance requirement	0	50ms
$P2^*_{CAN_ECU}$	Performance requirement for the ECU to start with a further response message after the transmission of a negative response message (indicated via N_USData.con) with response code 78 hex (enhanced response timing).	Performance requirement	0	5000ms

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Timing Parameter	Description	Type	Min.	Max.
	Name of Parameter in ISO 15765-3: $P2^*_{CAN_Server}$			
$P3_{CAN_Tester_Phys}$	Minimum time for the tester to wait after the successful transmission of a physically addressed request message (indicated via N_USData.con) with no response required (Suppress Pos Rsp Msg Indication Bit = TRUE) before it is allowed to transmit the next physically addressed request Message. Name of Parameter in ISO 15765-3: $P3_{CAN_Client_Phys}$	Timer reload value	$P2_{CAN_ECU_max}$[1]	N/A[1]
$P3_{CAN_Tester_Func}$	Minimum time for the tester to wait after the successful transmission of a functionally addressed request message (indicated via N_USData.con) before it is allowed to transmit the next functionally addressed request message. Name of Parameter in ISO 15765-3: $P2_{CAN_Client_Func}$	Timer reload value	$P2_{CAN_ECU_max}$[1]	N/A[1]
1. For details see ISO 15765-3.				

For the application layer timing parameters, other more detailed requirements, with which the tester and the ECUs shall comply, can be found in ISO 15765-3.

Table 8-4 应用层时间参数定义

定时参数	描述	类型	最小值	最大值
$P2_{CAN_Tester}$	从诊断仪成功发送请求消息（通过N_USData.con指示）到开始接收相应的响应消息（多帧消息的N_USDataFF.ind或者单帧消息的N_USData.ind）之间的定时器超时数值。 注：ISO 15765-3中的参数名称为：$P2_{CAN_Client}$	定时器重载值	$P2_{CAN_ECU_max}$ +$P2_{CAN}$[1]	N/A[1]
$P2^*_{CAN_Tester}$	从诊断仪接收到NRC为0x78的否定响应消息（通过N_USData.ind指示）到开始接收下一个响应消息（多帧消息的N_USDataFF.ind或者单帧消息的N_USData.ind）之间的增强型定时器超时数值。 注：ISO 15765-3中的参数名称为：$P2^*_{CAN_Client}$	定时器重载值	$P2^*_{CAN_ECU_max}$ +$P2_{CAN_rsp}$[1]	N/A1
$P2_{CAN_ECU}$	从ECU接收到请求消息（通过N_USData.ind指示）到开始发送响应消息之间的定时器性能要求数值。ECU必须确保一个单帧响应消息或者多帧响应消息的第一帧消息在$P2_{CAN_ECU}$ 时间内完成（成功发送到CAN网络上）。 注：ISO 15765-3中的参数名称为：$P2_{CAN_Server}$	性能要求	0	50ms
$P2^*_{CAN_ECU}$	从ECU发送了NRC为0x78的否定响应消息（通过N_USData.con指示）到开始发送下一个响应消息之间的增强型定时器性能要求数值。 注：ISO 15765-3中的参数名称为： $P2^*_{CAN_Server}$	性能要求	0	5000ms
$P3_{CAN_Tester_Phys}$	从诊断仪成功发送一个不要求肯定响应（抑制肯定响应消息指示位为真）的物理寻址请求消息（通过N_USData.con确认）到允许发送下一个物理寻址请求消息之间的最小间隔时间。 注：ISO 15765-3中的参数名称为：$P3_{CAN_Client_Phys}$	定时器重载值	$P2_{CAN_ECU_max}$[1]	N/A[1]
$P3_{CAN_Tester_Func}$	从诊断仪成功发送一个功能寻址请求消息（通过N_USData.con确认）到允许发送下一个功能寻址请求消息之间的最小间隔时间。	定时器重载值	$P2_{CAN_ECU_max}$[1]	N/A[1]

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定时参数	描述	类型	最小值	最大值
	注：ISO 15765-3中的参数名称为：$P2_{CAN_Client_Func}$			
1. 详细内容请见 ISO 15765-3。				

诊断仪和 ECU 需要遵从的其它详细的应用层时间参数要求，请参考 ISO 15765-3。

8.2.2.2 Session layer timing parameter definitions/会话层时间参数定义

The following table (Table 8-5) defines session layer timing parameter, which shall be used by the tester and the ECUs.
诊断仪和 ECU 都应（shall）遵从 Table 8-5 所定义的会话层时间参数。

Table 8-5 Session layer timing parameter definitions

Timing Parameter	Description	Type	Recommended timeout	Timeout
$S3_{Tester}$	Time between functionally addressed TesterPresent (3E hex) request messages transmitted by the tester to keep a diagnostic session other than the defaultSession active in multiple ECUs (functional communication) or maximum time between physically transmitted request messages to a single ECU (physical communication). Name of Parameter in ISO 15765-3: $S3_{Client}$	Timer reload value	2000ms	4000ms
$S3_{ECU}$	Time for the ECU to keep a diagnostic session other than the defaultSession active while not receiving any diagnostic request message. Name of Parameter in ISO 15765-3: $S3_{server}$	Timer reload value	N/A	5000ms

Table 8-5 会话层时间参数定义

定时参数	描述	类型	推荐超时	超时
$S3_{Tester}$	在功能通信方式下，为了使得多个ECU保持在非默认会话状态，诊断仪发送功能寻址的“诊断仪在线”请求消息之间的时间为$S3_{Tester}$。 或者，在物理通信方式下，针对单一ECU，诊断仪发送的物理寻址请求消息之间的最大间隔时间为$S3_{Tester}$。 注：ISO 15765-3中的参数名称为：$S3_{Client}$	定时器重载值	2000ms	4000ms
$S3_{ECU}$	在没有接收到任何诊断请求信息时，ECU保持在非默认会话状态下的时间为$S3_{ECU}$。 注：ISO 15765-3中的参数名称为：$S3_{server}$	定时器重载值	N/A	5000ms

For the session layer timing parameters, other more detailed requirements, with which the tester and the ECUs shall comply, can be found in ISO 15765-3.
诊断仪和 ECU 需要遵从的其它详细的会话层时间参数要求，请参考 ISO 15765-3。

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8.2.2.3 Tester and ECU timing resource requirements/诊断仪与 ECU 定时器资源要求

The tester and the ECUs shall provide the respective timer resources listed in the following table (Table 8-6) in order to meet the application layer timing requirements given in section above.
诊断仪和 ECU 应当提供各自如 Table 8-6 所示的定时器资源，用于满足上述章节定义的应用层定时要求。

Table 8-6 Tester and ECU timing resource requirements (for application layer)

Timing Parameter	Tester	ECU
$P2_{CAN_Tester}$	A single timer is required for each logical communication channel (physical and functional communication).	N/A
$P2_{CAN_ECU}$	N/A	An optional timer might be required for the enhanced response timing in order to ensure that subsequent negative response messages with response code 78 hex are transmitted prior to the expiration of $P2^*_{CAN_ECU}$.
$P3_{CAN_Physical}$	A single timer is required per logical physical communication channel.	N/A
$P3_{CAN_functional}$	A single timer is required per logical functional communication channel.	N/A

Table 8-6 诊断仪与 ECU 定时器资源要求（对应用层来说）

定时参数	诊断仪	ECU
$P2_{CAN_Tester}$	需要为每一个逻辑通信通道（物理和功能通信）设置一个单独的定时器。	N/A
$P2_{CAN_ECU}$	N/A	为了确保随后的NRC为$78（hex）的否定响应消息在增强型定时器计时超时前发送，可能需要一个可选定时器用于增强型响应消息的定时。
$P3_{CAN_Physical}$	需要为每一个物理通信口提供单独的定时器。	N/A
$P3_{CAN_functional}$	需要为每一个功能通信口提供单独的定时器。	N/A

The tester and the ECUs shall provide the respective timer resources listed in the following table (Table 8-7) in order to meet the session layer timing requirements given in section above.
诊断仪和 ECU 应（shall）提供各自如 Table 8-7 所示的定时器资源，用于满足上述章节定义的会话层定时要求。

Table 8-7 Tester and ECU timing resource requirements (for session layer)

Timing Parameter	Tester	ECU
$S3_{Tester}$	A single timer is required when using a periodically transmitted, functionally addressed TesterPresent (3E) hex request message to keep the ECUs in a non-defaultSession. There is no need for additional timers per activated diagnostic sessions.	N/A

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	A single timer is required for each point-to-point communication channel when using a sequentially transmitted, physically addressed TesterPresent (3E) hex request message to keep a single ECU in a non-defaultSession in case of the absence of another diagnostic request message .	
$S3_{ECU}$	N/A	A single timer is required in the ECU, because only a single diagnostic session can be active at a time in a single ECU.

Table 8-7 诊断仪与 ECU 定时器资源要求（对会话层来说）

定时参数	诊断仪	ECU
$S3_{Tester}$	当周期性发送功能寻址的“诊断仪在线”请求消息以使多个ECU保持在非默认会话状态时，需要诊断仪提供一个单独的定时器。不需要为每一个激活的诊断会话提供另外的定时器。 在无其它诊断请求消息的情况下，通过连续地发送物理寻址“诊断仪在线”请求消息以使单个ECU保持在非默认状态，这就需要诊断仪为每一个点对点的通信通道分别设置单独的定时器。	N/A
$S3_{ECU}$	N/A	在某一时刻，一个ECU仅保持在一种激活诊断会话状态，所以仅需要ECU提供一个单独的定时器。

For the application and session layer timing resource, other more detailed requirements, with which the tester and the ECUs shall comply, can be found in ISO 15765-3.
诊断仪和 ECU 需要遵从的其它详细的应用层和会话层定时资源要求，请参考 ISO 15765-3。

8.2.2.4 Detailed timing parameter descriptions/详细定时参数描述

For detailed application and session layer timing parameter descriptions, with which the tester and the ECUs shall comply, can be found in ISO 15765-3.
诊断仪和 ECU 需要遵从的其它详细的应用层和会话层定时参数描述，请参考 ISO 15765-3。

8.2.2.5 Error handling/错误处理

For error handling of the application and session layer timing parameter, with which the tester and the ECUs shall comply, can be found in ISO 15765-3.
诊断仪和 ECU 需要遵从的其它详细的应用层和会话层定时参数的错误处理要求，请参考 ISO 15765-3。

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9 Diagnostic service documentation structure/诊断服务文档结构

Each diagnostic service is documented and defined in this specification within a specific sub-section of section 10 to 15.
本规范中记录和定义的每一个诊断服务包含在第 10 到 15 章节中。

The diagnostic service definitions are structured as follows:

1. A **Service description** section which provides text describing the general functionality of the diagnostic service and how it should be used (and implemented).
2. A **Request message section** which includes:
   - a **Request message definition** section with a table(or tables) to define the structure of the request message.
   - a **Sub-function parameter definition** section with text and table descriptions of any sub level functions within the diagnostic service.
   - a **Request message data parameter definition** section with text and table descriptions of any additional data bytes used in the request message.
3. A **Positive response message** section which includes:
   - a **Positive response message definition** section with a table(or tables) to define the structure of the positive response message.
   - a **Positive response message data parameter definition** section with text and table descriptions of any additional data bytes used in the response message.
4. A **Supported negative response codes** section which defines specific negative response codes supported by the service.

诊断服务的定义被分为以下几部分：

1. 诊断服务描述部分：描述了诊断服务的一般功能以及如何使用（和执行）。
2. 请求消息部分包含：
   - 请求消息定义部分：以表格的形式定义了请求消息的结构。
   - 子功能参数定义部分：以文字和表格的形式描述了诊断服务的所有子功能。
   - 请求消息数据参数定义部分：以文字和表格的形式描述了请求消息中使用的所有额外数据字节。
3. 肯定响应消息部分包含：
   - 肯定响应消息定义部分：以表格的形式定义了肯定响应消息的结构。
   - 肯定响应消息数据参数定义部分：以文字和表格的形式描述了响应消息中使用的所有额外数据字节。
4. 支持的否定响应代码部分：定义了服务支持的具体否定响应码。

The following provides more detailed descriptions and examples of specific diagnostic service documentation sections and/or table examples.
下面章节针对具体诊断服务以文字和/或表格的形式提供了更为详细的描述。

9.1 Service description/服务描述

The service description gives a brief outline of the functionality of the service. Each diagnostic service specification starts with a description of the actions performed by the tester and the ECU(s).
服务描述部分给出了服务功能的简要概述。每个诊断服务都是以对诊断仪和 ECU(s)需要执行的动作描述来开始的。

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This section is based on ISO 15765-3, which defines the implementation of unified diagnostic services (UDS) on CAN bus. For this, the on-vehicle ECUs shall comply with the requirement of server in ISO 15765-3; correspondingly, the tester shall comply with the requirement of the client in ISO 15765-3.
本章节是基于 ISO 15765-3（该标准规定了统一诊断服务在 CAN 总线上的实现）制定的。因此，车载 ECU 应（shall）遵从 ISO 15765-3 中对服务器的要求，同时，诊断仪应（shall）遵从 ISO 15765-3 中对客户端的要求。

For all the supported services, other detailed requirements than that given in the section 10 to 15, with which the tester and the ECUs shall comply, can be found in ISO 14229-1 and ISO 15765-3.
对于所有支持的服务，除了 10 到 15 章节中规定的要求外，诊断仪和 ECU 需遵从的其它更详细要求请参考 ISO 14229-1 和 ISO 15765-3。

9.2 Request message/请求消息

9.2.1 Request message definition/请求消息定义

The description of each service includes a table (e.g. Table 9-1) which shows the appropriate structure for the request messages. The table defines which data byte(s) are applicable for the particular diagnostic service.
每个服务的描述包含一个表格，在该表格（e.g. Table 9-1）中给出了请求消息的结构。这些表格定义了某一指定诊断服务的哪些数据字节是适用的。

Table 9-1 Request message definition/请求消息定义

A_Data Byte	Parameter Name	Cvt	Hex value
#1	< Service Name > Request Service Id	M	xx
#2	sub-function(where applicable)	U	xx
#2 or #3 : #n	data (where applicable) = [data#1 : data#m]	 C : C	 xx xx xx

In the convention column (Cvt) the following legend applies:

- M mandatory, shall always be present.
- U The sub-function parameter may or may not be supported by the ECU, depending on the usage of the service.
- C conditional, the presence of the parameter depends upon other parameters within the service or if the ECU requires the use of the parameter in order to fulfill other diagnostic requirements (e.g., the use of negative response code 78 hex in order to meet P_2 timing requirements.).

在参数约定列的参数约定说明如下：

- M 强制的，总是存在的。
- U ECU 可能支持也可能不支持子功能参数，这取决于具体的服务用法。
- C 有条件的，参数的出现取决于服务中的其它参数或者如果 ECU 需要使用此参数以完成其它诊断请求（例如，否定响应码 78 hex 的使用就是为了满足 P2 定时参数的要求）。

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9.2.2 Request message sub-function parameter definition/请求消息子功能参数定义

A sub-function parameter is used when a diagnostic service supports multiple levels of functionality. The sub-function parameter tells the diagnostic application which functionality is being requested. A sub-function parameter is used in a tester to ECU request message and when used, shall always occupy the first byte after the Service Identifier. The sub-function parameter byte is divided into two parts (on bit-level) as defined in Table 9-2.
当诊断服务支持多级功能时就会使用子功能参数，子功能参数通知诊断应用层正在请求哪一个功能。子功能参数用于诊断仪向 ECU 请求消息，当使用时它应（shall）总是占据服务标识符之后的第一个字节。子功能参数被划分为两部分，如 Table 9-2 所示：

Table 9-2 Request message sub-function parameter definition

Bit Position	Description
7	**suppressPosRspMsgIndicationBit** This bit indicates if a positive response message shall be suppressed by the ECU. '0' = FALSE, do not suppress a positive response message (a positive response message is required). '1' = TRUE, suppress response message (a positive response message shall not be sent; the ECU being addressed shall not send a positive response message). Independent of the suppressPosRspMsgIndicationBit, negative response messages are sent by the ECU(s) according to the restrictions specified in section 7.5 of ISO 14229.
6-0	**sub-function parameter value** The bits 0-6 of the sub-function parameter contain the sub-function parameter value of the service (00 - 7F hex). Each service utilizing the sub-function parameter byte, but only supporting the SuppressPosRspMsgIndicationBit has to support the zeroSubFunction sub-function parameter value (00 hex).

Table 9-2 请求消息子功能参数定义

位	描述
7	**抑制肯定响应指示位** 这一位用来表明ECU是否需要抑制肯定响应消息。 '0' = FALSE，不抑制肯定响应消息（需要肯定响应消息）。 '1' = TRUE，抑制肯定响应消息（不发送肯定响应消息；被寻址的ECU不应（shall not）发送肯定响应消息）。 与抑制肯定响应信息指示位无关，ECU根据ISO 14229的7.5章节的规定正常发送否定响应消息。
6-0	**子功能参数值** 子功能参数的第0-6位包含了服务的子功能参数值（00 - 7F hex）。 每个使用子功能参数，但仅支持抑制肯定响应消息指示位的服务必须支持数值为00 hex的子功能参数。

The sub-function parameter value is a 7-bit value (bits 6-0 of the sub-function parameter byte) that can have multiple values to further specify the service behavior.
Each service only supporting the suppressPosRspMsgIndicationBit has to support the zeroSubFunction (00 hex).
Services supporting sub-function parameter values in addition to the suppressPosRspMsgIndicationBit shall support the sub-function parameter values as defined in the sub-function parameter value table.
Each service contains a table that defines values for the sub-function parameter values, taking into

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account only the bits 0-6 as defined in Table 9-3.
子功能参数值占据 7 个位的数据长度（子功能参数字节的 0-6 位），这样就为将来定义服务行为预留了多个数值。
仅支持抑制肯定响应消息指示位的服务必须支持数值为 00 hex 的子功能参数。
除了支持抑制肯定响应消息指示位以外，还支持其他子功能参数值的服务应（shall）支持子功能参数值表中定义的子功能参数值。
每一个服务都包含一个定义子功能参数值的表格，Table 9-3 中只定义了第 0-6 位。

Table 9-3 sub-function parameter (bit 6-0)/子功能参数(bit 6-0)

Hex(bit 6-0)	Description/描述	Cvt
xx	**sub-function#1** description of sub-function parameter#1 子功能参数#1 的描述	M/U
:	**:**	
xx	**sub-function#m** description of sub-function parameter 子功能参数#1 的描述	M/U

In the convention column (Cvt) the following legend applies:

- M mandatory, shall always be present.
- U The sub-function parameter may or may not be supported by the ECU, depending on the usage of the service.

在参数约定列的参数约定说明如下：

- M 强制的，总是存在的。
- U ECU 可能支持也可能不支持子功能参数，这取决于具体的服务用法。

9.2.3 Request message data parameter definition/请求消息数据参数定义

A request message data parameter is used to indicate to the ECU diagnostic application which specific information is being requested (e.g. specific ECU input or output status variables, calculated variables, etc). A request message data parameter can affect the data contained in a response message but it does not affect the functionality of the diagnostic service used to retrieve it.
请求消息数据参数用来向 ECU 诊断应用层表明具体请求的信息（例如：具体的 ECU 输入输出状态变量值与计算出的变量值）。请求的消息数据参数可以影响响应消息中的数据，但不能影响用来获取数据的诊断服务的功能。

Example: If a diagnostic service is used to provide ECU input statuses, output statuses, etc. then a data parameter would be passed in with the diagnostic request message to identify which inputs, outputs, etc. that the tester is requesting. A sub-function parameter would be contained in the request message for this service to tell the diagnostic application if the data is to be sent as a one time response, sent periodically, or to stop sending previously requested data.
例如：如果一个诊断服务用来提供 ECU 的输入输出状态等，那么会有一个数据参数通过诊断请求消息发送出去来明确诊断仪请求哪些输入与输出。子功能参数会包含在服务的请求消息内，以此来通知诊断应用层，该数据的传输是一次性响应、还是周期发送，亦或者是停止发送上一次请求的数据。

This sub-clause of section 10 to 15 does not contain any definition if the described service does not use any data parameter. The data parameter portion can contain multiple bytes. This sub-clause of section 10 to 15 provides a generic description of each data parameter as defined in the following Table (Table 9-4), and detailed definitions can be found in the annexes of this document. The annexes

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also specify whether a data parameter shall be supported or is user-optional to be supported if the ECU supports the service.
如果在第 10 到 15 章节内描述的服务没有使用任何数据参数，则在相应的章节没做任何数据参数的定义。数据参数部分可以包含多个数据字节。
下述第 10 到 15 章节依照 Table 9-4 所示样式对每个数据参数进行了通用性的描述，详细的定义可参照本规范的附录。对于 ECU 支持的服务，附录中还详述了其数据参数是强制支持的还是可选支持的。

Table 9-4 Request message data parameter definition/请求消息数据参数定义

Definition/描述
data-parameter#1/数据参数#1 description of data-parameter#1 数据参数#1 的描述
:
data-parameter#m/数据参数#m description of data-parameter#m 数据参数#m 的描述

9.3 Positive response message/肯定响应消息

9.3.1 Positive response message definition/肯定响应消息定义

The description of each service includes a table (e.g. Table 9-5) which shows the appropriate structure for the positive response messages. The table defines which data byte(s) are applicable for the particular diagnostic service.
每个服务的描述部分都包含一个用来描述肯定响应消息结构的表格。这些表格（e.g. Table 9-5）定义了某一指定诊断服务的哪些数据字节是适用的。

There might be a separate table for each sub-function parameter $Level when the response messages of the different sub-function parameters $Level differ in the structure of the A_Data parameters.
当不同子功能参数值的响应消息使用不同的应用层数据参数结构时，每个子功能参数值的响应消息均会有一个单独的表格。

The positive response message of a diagnostic service (if required) shall be sent after the execution of the diagnostic service. If a diagnostic service requires different handling (e.g. ECUReset service), the appropriate description when to send the positive response message can be found in the service description of the diagnostic service.
诊断服务的肯定响应消息（如果需要的话）应（shall）在执行该诊断服务后发送。如果一诊断服务需要特殊不同的处理方式（例如：ECU 复位服务），那么在诊断服务的服务描述中会定义发送肯定响应消息时的确切描述。

Table 9-5 Positive response message definition/肯定响应消息定义

A_Data Byte	Parameter Name	Cvt	Hex value
#1	< Service Name > Response Service Id	M	xx
#2	sub-function echo (only where applicable)	C	xx
#2 or #3 : #n	data (where applicable) = [data#1 :	 C :	 xx :

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	data#m]	C	xx

In the convention column (Cvt) the following legend applies:

- M mandatory, shall always be present.
- C conditional, the presence of the parameter depends upon other parameters within the service or if the ECU requires the use of the parameter in order to fulfill other diagnostic requirements (e.g., the use of negative response code 78 hex in order to meet P2 timing requirements.).

在参数约定列的参数约定说明如下：

- M 强制的，总是存在的。
- C 有条件的，参数的出现取决于服务中的其余参数或者如果 ECU 需要使用此参数以完成其它诊断请求（例如，否定响应码 78 hex 的使用就是为了满足 P2 定时参数的要求）。

9.3.2 Positive response message data parameter definition/肯定响应消息数据参数定义

A data parameter in a positive message is used to provide the information to the tester that was requested via a diagnostic request message.
肯定响应消息中的数据参数用于向诊断仪提供其通过诊断请求消息请求的信息。

It does not contain any definition if the described service does not use any data parameter. The data parameter portion can contain multiple bytes. This sub-clause provides a generic description of each data parameter, detailed definitions can be found in the annexes of this document. The annexes also specify whether a data parameter will be supported or is user-optional to be supported if the ECU supports the service.
如果服务描述部分没有使用任何数据参数，那么肯定响应消息也不需要定义数据参数。数据参数部分可以包含多个字节。本章节对每个数据参数进行了通用性的描述，详细的定义可参考本规范附录。对于 ECU 支持的服务，附录中还详述了其数据参数是强制支持的还是可选支持的。

Table 9-6 Positive response message data parameter definition/肯定响应消息数据参数定义

Definition/描述
data-parameter#1/数据参数#1 description of data-parameter#1 If the request supports a sub-function parameter byte then this parameter is an echo of the 7-bit sub-function parameter value contained within the sub-function parameter byte from the request message with bit 7 set to zero. The suppressPosRspMsgIndicationBit from the sub-function parameter byte is not echoed. 数据参数#1 的描述 如果请求消息支持子功能参数字节，那么该参数就是请求消息子功能参数中第 0-6 位数值的映射，第 7 位数值设置为 0。请求消息子功能参数中第 7 位数值不进行映射。
:
data-parameter#m/数据参数#m description of data-parameter#m 数据参数#m 的描述

9.4 Supported negative response codes/支持的否定响应码

The negative response message of a diagnostic service shall be used by an ECU to indicate that a

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diagnostic service request message is either invalid, being terminated, or currently cannot be acted upon at the time of the request. Included in the negative response message is a return code to indicate to the tester the reason the negative response message was sent.
ECU 使用诊断服务否定响应消息向诊断仪表示了该诊断服务请求消息是无效的、或者被异常中止、或者当前不能执行该请求。否定响应消息内有一个否定响应码，它向诊断仪表示了发送否定响应消息的原因。

The following table (Table 9-7) defines the format of negative response message.
Table 9-7 定义了否定响应消息的格式。

Table 9-7 The format of negative response message/否定响应消息格式

A_Data Byte	Parameter Name	Cvt	Hex value
#1	Negative Response Service Id	M	7F
#2	Request Service Id	M	xx
#3	Negative response code (NRC)	M	xx

NOTE: The format of every service is fixed as defined in the table above, so this sub-clause of section 10 to 15 do NOT provide any description or table for the negative response messages of all the services.
注：对于每个服务而言，上表所定义的否定响应消息格式是固定一致的。因此，针对所有服务的否定响应消息，第10到-15章节均没有做任何文字或表格形式的描述。

For this specification, the requirement of annex A of ISO 14229 is normative, only the negative response codes listed in annex A.1 of ISO 14229 shall be used. Table A.1 lists all negative response codes used within this specification.
就本规范而言，ISO 14229-1中附录A的要求是规范性的，只有ISO 14229-1附录A.1中的否定响应码可以使用，表A.1列出了本规范使用的所有否定响应码。

Annex A defines the possible negative response codes that will be implemented and the conditions that each response code would occur for all service defined in this document
附录 A 定义了本规范定义的所有诊断服务可能使用的否定响应代码。下表定义了每个否定响应代码的产生条件（即：什么情况下使用什么否定响应码）。

Table 9-8 negative response codes/否定响应码

Hex	Description	Cvt
xx	**NegativeResponseCode#1** 1. condition#1 : m. condition#m	M
:	:	M
xx	**NegativeResponseCode#n** 1. condition#1 : k. condition#k	M

In the convention column (Cvt) the following legend applies:

- M mandatory, shall always be present.

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在参数约定列的参数约定说明如下：

- M　强制的，总是存在的。

NOTE: If necessary, the RCRRP (requestCorrectlyReceived-ResponsePending, the value is 78(hex)) negative response code can be supported by all the ECUs, so it is not listed in the list of applicable response codes of the diagnostic services defined in the section 10 to 15.
The RCRRP NRC has the lower priority than all the NRCs listed in the list of applicable response codes of the diagnostic services defined in the section 10 to 15.
The maximum number of the RCRRP NRC used for one request shall be limited to 3.
注：如必要的话，那么所有 ECU 均能支持 78（hex）否定响应码。因此，在第 10-15 章节定义的诊断服务中没有列出该否定响应码。
RCRRP 否定响应码的优先级低于第 10 到 15 章节中定义的诊断服务所使用的所有否定响应码。
一次请求所允许使用的 RCRRP 否定响应码最大数量为 3。

NOTE: The SNS (serviceNotSupported, the value is 11(hex)) negative response code shall be used by the ECU, when the requested service is NOT supported in any session.
The GR (generalReject, the value is 10(hex)) negative response code shall NOT be used by the ECU for all the services, unless SGMW and the ECU supplier define the detailed condition for this NRC.
注：在任何会话模式下，当不支持请求的服务时，ECU 应（shall）使用 11（hex）否定响应码。
对于所有的服务，不允许 ECU 使用否定响应码 10（hex），除非 SGMW 与 ECU 供应商对这个否定响应码定义了详细的条件。

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10 Diagnostic and communication management functional unit/诊断与通信管理功能单元

10.1 DiagnosticSessionControl (10 hex)/诊断会话控制

10.1.1 Service description/服务描述

The DiagnosticSessionControl service is used to enable different diagnostic sessions in the ECU(s).
诊断会话控制服务用来使能ECU的不同诊断会话。

A diagnostic session enables a specific set of diagnostic services and/or functionality in the ECU(s). It can, furthermore, enable a data link layer dependent set of timing parameters applicable for the started session.
某一特定的诊断会话能够使能ECU内的一组特定的诊断服务和/或功能。此外，它还可以使能用于启动会话的数据链路层相关的时间参数设置。

Figure 10-1 provides an overview about the diagnostic session transition and what the ECU will do when it transitions to another session.
Figure 10-1说明了不同诊断会话模式之间相互转换的大概流程以及从一个模式转换到另一个模式时ECU的动作。

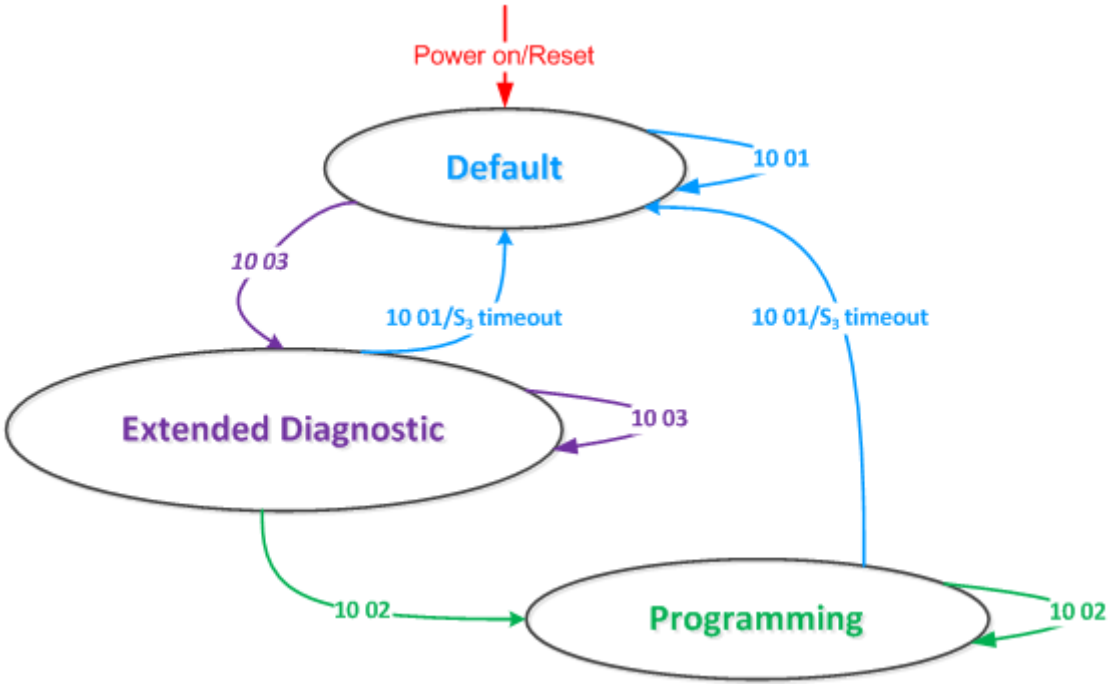

Figure 10-1 ECU diagnostic session state diagram/ECU 诊断会话状态转换图

NOTE: There shall always be exactly one diagnostic session active in the ECU. The ECU shall always start the default diagnostic session when powered up. If no other diagnostic session is started, then the default diagnostic session shall be running as long as the ECU is powered.

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注：ECU 内部应（shall）始终且仅有一个激活的诊断会话。当上电时，ECU 应（shall）总是首先启动默认会话。如果没有启动其它的诊断会话，那么 ECU 上电后应（shall）一直处于默认会话模式。

NOTE: If the tester has requested a diagnostic session which is already running, then the ECU shall send a positive response message and behave as shown in Figure 10-1.
注：如果诊断仪请求一个已经运行的诊断会话，那么 ECU 应发送一肯定响应消息且其行为如图 10-1 所示。

NOTE: Whenever the tester requests a new diagnostic session, the ECU shall send the DiagnosticSessionControl positive response message before the timings of the new session become active in the ECU. If the ECU is not able to start the requested new diagnostic session, then it shall respond with a DiagnosticSessionControl negative response message and the current session shall continue.
注：每当诊断仪请求一个新的诊断会话时，在内部新的诊断会话定时参数激活之前，ECU 应（shall）发送诊断会话控制(DiagnosticSessionControl)肯定响应消息。如果不能启动诊断仪请求的新的诊断会话，那么 ECU 应（shall）反馈诊断会话控制（DiagnosticSessionControl）否定响应消息，并且当前激活的诊断会话应（shall）保持不变。

NOTE: When the ECU is in the defaultSession and the tester requests to start the defaultSession, then the ECU shall re-initialize the defaultSession completely. The ECU shall reset all activated/initiated/changed settings/controls during the activated session. This does not include long-term changes programmed into non-volatile memory.
注：当 ECU 处于默认会话并且诊断仪请求启动默认会话时，ECU 应（shall）完全重新初始化默认会话。ECU 应（shall）复位所有在之前诊断会话中激活的/初始化的/更改的设置/控制。这不包括编程到非易失性存储器中的长时调整参数。

NOTE: The non-programmable ECU is allowed not to support programming session.
注：不可软件刷新的 ECU 可不支持编程会话。

10.1.2 Request message/请求消息

10.1.2.1 Request message definition/请求消息定义

Table 10-1 Request message definition

A_Data Byte	Parameter Name	Cvt	Hex value
#1	DiagnosticSessionControl Request Service Id	M	10
#2	sub-function = [diagnosticSessionType]	M	01/02/03

10.1.2.2 Request message sub-function parameter definition/请求消息子功能参数定义

Table 10-2 Request message sub-function parameter definition

Hex(bit 6-0)	Description/描述	Cvt
01	DefaultSession/默认会话模式	M
02	ProgrammingSession/编程会话模式	M
03	ExtendedDiagnosticSession/扩展诊断会话模式	M

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10.1.2.3 Request message data parameter definition/请求消息数据参数定义

This service does not support data parameters in the request message.
本服务请求消息不支持数据参数。

10.1.3 Positive response message/肯定响应消息

10.1.3.1 Positive response message definition/肯定响应消息定义

Table 10-3 Positive response message definition

A_Data Byte	Parameter Name	Cvt	Hex value
#1	DiagnosticSessionControl Response Service Id	S	50
#2	diagnosticSessionType	M	01/02/03
	sessionParameterRecord[] #1 = [		
#3	$P2_{CAN_ECU_max}$(high byte)	M	00
#4	$P2_{CAN_ECU_max}$(low byte)	M	32
#5	$P2^*_{CAN_ECU_max}$(high byte)	M	01
#6	$P2^*_{CAN_ECU_max}$(low byte)]	M	F4

10.1.3.2 Positive response message data parameter definition/肯定响应消息数据参数定义

Table 10-4 Positive response message data parameter definition

Description/描述
DiagnosticSessionType/诊断会话类型 This parameter is an echo of bits 6 - 0 of the sub-function parameter from the request message. 该数据参数是请求消息中子功能参数第6-0位的映射。
sessionParameterRecord $P2_{CAN_ECU_max}$ = 50ms (resolution: 1ms/bit, offset: 0.) $P2^*_{CAN_ECU_max}$ = 5000ms (resolution: 10ms/bit, offset: 0.)

NOTE: The definition about parameter $P2_{CAN_ECU_max}$ and $P2^*_{CAN_ECU_max}$ can be found in chapter 8.2.2 of this document and ISO 15765-3. These two parameters in positive response message are the ones that are used by ECU internally, if the value of this two parameters which is used by ECU are different from the definition in this document, it should be determined by SGMW and ECU Supplier together.
注：$P2_{CAN_ECU_max}$ 与 $P2^*_{CAN_ECU_max}$ 的定义见本文档第 8.2.2 章节与 ISO 15765-3。包含在肯定响应消息中的这两个参数就是 ECU 内部使用的 $P2_{CAN_ECU_max}$ 与 $P2^*_{CAN_ECU_max}$ 参数，如果 ECU 内部使用的这两个参数值与该文档定义不一致，则需要 SGMW 与 ECU 供应商共同沟通确定。

10.1.4 Supported negative response codes/支持的否定响应码

Table 10-5 Supported negative response codes

Hex	Description	Cvt
12	**subFunctionNotSupported/不支持子功能参数**	M
13	**incorrectMessageLengthOrInvalidFormat/不正确的消息长度或无效的格式**	M
22	**conditionsNotCorrect/条件不正确** 1. This response code shall be returned if the criteria for the request	M

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	DiagnosticSessionControl are not met. The correct condition for the ECU to implement the request message shall be defined in the individual ECU diagnostic specification. 当不满足请求诊断会话控制的操作条件时，ECU 应（shall）发送此否定响应码。ECU 执行该服务请求消息的正确条件应（shall）在 ECU 独立诊断规范内定义。 **2.** This response code shall be returned if the request message which requests ECU transfer its session from defaultSession to programmingSession or from programmingSession to extendedDiagnosticSession is received. 当接收到要求 ECU 从默认会话模式转换至编程会话模式或者从编程会话模式转换至扩展会话模式的请求消息时，ECU 应（shall）发送此否定响应码。	

NOTE: The strategy of negative response codes can be found in Figure A.2 of Annex A.
注：否定响应码的策略见附录 A 图 A.2。

10.2 ECUReset (11 hex) /ECU 复位

10.2.1 Service description/服务描述

The ECUReset service is used by the tester to request the ECU(s) reset. This service requests the ECU to effectively perform an ECU reset based on the content of the resetType parameter value embedded in the ECUReset request message. The ECUReset positive response message shall be sent before the reset is executed in the ECU(s). Before a Reset is performed a positive response has to be sent to the tester, The ECU must ensure that the response is put correctly onto the bus before performing the reset.
诊断仪使用ECU复位服务去请求ECU进行复位操作。该服务要求ECU依据ECU复位请求消息中复位类型去正确地执行ECU复位操作。肯定响应消息应（shall）在ECU执行复位之前发送。在复位之前肯定响应消息必须发送给诊断仪，ECU必须（must）确保复位之前把响应消息发送到总线上。

Following actions shall be performed if an ECU Reset request with reset type Hard Reset is received:
a.) immediately restart application program,
b.) re-initialize volatile memory which is also initialized during the power-up sequence,
c.) re-initialize non-volatile memory which is also initialized during the power-up sequence,
d.) re-initialize electronic sub-components directly connected to the ECU which are also initialized during the power-up sequence.

如果接收到复位类型为硬复位的ECU复位请求，以下动作应（shall）被执行：
a.) 立即重启执行程序，
b.) 重新初始化易失性存储器，此类存储器在上电过程中也会进行初始化，
c.) 重新初始化非易失性存储器，此类存储器在上电过程中也会进行初始化，
d.) 重新初始化与ECU直接连接的电子部件，这些电子部件在上电过程中也会进行初始化。

Following actions shall be performed if an ECU Reset request with reset type Soft Reset is received:
a.) save some volatile data (e.g. previously learned configuration data, adaptive factors and other long-term adjustments. which are needed beyond a reset),
b.) immediately restart application program.

如果接收到复位类型为软复位的ECU复位请求，以下动作应（shall）被执行：
a.) 保存一些（例如：之前学习的配置参数、自适应系数以及其他长时调整参数等）易失性数据，
b.) 立即重启应用程序。

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NOTE: After a successful ECU reset, the ECU shall activate the defaultSession.
注：ECU成功复位之后应（shall）激活默认会话模式。

10.2.2 Request message/请求消息

10.2.2.1 Request message definition/请求消息定义

Table 10-6 Request message definition

A_Data Byte	Parameter name	Cvt	Hex value
#1	ECUReset Request Service Id	M	11
#2	sub-function = [resetType]	M	01/03

10.2.2.2 Request message sub-function parameter definition/请求消息子功能参数定义

Table 10-7 Request message sub-function parameter definition

Hex(bit 6-0)	Description	Cvt
01	**hardReset/硬复位** This type of reset shall simulate a complete power on startup sequence same as performed when the ECU has been disconnected from its power supply. It shall result in the re-initialization of both volatile memory and non-volatile memory locations to default values. 该类型的复位应（shall）模拟一个完整的上电启动顺序，等同于ECU断电再上电。该复位操作应（shall）导致易失性存储器与非易失性存储器的数据重新初始化为默认值。	M
03	**softReset/软复位** Forces the ECU to restart its application and to store any data in non-volatile memory which might be lost on the application restart. It shall restart the application program without re-initializing of previously learned configuration data, adaptive factors and other long-term adjustments. 该类型的复位强迫ECU重启其应用程序，并且在非易失性存储器中存储所有在应用程序重启时可能会丢失的数据。该复位操作应（shall）导致应用程序的重启，但不对之前学习的配置参数、自适应系数以及其他长时调整参数等数据重新初始化。	M

10.2.2.3 Request message data parameter definition/请求消息数据参数定义

This service does not support data parameters in the request message.
本服务请求消息不支持数据参数。

10.2.3 Positive response message/肯定响应消息

10.2.3.1 Positive response message definition/肯定响应消息定义

Table 10-8 Positive response message definition

A_Data Byte	Parameter Name	Cvt	Hex value
#1	ECUReset Response Id	S	51
#2	resetType	M	01/03

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10.2.3.2 Positive response message data parameter definition/肯定响应消息数据参数定义

Table 10-9 Positive response message data parameter definition

Definition
resetType/复位类型 This parameter is an echo of bits 6 - 0 of the sub-function parameter from the request message. 该数据参数是请求消息中子功能参数第6-0位的映射。

10.2.4 Supported negative response codes/支持的否定响应码

Table 10-10Supported negative response codes

Hex	Description	Cvt
12	**subFunctionNotSupported/不支持子功能**	M
13	**incorrectMessageLengthOrInvalidFormat/不正确的消息长度或无效的格式**	M
22	**conditionsNotCorrect/条件不正确** This response code shall be returned if the criteria for the ECUReset request are not met. The correct condition for the ECU to implement the request message shall be defined in the individual ECU diagnostic specification. 当不满足 ECU 复位请求的操作条件时，应（shall）发送此否定响应码。ECU 执行该服务请求消息的正确条件应（shall）在 ECU 独立诊断规范内定义。	M

NOTE: The strategy of negative response codes can be found in Figure A.2 of Annex A.
注：否定响应码的策略见附录 A 图 A.2。

10.3 SecurityAccess (27 hex)/安全访问

10.3.1 Service description/服务描述

This service, as well as MEC that defined in the chapter **错误！未找到引用源。**, provides a protected mechanism to access the diagnostic services which have restricted access for security, emissions or safety reasons.
The security concept uses a seed and key relationship. The seed and key are each 32 bit numbers (4 bytes). ECU supplier takes the responsibility of realizing a random generator function, which generates a random seed for every time the tester requests the seed in its ECU software. SGMW takes the responsibility of providing a security algorithm, which describes how to use a certain seed to calculate a unique key, to the ECU supplier to implement the security access between the tester and the ECU. The security algorithm shall be stored in non-volatile memory during the manufacturing process of the ECU.
该服务与第**错误！未找到引用源。**章节中定义的MEC一起提供了一种保护机制，该机制用来保护因保密、排放或安全等原因而受到访问限制的诊断服务。
加密策略采用种子和密钥相关联的方法。种子与密钥长度均应为32位（4个字节）。ECU供应商负责实现产生随机数产生函数，即每次诊断仪请求种子时ECU软件内部就会产生一个随机的种子。SGMW负责提供一个安全算法给ECU供应商，用来执行诊断仪与ECU之间的安全访问。该算法描述了如何用某个种子计算得出唯一的密钥。在供应商开发ECU时，安全算法应（shall）存储在非易失性存储器内。

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10.3.2 Request message/请求消息

10.3.2.1 Request message definition 请求消息定义

Table 10-11Request message definition

A_DataByte	Parameter Name	Cvt	Hex value
#1	SecurityAccess Request Service Id	M	27
#2	Sub-function = [securityAccessType = requestSeed]	M	01/03/05

Table 10-12Request message definition

A_DataByte	Parameter Name	Cvt	Hex value
#1	SecurityAccess Request Service Id	M	27
#2	Sub-function = [securityAccessType = sendKey]	M	02/04/06
#3 #4 #5 #6	securityKey[] = [key #1(high byte) key #2 key #3 key#4(low byte)]	M M M M	00-FF : : 00-FF

10.3.2.2 Request message sub-function parameter definition/请求消息子功能参数定义

Table 10-13 Request message sub-function parameter definition

Level No. 级别编号	Security Level Description 安全级别描述	requestSeed 请求种子	SendKey 发送密钥	Applicable Session 适用的诊断会话	Cvt 约定
Level 1	general security 一般安全	01	02	extendedDiagnosticSession 扩展会话	M
Level 2	Immobilizer security 防盗安全	03	04	extendedDiagnosticSession 扩展会话	C[a]
Level 3	programming security 编程安全	05	06	programmingSession 编程会话	C[b]

a. Level 2 is only applicable for immobilizer system related ECUs.
级别 2 仅适用于与防盗系统相关的 ECU。

b. Level 3 is only applicable for ECU programming event, and shall only is supported in boot software.
级别 3 仅适用于 ECU 编程事件，并且应（shall）仅在引导程序下支持。

10.3.2.3 Request message data parameter definition/请求消息数据参数定义

Table 10-14 Request message data parameter definition

Definition
securityKey (high and low bytes)/安全密钥 The “key” parameter in the request message is the value generated by the security algorithm corresponding to a specific “seed” value. 请求消息内的密钥参数是依据一个具体的种子按照安全算法计算得出的。

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10.3.3 Positive response message/肯定响应消息

10.3.3.1 Positive response message definition/肯定响应消息定义

Table 10-15 Positive response message definition (sub-function = requestSeed)

A_DataByte	Parameter Name	Cvt	Hex value
#1	Security Access Response Service Id	S	67
#2	securityAccessType	M	01/03/05
	securitySeed[] = [		
#3	seed#1(high byte)	M	00-FF
#4	seed#2	M	:
#5	seed#3	M	:
#6	seed#4(low byte)]	M	00-FF

Table 10-16 Positive response message definition (sub-function = sendKey)

A_DataByte	Parameter Name	Cvt	Hex value
#1	Security Access Response Service Id	S	67
#2	securityAccessType	M	02/04/06

10.3.3.2 Positive response message data parameter definition/肯定响应消息数据参数定义

Table 10-17 Positive response message data parameter definition

Definition
securityAccessType/安全访问类型 This parameter is an echo of bits 6 - 0 of the sub-function parameter from the request message. 该数据参数是请求消息中子功能参数第6-0位的映射。
securitySeed (high and low bytes)/安全种子 The seed parameter is a data value sent by the ECU and is used by the tester when calculating the key needed to access security. 种子参数是ECU发送给诊断仪用于计算安全访问密钥的数值。

10.3.4 Supported negative response codes/支持的否定响应码

Table 10-18 Supported negative response codes

Hex	Description	Cvt
7F	**serviceNotSupportedInActiveSession/激活会话不支持该服务**	M
12	**subFunctionNotSupported/不支持子功能**	M
13	**incorrectMessageLengthOrInvalidFormat/不正确的消息长度或无效的格式**	M
22	**conditionsNotCorrect/条件不正确** This response code shall be returned if the criteria for the request SecurityAccess are not met. The correct condition for the ECU to implement the request message shall be defined in the individual ECU diagnostic specification. 当不满足请求安全访问的操作条件时，ECU 应（shall）发送此否定响应码。ECU 执行该服务请求消息的正确条件应（shall）在 ECU 独立诊断规范内定义。	M

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24	**requestSequenceError/请求序列错误** This response code shall be sent if the “sendKey” sub-function is received without first receiving a “requestSeed” request message. 如果在接收到一个子功能参数为“请求种子”的请求消息之前，首先接收到一个子功能参数为“发送密钥”的请求消息，ECU 应（shall）发送此否定响应码。	M
35	**invalidKey/密钥无效** This response code shall be sent if an expected “sendKey” sub-function value is received and the value of the key does not match the ECU's internally stored/calculated key. This counts as an attempt to gain security. The ECU shall remain locked and increment its internal SAFC. 如果已经接收到一个期望的子功能参数为“发送密钥”的请求消息，但是接收来的密钥与 ECU 内部存储/计算的密钥不匹配时，ECU 应（shall）发送此否定响应码。该操作算作一次安全访问的尝试。此时 ECU 应（shall）保持锁定状态，并且增加内部 SAFC 的计数。	M
36	**exceededNumberOfAttempts/超出尝试次数** This response code shall be sent when the forth SecurityAccess request is received after continuous false access attempts for 3 times. At that time the delay timer has been active due to exceeding the maximum number of allowed false access attempts. 在 3 次连续的错误访问尝试后，接收到第 4 次安全访问请求时，ECU 应（shall）发送此否定响应码。当时由于超过了所允许的最大错误访问尝试次数（3 次），延时定时器已经开始计时。	M
37	**requireTimeDelayNotExpired/所需时间延迟未到** This response code shall be sent if the fifth and successive SecurityAccess request is received during the delay timer is active. 在延时定时器计时过程中，如果接收到第五次及后续的安全访问请求消息，ECU应（shall）发送此否定响应码。	M

NOTE: The strategy of negative response codes can be found in Figure A.1 of Annex A.
注：否定响应码的策略见附录 A 图 A.1。

10.4 CommunicationControl (28 hex) /通信控制

10.4.1 Service description/服务描述

The purpose of this service is to switch on/off the transmission and/or the reception of certain messages of (a) ECU(s) (e.g. application communication messages).
本服务的目的是开启或关闭ECU对某一类特定报文（例如应用通信报文）的发送和/或接收。

After a reset of ECU, the status of communication shall return to the default status that is all the communications are enabled, whatever status before ECU reset.
无论复位之前状态如何，ECU复位之后其通信状态应（shall）返回到所有类型的通信均激活的默认状态。

10.4.2 Request message/请求消息

10.4.2.1 Request message definition 请求消息定义

Table 10-19 Request message definition

A_DataByte	Parameter Name	Cvt	Hex value
#1	CommunicationControl Request Service Id	M	28
	Sub-function = [		

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#2	controlType]	M	00/03
#3	communicationType	M	00-FF

10.4.2.2 Request message sub-function parameter definition/请求消息子功能参数定义

Table 10-20 Request message sub-function parameter definition

Hex(bit 6-0)	Description	Cvt
00	**enableRxAndTx/**使能收发	M
03	**disableRxAndTx/**禁止收发	M

10.4.2.3 Request message data parameter definition/请求消息数据参数定义

Table 10-21 Request message data parameter definition

Definition
communicationType/通信类型 This parameter is used to reference the kind of communication to be controlled. The communicationType parameter is a bit-encoded value which allows control of multiple communication types at the same time (see ISO 14229-1 Annex B.1 for the coding of the communicationType data parameter). 此参数用于描述进行控制的通信类型。通信类型（communicationType）参数是一个位编码类型的数值，该数值可以同时控制多种通信类型（见 ISO14229-1 附件 B.1 的 communicationType 数据参数的编码格式定义）。

10.4.3 Positive response message/肯定响应消息

10.4.3.1 Positive response message definition/肯定响应消息定义

Table 10-22 Positive response message definition

A_DataByte	Parameter Name	Cvt	Hex value
#1	CommunicationControl Response Service Id	S	68
#2	controlType	M	00/03

10.4.3.2 Positive response message data parameter definition/肯定响应消息数据参数定义

Table 10-23 Positive response message data parameter definition

Definition
controlType/控制类型 This parameter is an echo of bits 6 - 0 of the sub-function parameter from the request message. 该数据参数是请求消息中子功能参数第6-0位的映射。

10.4.4 Supported negative response codes/支持的否定响应码

Table 10-24 Supported negative response codes

Hex	Description	Cvt
7F	**serviceNotSupportedInActiveSession/**激活会话不支持该服务	M
12	**subFunctionNotSupported/**不支持子功能	M
13	**incorrectMessageLengthOrInvalidFormat/**不正确的消息长度或无	M

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	效的格式	
22	**conditionsNotCorrect/条件不正确** This code shall be returned when the ECU is in a critical normal mode activity and therefore cannot disable/enable the requested communication type. The correct condition for the ECU to implement the request message shall be defined in the individual ECU diagnostic specification. 当 ECU 处于一个不稳定的正常模式活动中因而无法禁止/使能请求的通信类型时，ECU 应（shall）发送此否定响应码。ECU 执行该服务请求消息的正确条件应（shall）在 ECU 独立诊断规范内定义。	M

NOTE: The strategy of negative response codes can be found in Figure A.2 of Annex A.
注：否定响应码的策略见附录 A 图 A.2。

10.5 TesterPresent (3E hex)/诊断仪在线

10.5.1 Service description/服务描述

This service is used to indicate to an ECU (or ECUs) that a tester is still connected to the vehicle and that certain diagnostic services and/or communications that have been previously activated are to remain active.
本服务被用来向ECU表明诊断仪仍然连接在车辆上，并且先前激活的特定诊断服务和/或通信仍然处于激活状态。

This service is used to keep one or multiple ECUs in a diagnostic session other than the defaultSession. This can either be done by transmitting the TesterPresent request message periodically or, in case of the absence of other diagnostic services, to prevent the ECU(s) from automatically returning to the defaultSession. The detailed session requirements that apply to the use of this service when keeping a single ECU or multiple ECUs in a diagnostic session other than the defaultSession can be found in section 8 of this specification and ISO 15765-3.
本服务用于保持单个或多个ECU处于非默认诊断会话状态下。为了防止ECU自动返回到默认会话状态下，可以周期性地或者在无其它诊断服务时，发送诊断仪在线请求消息。当为了保持单个或多个ECU处于非默认会话状态时，使用本服务所遵从的会话要求可详见本规范的第8章和ISO15765-3。

10.5.2 Request message/请求消息

10.5.2.1 Request message definition 请求消息定义

Table 10-25 Request message definition

A_DataByte	Parameter Name	Cvt	Hex value
#1	TesterPresent Request Service Id	M	3E
#2	Sub-function = [zeroSubFunction]	M	00/80

10.5.2.2 Request message sub-function parameter definition/请求消息子功能参数定义

Table 10-26 Request message sub-function parameter definition

Hex(bit 7-0)	Description	Cvt
00	zeroSubFunction (suppressPosRspMsgIndicationBit =0)	M

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80	zeroSubFunction (suppressPosRspMsgIndicationBit =1)	M

10.5.2.3 Request message data parameter definition/请求消息数据参数定义

This service does not support data parameters in the request message.
本服务请求消息不支持数据参数。

10.5.3 Positive response message/肯定响应消息

10.5.3.1 Positive response message definition/肯定响应消息定义

Table 10-27 Positive response message definition(sub-function = zeroSubFunction (suppressPosRspMsgIndicationBit =0))

A_DataByte	Parameter Name	Cvt	Hex value
#1	TesterPresent Response Service Id	S	7E
#2	zeroSubFunction	M	00

sub-function = zeroSubFunction (suppressPosRspMsgIndicationBit =1)

NOTE: There is no any response for the request message, because of suppressPosRspMsgIndicationBit =1.
注：该请求消息没有肯定响应消息，因为抑制肯定响应消息指示位为 1.

10.5.3.2 Positive response message data parameter definition/肯定响应消息数据参数定义

Table 10-28 Positive response message data parameter definition

Definition
ZeroSubFunction/零子功能 This parameter is an echo of bits 6 - 0 of the sub-function parameter from the request message. 该数据参数是请求消息中子功能参数第6-0位的映射。

10.5.4 Supported negative response codes/支持的否定响应码

Table 10-29 Supported negative response codes

Hex	Description	Cvt
12	**subFunctionNotSupported/**不支持子功能	M
13	**incorrectMessageLengthOrInvalidFormat/**不正确的消息长度或无效的格式	M

NOTE: The strategy of negative response codes can be found in Figure A.2 of Annex A.
注：否定响应码的策略见附录 A 图 A.2。

10.6 AccessTimingParameter (83 hex)/访问定时参数

N/A

10.7 SecuredDataTransmission (84 hex)/安全数据传输

N/A

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10.8 ControlDTCSetting (85 hex) /DTC 设置控制

10.8.1 Service description/服务描述

The ControlDTCSetting service shall be used by a tester to stop or resume the setting of diagnostic trouble codes (DTCs) in the ECU(s).
The ControlDTCSetting request message can be used to stop the setting of diagnostic trouble codes in an individual ECU or a group of ECUs. If the ECU being addressed is not able to stop the setting of diagnostic trouble codes, it shall respond with a ControlDTCSetting negative response message indicating the reason for the rejection.
诊断仪使用控制DTC设置服务停止或恢复ECU内部的诊断故障码设置功能。
控制DTC设置服务的请求消息可用来停止单独的ECU或一组ECU的诊断故障码设置功能。如果被寻址的ECU不能够停止诊断故障码的设置，那么它应（shall）通过控制DTC设置服务的否定响应消息进行响应，并说明拒绝的原因。

The update of the DTC status bit information shall continue once a ControlDTCSetting request is performed with sub-function set to “on” or a session layer timeout occurs (ECU transitions to defaultSession). The ECU shall still send a positive response if the service is supported in the active session with a requested sub-function set to either “on” or “off” even if the requested DTC setting state is already active.
If a ClearDiagnosticInformation (14 hex) service is sent by the tester, the ControlDTCSetting shall not prohibit resetting the ECU's DTC memory.
If a successful ECUReset is performed, then this re-enables the setting of DTCs.
一旦ECU接收到子功能参数设置为“使能”的控制DTC设置服务请求消息并执行该请求，或者会话层定时器计时超时（ECU转换到默认会话状态），ECU应（shall）继续DTC状态位信息的更新。如果在当前会话状态下支持所请求的控制DTC设置服务，那么无论子功能参数为“使能”还是“禁止”，ECU均应（shall）发送一个肯定响应消息，即使请求的DTC设置状态已经是激活的。
如果诊断仪发送清除诊断信息服务（14 hex）请求消息，那么控制DTC设置服务不应（shall not）阻止ECU诊断故障码内存的复位清零。
如果ECU成功执行复位操作，那么将重新使能诊断故障码的设置功能。

10.8.2 Request message/请求消息

10.8.2.1 Request message definition 请求消息定义

Table 10-30 Request message definition

A_DataByte	Parameter Name	Cvt	Hex value
#1	ControlDTCSetting Request Service Id	M	85
#2	sub-function = [DTCSettingType]	M	01/02

10.8.2.2 Request message sub-function parameter definition/请求消息子功能参数定义

Table 10-31 Request message sub-function parameter definition

Hex(bit 6-0)	Description	Cvt
01	**ON/使能** The ECU(s) shall resume the setting of diagnostic trouble codes according to normal	M

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	operating conditions. ECU应（shall）依照正常操作条件恢复诊断故障码的设置功能。	
02	**OFF/禁止** The ECU(s) shall stop the setting of diagnostic trouble codes. ECU应（shall）停止诊断故障码的设置功能。	M

10.8.2.3 Request message data parameter definition/请求消息数据参数定义

This service does not support data parameters in the request message.
本服务请求消息不支持数据参数。

10.8.3 Positive response message/肯定响应消息

10.8.3.1 Positive response message definition/肯定响应消息定义

Table 10-32 Positive response message definition

A_DataByte	Parameter Name	Cvt	Hex value
#1	ControlDTCSetting Response Service Id	S	C5
#2	DTCSettingType	M	01/02

10.8.3.2 Positive response message data parameter definition/肯定响应消息数据参数定义

Table 10-33 Positive response message data parameter definition

Definition
DTCSettingType/DTC 设置类型 This parameter is an echo of bits 6 - 0 of the sub-function parameter from the request message. 该数据参数是请求消息中子功能参数第6-0位的映射。

10.8.4 Supported negative response codes/支持的否定响应码

Table 10-34 Supported negative response codes

Hex	Description	Cvt
7F	**serviceNotSupportedInActiveSession/激活会话不支持该服务**	M
12	**subFunctionNotSupported/不支持子功能**	
13	**incorrectMessageLengthOrInvalidFormat/不正确的消息长度或无效的格式**	M
22	**conditionsNotCorrect/条件不正确** This code shall be returned when the ECU is in a critical normal mode activity and therefore cannot perform the requested DTC control functionality. The correct condition for the ECU to implement the request message shall be defined in the individual ECU diagnostic specification. 当ECU处于一个不稳定的正常模式活动中因而无法执行请求的DTC控制功能时，ECU应（shall）发送此否定响应码。ECU执行该服务请求消息的正确条件应（shall）在ECU独立诊断规范内定义。	M
31	**requestOutOfRange/请求超出范围**	M

NOTE: The strategy of negative response codes can be found in Figure A.2 of Annex A.
注：否定响应码的策略见附录 A 图 A.2。

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10.9 ResponseOnEvent (86 hex)/事件响应

N/A

10.10 LinkControl (87 hex)/链路控制

10.10.1 Service description/服务描述

The LinkControl service is used to control the communication link baud rate between the tester and the ECU (s) for the exchange of diagnostic data. This service optionally applies to those data link layers which allow for a baud rate transition during an active diagnostic session.
This service is used to transition the baud rate of the data link layer. To overcome functional communication, where the baud rate must be transitioned in multiple ECUs at the same time, the baud rate transition is split into two steps:
链路控制服务用来控制诊断仪与ECU之间交换诊断数据所使用的通信链路波特率。该服务可选择性地应用到那些在当前诊断会话状态下允许改变波特率的数据链路层上。
本服务用来改变数据链路层的波特率。考虑到功能寻址通信方式，所有ECU必须（must）在同一时间改变波特率，波特率的改变分为以下两步：

-**Step #1**: The tester verifies if the transition can be performed and informs the ECU(s) about the baud rate to be used. Each ECU shall respond positively (suppressPosRspMsgIndicationBit = FALSE) before the tester performs step #2. This step does not actually perform the baud rate transition.
-**Step #2**: The tester actually requests the transition of the baud rate. This step shall only be performed if it is verified that the baud rate transition can be performed (step #1 performed). In case of functional communication, it is recommended that there should not be any response from an ECU when the baud rate is transitioned (suppressPosRspMsgIndicationBit = TRUE) because one ECU might already have been transitioned to the new baud rate while others still need to transmit their response message(s) (baud rate mismatch avoidance).
-**步骤#1**：诊断仪验证改变能否执行，并且把将要使用的波特率发送给ECU。每个ECU应（shall）在执行步骤#2之前给出肯定响应（抑制肯定响应消息指示位为假）。实际上，这一步并不执行波特率的改变。
-**步骤#2**：诊断仪实际请求改变波特率。只有确认波特率改变能够执行（step #1执行后）之后，这一步才能执行。在采用功能寻址通信方式的情况下，推荐的做法是当波特率改变后，ECU不发送任何响应消息（抑制肯定响应消息指示位为真）。这是因为有的ECU可能改变为新的波特率，而其它ECU仍需要使用原来的波特率发送他们的响应消息（避免了波特率之间的不匹配）。

The linkControlType parameter in the request message, in conjunction with the conditional baudrateIdentifier/linkBaudrateRecord parameter, provides a mechanism to transition to a predefined or specifically defined baud rate.
Any baud rate transition shall occur as follows:
- suppressPosRspMsgIndicationBit = TRUE: after the successful transmission/reception of the tester request message, which requests the baud rate transition.
- suppressPosRspMsgIndicationBit = FALSE: after the successful transmission/reception of the ECU positive response message, which confirms the successful reception of the request, which requests the baud rate transition.

请求消息中的链路控制类型参数与条件参数（波特率标识符/链路波特率记录）一起，提供了一种改变预定义的或指定的波特率的机制。

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满足以下条件时，应（shall）改变波特率：
- 抑制肯定响应消息指示位为真：当成功发送/接收到诊断仪的请求改变波特率的请求消息之后。
- 抑制肯定响应消息指示位为假：当成功发送/接收到ECU的确认成功接收请求改变波特率的肯定响应消息之后。

NOTE: This service is tied to a non-defaultSession. A session layer timer timeout will transition the ECU(s) back to its (their) normal speed of operation. The same applies if an ECUReset service (11 hex) is performed. The transition into another non-defaultSession shall not influence the baud rate.
注: 该服务绑定一种非默认会话状态。会话层定时器的定时超时将使得ECU转换到其正常的通信波特率。同样地，当ECU执行了复位操作时，也会使得ECU转换到其正常的通信波特率。从一种非默认会话状态转换到另一个非默认会话状态时，不应（shall not）影响波特率。

10.10.2 Request message/请求消息

10.10.2.1 Request message definition 请求消息定义

Table 10-35 Request message definition (linkControlType =verifyBaudrateTransitionWithFixedBaudrate)

A_DataByte	Parameter Name	Cvt	Hex value
#1	LinkControl Request Service Id	M	87
#2	sub-function = [linkControlType]	M	01
#3	baudrateIdentifier	M	00-FF

Table 10-36 Request message definition (linkControlType = transitionBaudrate)

A_DataByte	Parameter Name	Cvt	Hex value
#1	LinkControl Request Service Id	M	87
#2	sub-function = [linkControlType]	M	83
NOTE: suppressPosRspMsgIndicationBit = TRUE.			

10.10.2.2 Request message sub-function parameter definition/请求消息子功能参数定义

Table 10-37 Request message sub-function parameter definition

Hex(bit 7-0)	Description	Cvt
01	**verifyBaudrateTransitionWithFixedBaudrate/确认使用固定波特率发送** This parameter is used to verify if a transition to a predefined baud rate, which is specified by the baudrateIdentifier data parameter, can be performed. 此子功能参数用来确认改变成通过波特率标识符数据参数定义的固定波特率是否可行执行。	M
83	**transitionBaudrate/发送波特率** This sub-function parameter requests the ECU(s) to transition the baud rate to the one that was specified in the preceding verification message. 此子功能参数请求ECU改变成给前一个确认请求消息中指定的波特率。	M

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10.10.2.3 Request message data parameter definition/请求消息数据参数定义

The following data parameters are defined for this service.
下面定义了本服务使用的数据参数。

Table 10-38 Request message data parameter definition

Definition
baudrateIdentifier/波特率标识符 This parameter references a fixed defined baud rate to transition to (see annex F.1). 此条件参数表示了要改变成一个固定的波特率（见附录B.1）。

10.10.3 Positive response message/肯定响应消息

10.10.3.1 Positive response message definition/肯定响应消息定义

Table 10-39 Positive response message definition (sub-function = verifyBaudrateTransitionWithFixedBaudrate (suppressPosRspMsgIndicationBit =0))

A_DataByte	Parameter Name	Cvt	Hex value
#1	LinkControl Response Service Id	S	C7
#2	linkControlType	M	01

sub-function = transitionBaudrate (suppressPosRspMsgIndicationBit =1)

NOTE: There is no any response for the request message, because of suppressPosRspMsgIndicationBit =1.
注：该请求消息没有肯定响应消息，因为抑制肯定响应消息指示位为 1.

10.10.3.2 Positive response message data parameter definition/肯定响应消息数据参数定义

Table 10-40 Positive response message data parameter definition

Definition
linkControlType This parameter is an echo of bits 6 - 0 of the linkControlType sub-function parameter from the request message. 该数据参数是请求消息中子功能参数第6-0位的映射。

10.10.4 Supported negative response codes/支持的否定响应码

Table 10-41 Supported negative response codes

Hex	Description	Cvt
7F	**serviceNotSupportedInActiveSession/激活会话不支持该服务**	M
12	**subFunctionNotSupported/不支持子功能**	M
13	**incorrectMessageLengthOrInvalidFormat/不正确的消息长度或无效的格式**	M
22	**conditionsNotCorrect/条件不正确** This code shall be returned if the criteria for the requested LinkControl are not met. The correct condition for the ECU to implement the request message shall be defined in the individual ECU diagnostic specification. 当不满足请求的链路控制的操作条件时，ECU应（shall）发送此否定响应码。ECU	M

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	执行该服务请求消息的正确条件应（shall）在ECU独立诊断规范内定义。	
24	**requestSequenceError/请求序列错误** This code shall be returned if the tester requests the transition of the baud rate without a preceding verification step which specifies the baud rate to transition to. 在发送确认要改变的波特率的请求消息之前，如果诊断仪发送了改变波特率的请求消息，ECU 应（shall）发送此否定响应码。	M
31	**requestOutOfRange/请求超出范围** This code shall be returned if the requested fixed baud rate (baudrateIdentifier) is invalid. 如果请求的固定波特率（波特率标识符）无效，ECU 应（shall）发送此否定响应码。	M

NOTE: The strategy of negative response codes can be found in Figure A.2 of Annex A.
注：否定响应码的策略见附录 A 图 A.2。

10.11 ProgrammingMode (A5 hex)/编程模式

N/A

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11 Data transmission functional unit/数据传输功能单元

11.1 ReadDataByIdentifier (22 hex)/通过标识符读数据

11.1.1 Service description/服务描述

The ReadDataByIdentifier service allows the tester to request data record values from the ECU identified by one or more dataIdentifiers.
The tester request message contains one or more two-byte dataIdentifier values that identify data record(s) maintained by the ECU.
From an attribute point of view, data records in the ECU are divided into two parts: one is the general data record for all the ECUs; the other is individual data record for each ECU.
通过标识符读数据服务允许诊断仪从一个或多个数据标识符（DID）的ECU中请求的数据记录的数值。
诊断仪请求消息包含一个或多个占2个字节的DID，ECU通过此标识符标识记录的数据。
从属性的角度来看，ECU中记录的数据可分为2部分：一部分是对所有ECU的通用性数据；另一部分是针对每个ECU的独立数据。

NOTE: The DID which used to identify the general data record for all the ECUs is called vehicle-manufacturer-specific DID, The DID which used to identify the individual data record for each ECU is called ECU-supplier-specific DID.
注：用来标识所有ECU的通用性数据的DID称为企业级DID，用来标识单个ECU的独立数据的DID称为ECU级DID。

The dataIdentifier values that identify the general dataRecord(s), as well as the format and definition of the dataRecord, are defined detailedly in Annex C, which is based on Annex C.1 of ISO 14229-1.
The dataIdentifier values that identify the individual dataRecord(s), as well as the format and definition of the dataRecord including analogue input and output signals, digital input and output signals, internal data and system status information if supported by the ECU, are defined detailedly in the individual diagnostic specification for each ECU, which comply with the range requirement of Annex C.1 of ISO 14229-1.
DID数值支持的通用性数据记录，以及格式和数据记录的定义在附录C有详细定义，附录C是基于ISO 14229-1附录C.1制定的。
DID数值支持的独立性数据记录，以及格式和数据记录的定义在ECU独立诊断规范内有详细定义，其中，数据记录包括模拟量输入输出信号，数字量输入输出信号，ECU支持的内部数据与系统状态信息。DID数值遵从ISO 14229-1附录C.1所要求的范围。

NOTE: The maximum number of dataIdentifier that can be simultaneously requested should be determined by SGMW and ECU Supplier together.
注：ECU所允许同时请求的DID最大数量应该（should）由SGMW与ECU供应商共同商定。

Upon receiving a ReadDataByIdentifier request, the ECU shall access the data elements of the records specified by the dataIdentifier parameter(s) and transmit their value in one single ReadDataByIdentifier positive response containing the associated dataRecord parameter(s). The request message may contain the same dataIdentifier multiple times. The ECU shall treat each dataIdentifier as a separate parameter and respond with data for each dataIdentifier as often as requested.
一旦收到通过标识符读数据服务请求，ECU应（shall）访问由DID参数定义的记录数据参数，并通过一个包含相关数据记录参数的肯定响应发送所请求数据记录的数值。一个请求消息中可能多次出现同一个

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DID，这种情况下，ECU应（shall）将每一个DID独立对待，并在肯定响应消息中对每一个DID给出相应的数据。

11.1.2 Request message/请求消息

11.1.2.1 Request message definition 请求消息定义

Table 11-1 Request message definition

A_DataByte	Parameter Name	Cvt	Hex value
#1	ReadDataByIdentifier Request Service Id	M	22
	dataIdentifier[] #1 = [		
#2	byte #1(MSB)	M	00-FF
#3	byte #2]	M	00-FF
:	:	:	:
	dataIdentifier[] #m[a] = [		
#n-1	byte #1(MSB)	U[b]	00-FF
#n	byte #2]	U	00-FF

a. The number of dataIdentifiers that can be simultaneously requested as agreed upon by SGMW and ECU supplier.
同时请求的 DID 数量由 SGWM 与 ECU 供应商共同商定。

b. The dataIdentifier parameter may or may not present, depending on the usage of the service.
依据服务的使用情况，该 DID 参数有可能出现也有可能不出现。

11.1.2.2 Request message sub-function parameter definition/请求消息子功能参数定义

This service does not use a sub-function parameter.
本服务不使用子功能参数。

11.1.2.3 Request message data parameter definition/请求消息数据参数定义

Table 11-2 Request message data parameter definition

Definition
dataIdentifier/数据标识符 This parameter identifies the ECU data record(s) being requested by the tester (see Annex C and the individual diagnostic specification for each ECU). 该参数表示诊断仪请求的ECU中的数据记录（参见附录C与ECU独立诊断规范）。

11.1.3 Positive response message/肯定响应消息

11.1.3.1 Positive response message definition/肯定响应消息定义

Table 11-3 Positive response message definition

A_DataByte	Parameter Name	Cvt	Hex value
#1	ReadDataByIdentifier Response Service Id	M	62
	dataIdentifier[] #1 = [		
#2	byte #1(MSB)	M	00-FF
#3	byte #2]	M	00-FF
	dataRecord[] #1 = [		

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#4 : #(k-1)+4	data #1 : data #k]	M : U1[a]	00-FF : 00-FF
:	:	:	:
#n-(o-1)-2 #n-(o-1)-1	dataIdentifier[] #m = [byte #1(MSB) byte #2]	U2[b] U2	00-FF 00-FF
#n-(o-1) : #n	dataRecord[] #m = [data #1 : data #o]	U2 : U3[a]	00-FF : 00-FF

c. The dataRecord parameter may or may not present, depending on the format and definition of the dataRecord. 数据记录参数能否出现取决于数据记录的格式与定义。
d. The dataIdentifier and dataRecord parameter may or may not present, depending on the usage of the service. DID 与数据记录参数能否出现取决于该服务的使用。

11.1.3.2 Positive response message data parameter definition/肯定响应消息数据参数定义

Table 11-4 Positive response message data parameter definition

Definition
dataIdentifier (#1 to #m) /数据标识符（#1 to #m） This parameter is an echo of bits 6 - 0 of the data parameter dataIdentifier from the request message. 该数据参数是请求消息中数据参数DID第6-0位的映射。
dataRecord (#1 to #k/o) /数据标识符（#1 to #k/o） This parameter is used by the ReadDataByIdentifier positive response message to provide the requested data record values to the tester. The content of the dataRecord is defined in Annex C and the individual diagnostic specification for each ECU. 该参数被通过标识符读数据的肯定响应消息使用，以向诊断仪提供请求的数据记录值。数据记录的内容在附录C及每个ECU独立诊断规范中有定义。

11.1.4 Supported negative response codes/支持的否定响应码

Table 11-5 Supported negative response codes

Hex	Description	Cvt
31	**requestOutOfRange/请求超出范围** This code shall be sent if: 如果出现以下任一条件，ECU应（shall）发送此否定响应码： 1) none of the requested dataIdentifier values are supported by the ECU; ECU不支持所请求的DID； 2) the tester exceeded the maximum number of dataIdentifiers allowed to be requested at a time. 在诊断仪发送的一次请求中超出了所允许的DID的最大数量。	M
13	**incorrectMessageLengthOrInvalidFormat/不正确的消息长度或无效的格式**	M
22	**conditionsNotCorrect/条件不正确** This code shall be sent if the operating conditions of the ECU to perform the required action are not met. The correct condition for the ECU to implement the request message shall be defined in the individual ECU diagnostic specification. 当不满足 ECU 执行诊断仪所请求动作的操作条件时，ECU 应（shall）发送此否定	M

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	响应码。ECU 执行该服务请求消息的正确条件应（shall）在 ECU 独立诊断规范内定义。	

NOTE: The strategy of negative response codes can be found in Figure A.3 of Annex A.
注：否定响应码的策略见附录 A 图 A.3。

11.2 ReadMemoryByAddress (23 hex)/通过地址读内存

N/A

11.3 ReadScalingDataByIdentifier (24 hex)/通过标识符读缩放数据

N/A

11.4 ReadDataByPeriodicIdentifier (2A hex)/通过周期标识符读数据

N/A

11.5 DynamicallyDefineDataIdentifier (2C hex)/动态定义数据标识符

11.5.1 Service description/服务描述

The DynamicallyDefineDataIdentifier service allows the tester to dynamically define in an ECU a data identifier that can be read via the ReadDataByIdentifier service at a later time.
The intention of this service is to provide the tester with the ability to group one or more data elements into a data superset that can be requested via the ReadDataByIdentifier or ReadDataByPeriodicIdentifier service. The data elements to be grouped together can be referenced by either

- a source data identifier, a position and size, or
- a memory address and a memory length, or
- a combination of the two methods listed above using multiple requests to define the single data element. The dynamically defined dataIdentifier will then contain a concatenation of the data parameter definitions.

动态定义DID服务（2C hex）允许诊断仪动态定义ECU内部的DID，之后，通过标识符读数据服务（22 hex）可以使用这些动态定义的DID读取相应的数据。
该服务的目的是使诊断仪具有把一个或多个数据元素组合成一个超级数据集合的能力，以便于可以使用通过标识符读数据服务（22 hex）或通过周期DID读数据服务（2A hex）读取该数据集合。可以通过以下任何一种方式进行获得需要组合的各个数据元素：

- 通过使用源DID，位置和大小的方式，
- 通过使用存储地址和存储长度的方式，
- 通过多次使用以上两种方法的请求消息来定义单个数据元素的组合方式。这种情况下，动态定义的DID包含一系列相互关联的数据参数定义。

This service allows greater flexibility in handling ad hoc data needs of the diagnostic application that extend beyond the information that can be read via statically defined data identifiers, and can also be used to reduce bandwidth utilization by avoiding overhead penalty associated with frequent request/response transactions.
在某些诊断应用场合下，需要的信息超出了通过静态定义的DID所能读取的信息范围。在处理这类诊断应用需要很多临时性数据时，对此，使用本服务可以具有很大的灵活性。因为能够采用避免频繁的去请求/响应的信息传输方式，所以，使用本服务也可以降低带宽的使用率。

The definition of the dynamically defined data identifier can either be done via a single request

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message or via multiple request messages. This allows for the definition of a single data element referencing source identifier(s) and memory addresses. The ECU shall concatenate the definitions for the single data element.
A redefinition of a dynamically defined data identifier can be achieved by clearing the current definition and starting over with the new definition.
动态定义DID可以通过单次或多次请求消息来定义。这样就允许通过源标识符和存储地址来定义单个数据元素。ECU应（shall）将通过单一数据元素定义方式定义的多个数据元素关联起来。动态定义DID的重定义可以通过清除当前的定义并重新开始新的定义来实现。

NOTE: Although this service does not prohibit such functionality, it is not recommended that the tester reference one dynamically defined data record from another, because deletion of the referenced record could create data consistency problems within the referencing record.
注：虽然该服务不禁止使用重定义功能，但是诊断仪重新定义或更换DID是不推荐的，因为这样会引起获取的记录数据的一致性问题。

This service also provides the ability to clear an existing dynamically defined data record. Requests to clear a data record shall be responded to positively if the specified data record identifier is within the range of valid dynamic data identifiers supported by the ECU (see A.1 for more details).
The ECU shall maintain the dynamically defined data record until it is cleared or as specified by SGMW (e.g. deletion of dynamically defined data records upon session transition or upon power down of the ECU).
该服务还提供了清除已有的动态定义数据记录的功能。如果指定的数据记录标识符在ECU支持动态DID的可用范围内，那么当诊断仪请求清除数据记录时，ECU应（shall）发送肯定响应消息（见附录C.1）。在SGMW清除或重新指定（例如由于ECU会话状态改变或者断电而导致动态定义数据记录的删除）之前，ECU应（shall）保持动态定义的数据记录。

The ECU can implement data records in two different ways:
-composite data records containing multiple elemental data records which are not individually referenced;
-unique two-byte identification “tags” or parameter identifier (PID) values for individual, elemental data records supported within the ECU (an example elemental data record, or PID, is engine speed or intake air temperature): this implementation of data records is a subset of a composite data record implementation because it only references a single elemental data record instead of a data record including multiple elemental data records.
ECU可以通过以下两种不同的方式实现动态定义DID的数据记录：
-包含多个不能单独获取的基本数据记录的混合数据记录方式；
-包含ECU支持的独立基本数据记录的2个字节的标识数据或PID方式（一个基本数据记录或PID的例子就是发动机转速或进气温度）：这种数据记录的实现方式是一个混合数据记录实现的子集，因为它仅仅获取了一个基本数据记录，而不是获取了一个包含多个基本数据记录的数据记录。

Both types of implementing of data records are supported by the DynamicallyDefineDataIdentifier service to define a dynamic data identifier.
- Composite block of data: the position parameter shall reference the starting point in the composite block of data and the size parameter shall reflect the length of data to be placed in the dynamically defined data identifier. The tester is responsible for including only a portion of an elemental data record of the composite block of data in the dynamic data record.
- Two-byte PID: the position parameter shall be set to one (1) and the size parameter shall reflect the length of the PID (length of the elemental data record). The tester is responsible to not include only a

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portion of the two-byte PID value in the dynamic data record.
动态定义DID服务支持上述两种类型的数据记录实现方式，并用来定义动态DID。
- 混合数据模块：位置参数应（shall）标明混合数据模块的起始位置；长度大小参数应（shall）反映放置在动态定义DID内的数据长度。
- 2字节的PID：位置参数应（shall）设置为1，长度大小参数应（shall）反映PID的长度（基本数据记录的长度）。诊断仪有责任保证在动态数据记录中不能仅包含2个字节PID表示数据内容的一部分。

The ordering of the data within the dynamically defined data record shall be the same as specified in the tester request message(s). Also, the first position of the data specified in the tester's request shall be oriented such that it occurs closest to the beginning of the dynamic data record, in accordance with the ordering requirement mentioned in the preceding sentence.
肯定响应消息中动态定义的数据记录顺序应（shall）与诊断仪请求消息内定义的一致。另外，诊断仪请求消息中定义的第一个的数据应（shall）是导向性的，因此，它出现在距离动态数据记录的开始最近的地方，与上述段落提到的顺序要求一致。

11.5.2 Request message/请求消息

11.5.2.1 Request message definition 请求消息定义

Table 11-6 Request message definition (Request message definition — Sub-function = defineByIdentifier)

A_DataByte	Parameter Name	Cvt	Hex value
#1	DynamicallyDefineDataIdentifier Request Service Id	M	2C
#2	sub-function = [defineByIdentifier]	M	01
#3 #4	dynamicallyDefineDataIdentifier[] = [byte#1 (MSB) byte#2 (LSB)]	M M	F3 00-FF
#5 #6	sourceDataIdentifier[] #1 = [byte#1 (MSB) byte#2 (LSB)]	M M	00-FF 00-FF
#7	positionInSourceDataRecord #1	M	01-FF
#8	memorySize #1	M	00-FF
:	:	:	:
#n-3 #n-2	sourceDataIdentifier[] #m = [byte#1 (MSB) byte#2 (LSB)]	U U	00-FF 00-FF
#n-1	positionInSourceDataRecord #m	U	01-FF
#n	memorySize #m	U	00-FF

Table 11-7 Request message definition (Request message definition — Sub-function = clearDynamicallyDefinedDataIdentifier)

A_DataByte	Parameter Name	Cvt	Hex value
#1	DynamicallyDefineDataIdentifier Request Service Id	M	2C
#2	sub-function = [clearDynamicallyDefinedDataIdentifier]	M	03
#3	dynamicallyDefineDataIdentifier[] = [byte#1 (MSB)	C[a]	F3

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#4	byte#2 (LSB)]	C	00-FF
a. The presence of the C parameter requires the ECU to clear the dynamicallyDefineDataIdentifier included in byte#1 and byte#2.If the parameter is not present all dynamicallyDefineDataIdentifier in the ECU shall be cleared. 如果使用参数C，则要求ECU删除该参数中byte#1和byte#2指定的动态定义DID。如果不使用参数C，那么则要求ECU删除所有的动态定义DID。			

11.5.2.2 Request message sub-function parameter definition/请求消息子功能参数定义

Table 11-8 Request message sub-function parameter definition

Hex(bit 6-0)	Description	Cvt
01	**DefineByIdentifier/定义类型**	M
03	**ClearDynamicallyDefinedDataIdentifier/清除动态定义的DID**	M

11.5.2.3 Request message data parameter definition/请求消息数据参数定义

Table 11-9 Request message data parameter definition

Definition
dynamicallyDefineDataIdentifier/动态定义DID This parameter specifies how the dynamic data record which is being defined by the tester will be referenced in future calls to the service ReadDataByIdentifier or ReadDataByPeriodicDataIdentifier. The dynamicallyDefineDataIdentifier shall be handled as a dataIdentifier in the ReadDataByIdentifier service (see A for further details). It shall be handled as a periodicRecordIdentifier in the ReadDataByPeriodicDataIdentifier service. 该参数是诊断仪定义的用于动态数据记录的动态DID，定好该参数之后，通过标识符读数据服务（22 hex）或通过周期性DID读数据服务（2A hex）可以使用该动态定义的DID请求相应的动态数据记录内容。 在通过标识符读数据服务（22 hex）中，动态定义的DID应（shall）作为普通DID（详请见附录C）使用。在通过周期性DID读数据服务（2A hex）中，动态定义的DID应（shall）作为周期性DID使用。
sourceDataIdentifier/源DID This parameter is only present for sub-function = defineByIdentifier. This parameter logically specifies the source of information to be included into the dynamic data record. For example, this could be a 2-byte PID identifier used to reference engine speed or a 2-byte data record identifier used to reference a composite block of information containing engine speed, vehicle speed, intake air temperature, etc. (see A for further details). 该参数仅在子功能参数为通过标识符定义（defineByIdentifier）的请求消息中使用。该参数从逻辑上指明了动态数据记录中的信息源。例如：该参数可以是一个用于表示发动机转速的2个字节的PID标识符，也可以是一个用于表示发动机转速、车速以及进气温度等混合信息块的2个字节的数据记录标识符（详请见附录C）。
positionInSourceDataRecord/源数据记录中位置信息 This parameter is only present for sub-function = defineByIdentifier. This one-byte parameter is used to specify the starting byte position of the excerpt of the source data record to be included in the dynamic data record. A position of one (1) shall reference the first byte of the data record referenced by the sourceDataIdentifier. 该参数仅在子功能参数为通过标识符定义（defineByIdentifier）的请求消息中使用。该参数长度为1个字节，用来定义动态数据记录中的源数据记录起始字节位置。一个位置为“1”该参数表示源DID所表示数据记录的第一个字节。
memorySize/存储大小 This parameter is used to specify the total number of bytes from the source data record that are to be included in the dynamic data record. In case of sub-function = defineByIdentifier, then the positionInSourceDataRecord parameter is used in addition to specify the starting position in the source data identifier from which the memorySize applies. The number of bytes used for this size is one (1) byte. 该参数用来定义动态数据记录中源数据记录的所有字节数。 如果请求消息中的子功能参数为通过标识符定义（defineByIdentifier），那么请求消息中的“源数据记录中位置信息”参数同时也定义了使用存储大小参数表示的源DID的起始位置，该位置就是。这个表示存储大小的参数长度为1个字节。

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11.5.3 Positive response message/肯定响应消息

11.5.3.1 Positive response message definition/肯定响应消息定义

Table 11-10 Positive response message definition

A_DataByte	Parameter Name	Cvt	Hex value
#1	DynamicallyDefineDataIdentifier Response Service Id	M	6C
#2	definitionType	M	01,03
	dynamicallyDefineDataIdentifier [] = [		
#3	byte#1 (MSB)	C[a]	F3
#4	byte#2 (LSB)]	C	00-FF
a. The presence of the C parameter is required if the dynamicallyDefineDataIdentifier parameter is present in the request message, otherwise the parameter shall not be included. 如果请求消息中存在动态定义DID，那么参数C就需要存在，否则该参数C不应（shall not）存在。			

11.5.3.2 Positive response message data parameter definition/肯定响应消息数据参数定义

Table 11-11 Positive response message data parameter definition

Definition
definitionType/定义类型 This parameter is an echo of bits 6 - 0 of the sub-function parameter from the request message. 该数据参数是请求消息中子功能参数第6-0位的映射。
dynamicallyDefineDataIdentifier/动态定义DID This parameter is an echo of the data parameter dynamicallyDefineDataIdentifier from the request message. 该参数是请求消息中数据参数（动态定义DID）的映射。

11.5.4 Supported negative response codes/支持的否定响应码

Table 11-12 Supported negative response codes

Hex	Description	Cvt
7F	**serviceNotSupportedInActiveSession/激活会话不支持该服务**	M
12	**subFunctionNotSupported/不支持子功能**	M
13	**incorrectMessageLengthOrInvalidFormat/不正确的消息长度或无效的格式**	M
22	**conditionsNotCorrect/条件不正确** This code shall be sent if the operating conditions of the ECU to perform the required action are not met. The correct condition for the ECU to implement the request message shall be defined in the individual ECU diagnostic specification. 当不满足ECU执行诊断仪所请求动作的操作条件时，ECU应（shall）发送此否定响应码。ECU执行该服务请求消息的正确条件应（shall）在ECU独立诊断规范内定义。	M
31	**requestOutOfRange/请求超出范围** This response code shall be sent if: 如果出现以下任一条件，ECU应（shall）发送此否定响应码： 1) any data identifier (dynamicallyDefineDataIdentifier or any sourceDataIdentifier) in the request message is not supported/invalid; 不支持请求消息中的任何DID（动态定义的DID或者任何的源DID）。 2) the positionInSourceDataRecord is incorrect (less than 1 or greater than the	M

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	maximum allowed by the ECU); “源数据记录中位置信息”参数不正确（小于1或大于ECU允许的最大值）； 3) the specified memorySize is invalid; 定义的“存储大小”参数无效； 4) the amount of data to be packed into the dynamic data identifier exceeds the maximum allowed by the ECU. 封装成动态DID的数据超过了ECU允许的最大数量。 **NOTE:** The maximum number of data that are to be packed into the dynamic data identifier allowed by the ECU shall be defined in the individual ECU diagnostic specification. **注：** ECU所允许的封装成动态DID内数据记录的最大数量应（shall）在ECU独立诊断规范内定义。	

NOTE: The strategy of negative response codes can be found in Figure A.2 of Annex A.
注：否定响应码的策略见附录 A 图 A.2。

11.6 WriteDataByIdentifier (2E hex)/通过标识符写数据

11.6.1 Service description/服务描述

The WriteDataByIdentifier service allows the tester to write information into the ECU at an internal location specified by the provided data identifier.
通过标识符写数据服务允许诊断仪将相关信息写入到数据标识符规定的ECU内部位置。

The ECU must be “unlocked” through the SecurityAccess service ($27), before performing this service. Other condition(s) that shall be met is/are defined in the individual diagnostic specification for each ECU. Possible uses for this service are, and not limit to:

- writing configuration information into the ECU (e.g. VIN number);
- resetting learned values; and
- setting option content.

执行此服务之前，ECU必须（must）通过安全访问服务（$27）使其解锁。需要满足的其它条件在每个ECU独立诊断规范内定义。可能应用此服务的情况如下（但不限于）所示：

- 向ECU中写入配置信息（例如VIN码）；
- 复位已学过的数值；以及
- 设置操作内容。

The ECU shall restrict or prohibit write access to the dataIdentifier values that is defined as read-only identifiers in Annex C and/or the individual diagnostic specification for each ECU.
对于附录C中定义的只读DID的数据记录和/或ECU独立诊断规范中定义的只读DID数记录，ECU应（shall）限制或禁止对其进行的写操作。

11.6.2 Request message/请求消息

11.6.2.1 Request message definition 请求消息定义

Table 11-13 Request message definition

A_DataByte	Parameter Name	Cvt	Hex value
#1	WriteDataByIdentifier Request Service Id	M	2E
#2	dataIdentifier[] = [byte #1(MSB)	M	00-FF

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#3	byte #2]	M	00-FF
#4 : #m+3	dataRecord[] = [data #1 : data #m]	M : U[a]	00-FF : 00-FF
a. The dataIdentifier parameter may or may not present, depending on the format and definition of the dataRecord. DID 参数是否出现，取决于于数据记录的格式和定义。			

11.6.2.2 Request message sub-function parameter definition/请求消息子功能参数定义

This service does not use a sub-function parameter.
本服务不使用子功能参数。

11.6.2.3 Request message data parameter definition/请求消息数据参数定义

Table 11-14 Request message data parameter definition

Definition
dataIdentifier/数据标识符 This parameter identifies the ECU data record(s) being requested by the tester (see and the individual diagnostic specification for each ECU). 该参数表示诊断仪请求的ECU中的数据记录（参见ECU独立诊断规范）。
dataRecord/数据记录 This parameter provides the data record associated with the dataIdentifier that the tester is requesting to write to. The content of the dataRecord is defined in Annex C or the individual diagnostic specification for each ECU. 该参数提供了与诊断仪请求写入的DID有关的数据记录。该数据记录内容定义在附录C或每个ECU独立诊断规范中。

11.6.3 Positive response message/肯定响应消息

11.6.3.1 Positive response message definition/肯定响应消息定义

Table 11-15 Positive response message definition

A_DataByte	Parameter Name	Cvt	Hex value
#1	WriteDataByIdentifier Response Service Id	M	6E
#2 #3	dataIdentifier[] = [byte #1(MSB) byte #2]	M M	00-FF 00-FF

11.6.3.2 Positive response message data parameter definition/肯定响应消息数据参数定义

Table 11-16 Positive response message data parameter definition

Definition
dataIdentifier/数据标识符 This parameter is an echo of bits6-0 of the data parameter dataIdentifier from the request message. 该参数是请求消息中数据参数（DID）第6-0位的映射。

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11.6.4 Supported negative response codes/支持的否定响应码

Table 11-17 Supported negative response codes

Hex	Description	Cvt
7F	**serviceNotSupportedInActiveSession/激活会话不支持该服务**	M
33	**securityAccessDenied/安全访问拒绝** This code shall be sent if the ECU is not in an unlocked state. 如果处于锁闭状态，ECU应（shall）发送此否定响应码。	M
31	**requestOutOfRange/请求超出范围** This response code shall be sent if: 如果出现以下任一条件，ECU应（shall）发送此否定响应码： 1) the dataIdentifier in the request message is not supported in the ECU or the dataIdentifier is supported for read only purpose; ECU不支持请求消息中的DID或者支持的DID为只读的。 2) any data transmitted in the request message after the dataIdentifier is invalid (if applicable to the). 请求消息中DID之后发送的数据（如果适用于该节点的话）为无效的。	M
13	**incorrectMessageLengthOrInvalidFormat/不正确的消息长度或无效的格式**	M
22	**conditionsNotCorrect/条件不正确** This code shall be sent if the operating conditions of the ECU to perform the required action are not met. The correct condition for the ECU to implement the request message shall be defined in the individual ECU diagnostic specification. 当不满足 ECU 执行诊断仪所请求动作的操作条件时，ECU 应（shall）发送此否定响应码。ECU 执行该服务请求消息的正确条件应（shall）在 ECU 独立诊断规范内定义。	M
72	**generalProgrammingFailure/一般编程失败** This return code shall be sent if the ECU detects an error when writing to a memory location. 当向存储区域内写数据时，如果检测到一个错误，ECU应（shall）发送此否定响应码。	M

NOTE: The strategy of negative response codes can be found in Figure A.3 of Annex A.
注：否定响应码的策略见附录 A 图 A.3。

11.7 WriteMemoryByAddress (3D hex)/通过地址写内存

N/A

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12 Stored data transmission functional unit/存储数据传输功能单元

12.1 ClearDiagnosticInformation (14 hex)/清除故障信息

12.1.1 Service description/服务描述

The ClearDiagnosticInformation service is used by the tester to clear diagnostic information in one ECU's or multiple ECUs' memory.
诊断仪使用清除故障信息服务（14 hex）来清除一个或多个ECU内存中的诊断信息。

12.1.2 Request message/请求消息

12.1.2.1 Request message definition 请求消息定义

Table 12-1 Request message definition

A_DataByte	Parameter Name	Cvt	Hex value
#1	ClearDiagnosticInformation Request Service Id	M	14
	groupOfDTC[] =[		
#2	groupOfDTCHighByte	M	00-FF
#3	groupOfDTCMiddleByte	M	00-FF
#4	groupOfDTCLowByte]	M	00-FF

12.1.2.2 Request message sub-function parameter definition/请求消息子功能参数定义

This service does not use a sub-function parameter.
本服务不使用子功能参数。

12.1.2.3 Request message data parameter definition/请求消息数据参数定义

Table 12-2 Request message data parameter definition

Definition
groupOfDTC/DTC类别 This parameter contains a three-byte value indicating the group of DTCs (e.g. powertrain, body, and chassis) or the particular DTC to be cleared. The definition of values for each value/range of values is included in Annex D.1. 该参数长度为3个字节，用来表示要清除的诊断故障码属于哪一个类别（例如动力、车身以及底盘）或者哪一个特殊的故障码。附录D.1定义了该参数的数值及其表示的范围。

12.1.3 Positive response message/肯定响应消息

12.1.3.1 Positive response message definition/肯定响应消息定义

Table 12-3 Positive response message definition

A_DataByte	Parameter Name	Cvt	Hex value
#1	ClearDiagnosticInformation Response Service Id	M	54

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12.1.3.2 Positive response message data parameter definition/肯定响应消息数据参数定义

There are no data parameters used by this service in the positive response message.
本服务的肯定响应消息不使用数据参数。

12.1.4 Supported negative response codes/支持的否定响应码

Table 12-4 Supported negative response codes

Hex	Description	Cvt
7F	**serviceNotSupportedInActiveSession/激活会话不支持该服务**	M
31	**requestOutOfRange/请求超出范围** This return code shall be sent if the specified groupOfDTC parameter is not supported. 如果不支持指定的 DTC 类别，ECU 应（shall）发送此否定响应码。	M
13	**incorrectMessageLengthOrInvalidFormat/不正确的消息长度或无效的格式**	M
22	**conditionsNotCorrect/条件不正确** This code shall be sent if the operating conditions of the ECU to perform the required action are not met. The correct condition for the ECU to implement the request message shall be defined in the individual ECU diagnostic specification. 当不满足 ECU 执行诊断仪所请求动作的操作条件时，ECU 应（shall）发送此否定响应码。ECU 执行该服务请求消息的正确条件应（shall）在 ECU 独立诊断规范内定义。	M

NOTE: The strategy of negative response codes can be found in Figure A.3 of Annex A.
注：否定响应码的策略见附录 A 图 A.3。

12.2 ReadDTCInformation (19 hex) /读取诊断故障代码信息

12.2.1 Service description/服务描述

12.2.1.1 General description/概述

This service allows a tester to read the status of ECU-resident Diagnostic Trouble Code (DTC) information from any ECU or group of ECUs within a vehicle. Unless otherwise stated, the ECU shall return both emissions-related and non-emissions-related DTC information.
诊断仪通过该服务读取任一车载ECU或一组ECU内存储的诊断故障码（DTC）状态。ECU应（shall）返回与排放相关以及与排放不相关的的DTC信息，除非另有规定。

NOTE: The maximum number of DTC that ECU can store simultaneously should be determined by SGMW and ECU Supplier together.
注：ECU 同时可存储 DTC 的最大数量应该（should）由 SGMW 与 ECU 供应商共同确定。

NOTE: If a DTC only supports a group snapshot data record, the maximum number of DTC snapshot record that ECU can store simultaneously should the same as the maximum number of DTC; If a DTC can support several groups snapshot data record, the corresponding relationship between stored DTC and snapshot data record should be determined by SGMW and ECU Supplier together.
注：如果一个 DTC 仅支持一组快照数据记录，那么 ECU 同时可存储快照数据记录的最大数量应该

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（should）与 DTC 的最大数量相同；如果一个 DTC 可支持多组快照数据记录，那么 ECU 已存储 DTC 与其快照数据记录的对应关系应该（should）由 SGMW 与 ECU 供应商共同制定。

NOTE: From an attribute point of view, snapshot is divided into two parts: one is the general snapshot with one or more group data records for all the ECUs, The format and content defined in chapter 12.2.3.2; the other is local snapshot with one or more group data records for each specified ECU. The format and content about snapshot should be determined by SGMW and ECU Supplier together.
注：从属性的角度来说，快照可分为全局快照与局部快照。全局快照是指适用于所有 ECU 的一组或多组快照记录数据，其格式与内容的定义在 12.2.3.2 章节；局部快照是指适用于特定 ECU 的一组或多组快照记录数据。其格式与内容应该（should）由 SGMW 与供应商共同确定，并在独立诊断规范内明确。

NOTE: If all fault memories are occupied, while the new DTC is generated, the Overflow strategy about stored DTCs and snapshot data records should be determined by SGMW and ECU Supplier together.
注：当所有的故障存储内存已被占用时，如果产生新的 DTC，那么已存储的 DTC 及其快照数据记录的溢出策略应该（should）由 SGMW 与 ECU 供应商共同制定。

12.2.2 Request message/请求消息

12.2.2.1 Request message definition 请求消息定义

Table 12-5 Request message definition (sub-function = reportNumberOfDTCByStatusMask, reportDTCByStatusMask)

A_DataByte	Parameter Name	Cvt	Hex value
#1	ReadDTCInformation Request Service Id	M	19
#2	Sub-function = [reportNumberOfDTCByStatusMask reportDTCByStatusMask]	M	 01 02
#3	DTCStatusMask	M	00-FF

Table 12-6 Request message definition (sub-function = reportDTCSnapshotIdentification,reportDTCSnapshotRecordByDTCNumber)

A_DataByte	Parameter Name	Cvt	Hex value
#1	ReadDTCInformation Request Service Id	M	19
#2	Sub-function = [reportDTCSnapshotIdentification reportDTCSnapshotRecordByDTCNumber]	M	 03 04
 #3 #4 #5	DTCMaskRecord[] = [DTCHighByte DTCMiddleByte DTCLowByte]	 C[a] C C	 00-FF 00-FF 00-FF
#6	DTCSnapshotRecordNumber	C	00-FF

a. The parameter C DTCMaskRecord and DTCSnapshotRecordNumber parameters are only present if the sub-function parameter is equal to reportDTCSnapshotRecordByDTCNumber.
只有当子功能参数为“通过DTC报告DTC快照记录”（reportDTCSnapshotRecordByDTCNumber）时，才存在参数C（DTC掩码记录与DTC快照记录码）。

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Table 12-7 Request message definition (sub-function = reportDTCExtendedDataRecordByDTCNumber)

A_DataByte	Parameter Name	Cvt	Hex value
#1	ReadDTCInformation Request Service Id	M	19
#2	Sub-function = [reportDTCExtendedDataRecordByDTCNumber]	M	06
	DTCMaskRecord[] = [		
#3	DTCHighByte	M	00-FF
#4	DTCMiddleByte	M	00-FF
#5	DTCLowByte]	M	00-FF
#6	DTCExtendedDataRecordNumber	M	00-FF

Table 12-8 Request message definition (Sub-function = reportSupportedDTC)

A_DataByte	Parameter Name	Cvt	Hex value
#1	ReadDTCInformation Request Service Id	M	19
#2	Sub-function = [reportSupportedDTC]	M	0A

12.2.2.2 Request message sub-function parameter definition/请求消息子功能参数定义

Table 12-9 Request message sub-function parameter definition

Hex(bit 6-0)	Description	Cvt
01	reportNumberOfDTCByStatusMask/通过状态掩码报告DTC数量	M
02	reportDTCByStatusMask/通过状态掩码报告DTC	M
03	reportDTCSnapshotIdentification/报告 DTC 快照标识符	M
04	reportDTCSnapshotRecordByDTCNumber/通过DTC报告DTC快照记录	M
06	reportDTCExtendedDataRecordByDTCNumber/通过DTC报告DTC扩展数据记录	M
0A	reportSupportedDTC/报告支持的DTC	M

12.2.2.3 Request message data parameter definition/请求消息数据参数定义

Table 12-10 Request message data parameter definition

Definition
DTCStatusMask/DTC状态掩码 The DTCStatusMask contains eight (8) DTC status bits. The definitions for each of the eight (8) bits can be found in D.2 of ISO 14229-1. This byte is used in the request message to allow a tester to request DTC information for the DTC's whose status matches the DTCStatusMask. A DTC's status matches the DTCStatusMask if any one of the DTCs actual status bits is set to 1 and the corresponding status bit in the DTCStatusMask is also set to 1 (i.e. if the DTCStatusMask is bit-wise logically ANDed with the DTC's actual status and the result is non-zero, then a match has occurred). If the tester specifies a status mask that contains bits that the ECU does not support, then the ECU shall process the DTC information using only the bits that it does support. DTC状态掩码参数包含8个DTC状态位，每一位的定义见ISO 14229-1。通过该参数诊断仪能够请求ECU中DTC状态与DTC状态掩码相匹配的故障码信息。如果某一个故障码的实际状态位为1，并且DTC状态掩码中的相应位也为1，那么就认为该故障码的状态与DTC状态掩码相匹配（即：如果DTC状态掩码字节与DTC实际状态字节进行逻辑“位与”运算后的结果为非零值，那么两者就相匹配）。如果诊断仪定义了一个状态掩码，其中包含ECU不支持的位，那么ECU应（shall）仅使用本身支持的位进行处理故障信息。

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DTCMaskRecord [DTCHighByte, DTCMiddleByte, DTCLowByte]/DTC掩码记录

DTCMaskRecord is a three-byte value containing DTCHighByte, DTCMiddleByte and DTCLowByte, which together represent a unique identification number for a specific diagnostic trouble code supported by an ECU.
The definition of the three-byte DTC number allows for the only one way of coding DTC information. It shall be done by using the decoding of the DTCHighByte, DTCMiddleByte and DTCLowByte according to the ISO 15031-6 specification (this format is identified by the DTCFormatIdentifier = ISO15031-6DTCFormat).
DTC掩码记录参数长度为3个字节，包括高字节、中间字节与低字节。这三个字节一起代表ECU所支持的规定故障码的唯一标识信息。
这三个字节的DTC定义规定DTC编码只能采用一种方式。这就要求设计者应（shall）参照ISO 15031-6中DTC高字节、中间字节与低字节的方式（该格式是在DTC格式标识符参数为“ISO15031-6DTC格式”的情况下标识的）进行DTC编码。

DTCSnapshotRecordNumber/DTC快照记录码

DTCSnapshotRecordNumber is a one-byte value indicating the number of the specific DTCSnapshot data records requested for a tester-defined DTCMaskRecord via the reportDTCSnapshotByDTCNumber sub-function. For emissions-related ECUs (OBD-compliant ECUs), the DTCSnapshot data record number 00 hex shall be the equivalent data record as specified in ISO 15031-5 service 02 hex frame number 00 hex. If the ECU supports multiple DTCSnapshot data records, the range of 01 hex through FE hex shall be used. A value of FF hex requests the ECU to report all stored DTCSnapshot data records at once.
DTC快照记录码占一个字节，表示特定的DTC快照数据记录编号，通过“通过DTC报告DTC快照记录（reportDTCSnapshotRecordByDTCNumber）”子功能可以请求一诊断仪定义的DTC掩码记录（DTCMaskRecord）。对于排放相关的ECU来说，DTC快照数据记录码00 hex应（shall）等同于（ISO 15031-5中描述的）请求动力系统冻结帧数据（02 hex）服务中编号为00 hex的冻结帧定义的数据记录。如果ECU支持多个DTC快照数据记录，那么该纪录码应（shall）使用01 hex ~ FE hex范围内的数值。当该参数值为FF hex时，要求ECU一次性报告所有存储的DTC快照数据记录。

DTCExtendedDataRecordNumber/DTC扩展数据记录码

DTCExtendedDataRecordNumber is a one-byte value indicating the number of the specific DTCExtendedData record requested for a tester-defined DTCMaskRecord via the reportDTCExtendedDataRecordByDTCNumber sub-function. For emissions-related ECUs (OBD-compliant ECUs), the DTCExtendedDataRecordNumber 00 hex shall be reserved for future OBD use.
DTC扩展数据记录码参数长度为1个字节，该数值表示诊断仪请求的指定故障码扩展数据记录的编号，该请求消息中的子功能参数为“通过DTC上报DTC扩展数据记录”。对于与排放相关的ECU（符合OBD标准的ECU），DTC扩展数据记录码参数为0的情况应（shall）预留给将来的OBD功能使用。
The following DTCExtendedDataRecordNumber ranges are used for this document:
本文档使用的DTC扩展数据记录码的范围如下：

- A value of 01 hex - 8F hex requests the ECU to report the stored DTCExtendedData records, which are defined detailedly in the individual diagnostic specification for each ECU.
  DTC扩展数据记录码数值为01 hex - 8F hex，诊断仪向ECU请求其存储的故障码扩展数据记录信息，这些记录信息在ECU独立诊断规范内有详细定义。
- A value of 90 hex - EF hex requests the ECU to report legislated OBD stored DTCExtendedData records. This is only valid for OBD ECU.
  DTC扩展数据记录码数值为90 hex - EF hex，诊断仪向ECU请求其存储的OBD法规要求的故障码扩展数据记录信息，该数值仅对OBD ECU有效。
- A value of FE hex requests the ECU to report all legislated OBD stored DTCExtendedData records in a single response message. This is only valid for OBD ECU.
  DTC扩展数据记录码数值为FE hex，诊断仪向ECU请求其存储的所有OBD法规要求的故障码扩展数据记录信息，这些信息通过一个响应消息全部发出，该数值仅对OBD ECU有效。
- A value of FF hex requests the ECU to report all stored DTCExtendedData records in a single response message.
  DTC扩展数据记录码数值为FF hex，诊断仪向ECU请求其存储的所有故障码扩展数据记录信息，这些信息通过一个响应消息全部发出。

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12.2.3 Positive response message/肯定响应消息

12.2.3.1 Positive response message definition/肯定响应消息定义

Table 12-11 Positive response message definition (sub-function = reportNumberOfDTCByStatusMask)

A_DataByte	Parameter Name	Cvt	Hex value
#1	ReadDTCInformation Positive Response Service Id	M	59
#2	reportType = [reportNumberOfDTCByStatusMask]	M	01
#3	DTCStatusAvailabilityMask	M	00-FF
#4	DTCFormatIdentifier = [ISO 15031-6 DTCFormat]	M	00
#5 #6	DTCCount[] = [DTCCountHighByte DTCCountLowByte]	M M	00-FF 00-FF

Table 12-12 Positive response message definition (sub-function = reportDTCByStatusMask, reportSupportedDTC)

A_DataByte	Parameter Name	Cvt	Hex value
#1	ReadDTCInformation Positive Response Service Id	M	59
#2	reportType = [reportDTCByStatusMask reportSupportedDTC]	M	02 0A
#3	DTCStatusAvailabilityMask	M	00-FF
#4	DTCAndStatusRecord[] = [DTCHighByte #1	C_1[a]	00-FF
#5	DTCMiddleByte #1	C_1	00-FF
#6	DTCLowByte #1	C_1	00-FF
#7	statusOfDTC #1	C_1	00-FF
#8	DTCHighByte #2	C_2[b]	00-FF
#9	DTCMiddleByte #2	C_2	00-FF
#10	DTCLowByte #2	C_2	00-FF
#11	statusOfDTC #2	C_2	00-FF
:	:	:	:
#n-3	DTCHighByte #m	C_2	00-FF
#n-2	DTCMiddleByte #m	C_2	00-FF
#n-1	DTCLowByte #m	C_2	00-FF
#n	statusOfDTC #m]	C_2	00-FF

a. The C_1 parameter is only present if DTC information is available to be reported.
只有在上报的DTC信息是可用的情况下，才存在参数C_1。

b. The C_2 parameter is only present if more than one set of DTC information is available to be reported.
只有在上报的DTC信息是可用的，并且可用的DTC信息多于一组的情况下，才存在参数C_2。

Table 12-13 Positive response message definition (sub-function = reportSnapshotIdentification)

A_DataByte	Parameter Name	Cvt	Hex value
#1	ReadDTCInformation Positive Response Service Id	M	59
#2	reportType = [reportSnapshotIdentification]	M	03
#3	DTCRecord[] #1 = [DTCHighByte#1	C_1[a]	00-FF

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#4 #5	DTCMiddleByte#1 DTCLowByte#1]	C_1 C_1	00-FF 00-FF
#6	DTCSnapshotRecordNumber#1	C_1	00-FF
:	:	:	:
#n-3 #n-2 #n-1	DTCRecord[] #m = [DTCHighByte#m DTCMiddleByte#m DTCLowByte#m]	C_2[b] C_2 C_2	00-FF 00-FF 00-FF
#n	DTCSnapshotRecordNumber#m	C_2	00-FF

a. For C_1, the DTCRecord and DTCSnapshotRecordNumber parameter is only present if at least one DTCSnapshot record is available to be reported.
只有在上报的DTC快照记录是可用的，并且至少存在一组可用的DTC快照记录的情况下，才存在参数C_1（DTC记录与DTC快照记录码）。

b. For C_2, the DTCRecord and DTCSnapshotRecordNumber parameter is only present if more than one DTCSnapshot record is available to be reported.
只有在上报的DTC快照记录是可用的，并且可用的DTC快照记录多于一组的情况下，才存在参数C_2（DTC记录与DTC快照记录码）。

Table 12-14 Positive response message definition (sub-function = reportDTCSnapshotRecordByDTCNumber)

A_DataByte	Parameter Name	Cvt	Hex value
#1	ReadDTCInformation Positive Response Service Id	M	59
#2	reportType = [reportDTCSnapshotRecordByDTCNumber]	M	04
#3 #4 #5 #6	DTCAndStatusRecord[]= [DTCHighByte DTCMiddleByte DTCLowByte statusOfDTC]	M M M M	00-FF 00-FF 00-FF 00-FF
#7	DTCSnapshotRecordNumber#1	M	00-FF
#8	DTCSnapshotRecordNumberOfIdentifiers#1	M	00-FF
#9 #10 #11 #12 #13 #14 #15 #16 #17 #18 #19 #20 #21 #22 : #22+(p-1) : #r-2	DTCSnapshotRecord[] #1 = [ECU Voltage DataIdentifier(MSB) ECU Voltage DataIdentifier(LSB) ECU Voltage Engine Speed DataIdentifier(MSB) Engine Speed DataIdentifier(LSB) Engine Speed (High Byte) Engine Speed (Low Byte) Vehicle Speed DataIdentifier(MSB) Vehicle Speed DataIdentifier(LSB) Vehicle Speed (High Byte) Vehicle Speed (Low Byte) dataIdentifier#4 byte #1 (MSB) dataIdentifier#4 byte #2 (LSB) snapshotData#4 byte #1 : snapshotData#4 byte #p : dataIdentifier#m byte #1 (MSB)	M_1[a] M_1 M_1 M_1 M_1 M_1 M_1 M_1 M_1 M_1 M_1 C_1[b] C_1 C_1 : C_1 : C_2[b,c]	F0 11 00-FF F0 12 00-FF 00-FF F0 13 00-FF 00-FF 00-FF 00-FF 00-FF : 00-FF : 00-FF

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#r-1 #r : #r+(u-1)	dataIdentifier#m byte #2 (LSB) snapshotData#m byte #1 : snapshotData#m byte #u]	C_2 C_2 : C_2	00-FF 00-FF : 00-FF
:	:	:	:
#t	DTCSnapshotRecordNumber#x	C_3[d]	00-FF
#t+1	DTCSnapshotRecordNumberOfIdentifiers#x	C_3	00-FF
	DTCSnapshotRecord[] #x = [		
# t+2	ECU Voltage DataIdentifier(MSB)	C_3	00-FF
# t+3	ECU Voltage DataIdentifier(LSB)	C_3	00-FF
# t+4	ECU Voltage	C_3	00-FF
# t+5	Engine Speed DataIdentifier(MSB)	C_3	00-FF
# t+6	Engine Speed DataIdentifier(LSB)	C_3	00-FF
# t+7	Engine Speed (High Byte)	C_3	00-FF
# t+8	Engine Speed (Low Byte)	C_3	00-FF
# t+9	Vehicle Speed DataIdentifier(MSB)	C_3	00-FF
# t+10	Vehicle Speed DataIdentifier(LSB)	C_3	00-FF
# t+11	Vehicle Speed (High Byte)	C_3	00-FF
# t+12	Vehicle Speed (Low Byte)	C_3	00-FF
# t+13	dataIdentifier#4 byte #1 (MSB)	C_4[b,d]	00-FF
# t+14	dataIdentifier#4 byte #2 (LSB)	C_4	00-FF
# t+15	snapshotData#4 byte #1	C_4	00-FF
:	:	C_4	00-FF
#t+15+(p-1)	snapshotData#4 byte #p	C_4	00-FF
:	:	.	00-FF
#n-(u-1)-2	dataIdentifier#m byte #1 (MSB)	C_5[b,c,d]	00-FF
#n-(u-1)-1	dataIdentifier#m byte #2 (LSB)	C_5	00-FF
#n-(u-1)	snapshotData#m byte #1	C_5	00-FF
:	:	C_5	00-FF
#n	snapshotData#m byte #u]	C_5	00-FF

a. For M_1, these are the general Snapshots for all ECUs, detailed definitions of general snapshot record can be found in chapter 12.2.3.2.
参数M_1为全局快照数据记录，所有ECU都需要支持，全局快照记录的详细定义在12.2.3.2中。

b. For C_1, C_2, C_4, C_5, these are the local Snapshots for individual ECU. The format and content about snapshot should be determined by SGMW and ECU Supplier together.
参数C_1，C_2，C_4，C_5为局部快照数据记录，其格式与内容应该（should）由SGMW与供应商共同确定，并在ECU独立诊断规范内明确。

c. For C_2, C_5, the more than one dataIdentifier/snapshotData combination in a single local DTCSnapshotRecord are only present if a single dataIdentifier only references an integral part of the local snap shot record data.
只有当一个单独的DID仅表示局部快照数据的某一个组成部分时，在一个DTC局部快照记录中，才出现多于一组DID/快照数据组合参数C_2，C_5。

d. For C_3, C_4, C_5, the DTCSnapshotRecordNumber, DTCSnapshotRecordNumberOfIdentifiers and the DTCSnapshotRecord parameter are only present if all records are requested to be reported (DTCSnapshotRecordNumber set to FF hex in the request) and more than one record is available to be reported.
只有在请求上报所有的记录（请求消息中DTC快照记录码参数设置为FF hex），并且上报的可用记录多于一组的情况下，才存在参数C_3 ,C_4, C_5,（DTC快照记录码、DTC快照记录码的标识符数量以及DTC快照记录（全局快照DID及数据记录、局部快照DID及数据记录））。

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Table 12-15 Positive response message definition (sub-function = reportDTCExtendedDataRecordByDTCNumber)

A_DataByte	Parameter Name	Cvt	Hex value
#1	ReadDTCInformation Positive Response Service Id	M	59
#2	reportType = [reportDTCExtendedDataRecordByDTCNumber]	M	06
	DTCAndStatusRecord[] = [		
#3	DTCHighByte	M	00-FF
#4	DTCMiddleByte	M	00-FF
#5	DTCLowByte	M	00-FF
#6	statusOfDTC]	M	00-FF
#7	DTCExtendedDataRecordNumber #1	C[a]	01
#8	Fault occurrence counter	C[a]	00-FF
#9	DTCExtendedDataRecordNumber #2	C[a]	02
#10	Fault pending counter	C[a]	00-FF
#11	DTCExtendedDataRecordNumber #3	C[a]	03
#12	DTC Aging counter	C[a]	00-FF
#13	DTCExtendedDataRecordNumber #4	C[a]	04
#14	DTC Aged counter	C[a]	00-FF

a. This part is the DTCExtendedDataRecord for all ECUs, detailed definitions can be found in chapter 12.2.3.2. Whether the parameter appeared or not and the number of these parameters that appeared depend on the specific service requested.
该部分为DTC扩展数据记录，所有ECU都需要支持, DTC扩展数据记录的详细定义在12.2.3.2中。这些参数是否出现及出现数目取决于具体的服务请求。

12.2.3.2 Positive response message data parameter definition/肯定响应消息数据参数定义

Table 12-16 Positive response message data parameter definition

Definition
reportType/上报类型 This parameter is an echo of bits 6 - 0 of the sub-function parameter from the request message. 该数据参数是请求消息中子功能参数第6-0位的映射。
StatusOfDTC/DTC状态 The status of a particular DTC. The definition of the bits contained in the statusOfDTC byte can be found in D.2 of ISO 14229. 该参数表示一特定DTC的状态。DTC状态（statusOfDTC）字节中的位定义参见ISO 14229-1附录D.2。
DTCStatusAvailabilityMask/DTC状态支持的掩码 A byte whose bits are defined as the same as statusOfDTC and represents the status bits that are supported by the ECU. Bits that are not supported by the ECU shall be set to 0. 该参数长度为1个字节，其位的定义与DTC状态（statusOfDTC）一致，表示了ECU所支持的DTC状态位。ECU不支持的位应（shall）设置为0。
DTCFormatIdentifier/DTC格式标识符 This one-byte parameter value defines the format of a DTC reported by the ECU: ISO15031-6DTCFormat. This parameter value identifies the DTC format reported by the ECU as defined in ISO 15031-6. 该参数长度为1个字节，定义了ECU上报的DTC格式：ISO 15031-6中DTC格式。该参数表示ECU上报的DTC格式符合ISO 15031-6中的相关定义。
DTCCount/DTC数量 This two-byte parameter refers collectively to the DTCCountHighByte and DTCCountLowByte parameters that are

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sent in response to a reportNumberOfDTCByStatusMask request. DTCCount provides a count of the number of DTCs that match the DTCStatusMask defined in the tester's request. 该参数长度为2个字节，同时使用了DTC数量高字节与低字节两个参数。针对子功能参数为“通过状态掩码上报DTC数量”的请求消息，这两个字节的参数表示的内容通过肯定响应消息中发送出去。该参数提供了与诊断仪请求消息中定义的DTC状态掩码相匹配的DTC的数量。
DTCAndStatusRecord/DTC及其状态 This parameter record contains one or more groupings of DTCHighByte, DTCMiddleByte, DTCLowByte and statusOfDTC of ISO15031-6 DTCFormat. 该参数记录包含了一组或多组符合ISO15031-6中格式求的故障码（DTC高字节、中间字节与低字节）及其状态（statusOfDTC）。
DTCRecord/DTC记录 This parameter record contains one or more groupings of DTCHighByte, DTCMiddleByte and DTCLowByte. The interpretation of the DTCRecord depends on the value included in the DTCFormatIdentifier parameter as defined in this table. DTC记录参数包含一组或多组DTC码（DTC高字节、中间字节与低字节）。如何解释该参数取决于本表格中定义的DTC格式标识符参数的数值。
DTCSnapshotRecordNumber/DTC快照记录码 Either the echo of the DTCSnapshotRecordNumber parameter specified by the tester in the reportDTCSnapshotRecordByDTCNumber request, or the actual DTCSnapshotRecordNumber of a stored DTCSnapshot record. 该参数为诊断仪发送的“通过DTC报告DTC快照记录（reportDTCSnapshotRecordByDTCNumber）”请求中定义的“DTC快照记录码”的映射，亦或是ECU内存储的某个DTC快照记录实际对应的记录码。
DTCSnapshotRecordNumberOfIdentifiers/DTC快照记录的标识符数量 This single-byte parameter shows the number of dataIdentifiers in the immediately following DTCSnapshotRecord. 该单字节参数表示紧接在它之后的DTC快照记录中数据标识符的数量。
DTCSnapshotRecord/DTC快照记录 The DTCSnapshotRecord contains a snapshot of data values from the time of the system malfunction occurrence. DTC快照记录包含系统故障发生时数据值的快照。
DTCExtendedDataRecordNumber/DTC扩展数据记录码 Either the echo of the DTCExtendedDataRecordNumber parameter specified by the tester in the reportDTCExtendedDataRecordByDTCNumber request, or the actual DTCExtendedDataRecordNumber of a stored DTCExtendedData record. 该参数为诊断仪规定的子功能参数为“通过DTC上报DTC扩展数据记录”的请求消息中“DTC扩展数据记录码”参数的映射，或者为一个已存储的DTC扩展数据记录中实际的“DTC扩展数据记录码”。
DTCExtendedDataRecord/DTC扩展数据记录 The DTCExtendedDataRecord is an ECU-specific block of information that may contain extended status information associated with a DTC. DTCExtendedData contains DTC parameter values, which have been identified at the time of the request. DTC扩展数据记录参数是一个ECU定义的信息模块，该模块可能包含与DTC相关的扩展状态信息。DTC扩展数据包含在诊断仪发出请求时刻已被识别的DTC参数值。

Table 12-17 definition of ECU Voltage of DTCSnapshotRecord

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
ECU Voltage	1	v	0.1v/bit	0	0	25.5

Table 12-18 definition of Engine Speed of DTCSnapshotRecord

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Engine Speed	2	rpm	0.25rpm/bit	0	0	16838.75

Table 12-19 definition of Vehicle Speed of DTCSnapshotRecord

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.

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Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Vehicle Speed	2	km/h	0.0078125km/bit	0	0	511.9921875

Table 12-20 definition of Fault occurrence counter of DTCExtendedDataRecord

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Fault occurrence counter 故障发生计数器	1		1/位	0	0	255

NOTE:

1. The fault occurrence counter indicates the number of operating cycle in which failed' testResult has been reported;
   故障发生计数器表示发生故障的操作循环次数；
2. If the DTC status bit1 (testFailedThisOperationCycle) is set from 0 to 1, the fault occurrence counter shall be incremented by one;
   如果DTC状态位1（本次操作循环测试失败）由0变为1，则故障发生计数器应该（shall）加1；
3. In case of an overflow, the value of the fault occurrence counter shall be kept in FF hex;
   若数值溢出，故障发生计数器数值应该（shall）保持为FF hex；
4. The fault occurrence counter can be reset to zero by both ClearDiagnosticInformation (14 hex) service and the DTC aging mechanism (see table 12-22).
   清除诊断信息（14 hex）服务与DTC老化机制(见表12-22)均可以将故障发生计数器清零。

Table 12-21 definition of Fault pending counter of DTCExtendedDataRecord

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
Fault pending counter 故障待定计数器	1		1/位	0	0	255

NOTE:

1. The fault pending counter indicates the number of operating cycle in which failed' testResult has been reported consecutively;
   故障待定计数器表示连续发生故障的操作循环次数；
2. If the DTC status bit2 (pendingDTC) is set to 1 in an operating cycle, the fault pending counter shall be set to 1 simultaneously; If the DTC status bit2 (pendingDTC) is always 1 in the consecutive operating cycle, the fault pending counter shall be incremented by one in the corresponding operating cycle. In case of an overflow, the value of the fault pending counter shall be kept in FF hex;
   如果在某个操作循环内，DTC 状态位 2（待定 DTC）设置为 1，则故障待定计数器应（shall）同时设置为 1；如果在紧接着后续的操作循环内，DTC 状态位 2（待定 DTC）一直为 1，则故障待定计数器应（shall）在该相应的操作循环内加 1。若数值溢出，故障待定计数器数值应该（shall）保持为 FF hex；
3. If the DTC status bit2 (pendingDTC) is set to 0 in an operating cycle, the fault pending counter shall be reset to zero simultaneously;
   如果在某个操作循环内，DTC 状态位 2（待定 DTC）设置为 0，则故障待定计数器应（shall）同时清零；
4. The fault pending counter can be reset to zero by both ClearDiagnosticInformation (14 hex) service and the DTC aging mechanism (see table 12-22).
   清除诊断信息（14 hex）服务与 DTC 老化机制(见表 12-22)均可以将故障待定计数器清零。

Table 12-22 definition of DTC Aging counter of DTCExtendedDataRecord

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
DTC Aging counter 老化计数器	1		1/位	0	0	255

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Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.

NOTE:

1. The DTC aging counter indicates the number of operating cycle in which 'failed' testResult has never been reported;
   老化计数器表示连续测试通过的操作循环次数；
2. If the fault pending counter is set from 1 to 0 in an operating cycle, the DTC Aging counter shall be set from 0 to 1 simultaneously. If the "failed" testResult has never been reported in the consecutive operating cycles (i.e. the DTC status bit0 (testFailed) has never been set to 1 in the consecutive operating cycles), the DTC aging counter shall be incremented by one in the corresponding operating cycle. In case of an overflow, the value of the DTC aging counter shall be kept in FF hex;
   如果在某个操作循环内，故障待定计数器从 1 设置为 0，则 DTC 老化计数器应（shall）同时从 0 设置为 1；如果在紧接着后续的操作循环内，"失败"的测试结果从未发生（即：在所有的后续循环内，DTC 状态位 0（测试失败）从未被设置为 1），则 DTC 老化计数器应（shall）在该相应的操作循环内加 1。若数值溢出，DTC 老化计数器数值应该（shall）保持为 FF hex；
3. If the "failed" testResult has been reported at least once in an operating cycle (i.e. the DTC status bit0 (testFailed) has been set to 1 at least once in the operating cycle), the DTC aging counter shall be reset to zero;
   如果在某个操作循环内，"失败"的测试结果至少发生了一次（即：在此操作循环内，DTC 状态位 0（测试失败）至少有一次被设置为 1），DTC 老化计数器应（shall）复位清零；
4. The DTC aging counter can be reset to zero by 'ClearDiagnosticInformation (14 hex)'service;
   清除诊断信息（14 hex）服务可以将 DTC 老化计数器清零；
5. If during the previous 40 consecutive operation cycles, the test did not fail in each operation cycle (i.e. the DTC aging counter of this DTC has reached 40), then a particular DTC and its some information will be allowed to be cleared. This strategy is called DTC aging mechanism in this specification;
   如果在之前连续 40 个操作循环中，每一个操作循环均无某个 DTC 的测试失败结果（即：该 DTC 的 DTC 老化计数器达到 40），那么将允许清除该 DTC 及其部分相关信息，本规范中将这种策略称为 DTC 老化机制；

Table 12-23 definition of Aged counter of DTCExtendedDataRecord

Data Name	#of bytes	Unit	Scaling	Offset	Min.	Max.
DTC Aged counter 已老去计数器	1		1/位	0	0	255

NOTE:

1. The aged counter indicates the aged times since the last clearance of DTC by the ClearDiagnosticInformation (14 hex) service;
   已老去计数器表示自上次通过"清除诊断信息（14 hex）服务"进行清除诊断故障代码后，诊断故障代码已老去的次数；
2. If a DTC is cleared once due to DTC aging mechanism, the aged counter shall be incremented by one;
   如果某一故障代码由于 DTC 老化机制而被清除一次，则已老去计数器应该（shall）加 1；
3. Only the ClearDiagnosticInformation (14 hex) service can clear the aged counter. When this diagnostic service request has been sent to ECU, the aged counter shall be reset to 0;
   只有清除诊断信息（14 hex）服务能够清除已老去计数器。当 ECU 接收到该服务请求时，已老去计数器应（shall）清零。

12.2.4 Supported negative response codes/支持的否定响应码

Table 12-24 Supported negative response codes

Hex	Description	Cvt
7F	**serviceNotSupportedInActiveSession/激活会话不支持该服务**	M
12	**subFunctionNotSupported/不支持子功能**	M
31	**requestOutOfRange/请求超出范围** This code is returned if: 如果出现下任一条件，ECU应（shall）发送此否定响应码： 1) the tester specified a DTCMaskRecord/DTCSeverityMaskRecord that was not recognized by the ECU;	M

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	ECU不识别诊断仪定义的DTC掩码记录参数/DTC严重度参数； 2) the tester specified an invalid DTCSnapshotRecordNumber or DTCExtendedDataRecordNumber. 诊断仪定义的DTC快照记录码或DTC扩展数据记录码无效。	
13	**incorrectMessageLengthOrInvalidFormat/不正确的消息长度或无效的格式**	M

NOTE: The strategy of negative response codes can be found in Figure A.2 of Annex A.
注：否定响应码的策略见附录 A 图 A.2。

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13 InputOutput control functional unit/输入输出控制功能单元

13.1 InputOutputControlByIdentifier (2F hex) /通过标识符进行输入输出控制

13.1.1 Service description/服务描述

The InputOutputControlByIdentifier service is used by the tester to substitute a value for an input signal, internal ECU function and/or control an output (actuator) of an electronic system.
诊断仪使用通过标识符控制输入输出服务来替代输入一个信号值、启动ECU内部功能以及/或者控制一个电子系统的输出（执行器）。

13.1.2 Request message/请求消息

13.1.2.1 Request message definition 请求消息定义

Table 13-1 Request message definition

A_Data Byte	Parameter Name	Cvt	Hex value
#1	InputOutputControlByIdentifier Request Service Id	M	2F
	dataIdentifier#1[] = [		
#2	byte #1(MSB)	M	00-FF
#3	byte #2]	M	00-FF
	controlOptionRecord#1[] = [		
#4	controlState#1/inputOutputControlParameter	M_1[a]	00-FF
:	:	:	:
#4+(m-1)	controlState#m]	C_1[b]	00-FF
	controlEnableMaskRecord#1[] = [		
#4+m	controlMask#1	C_2[c]	00-FF
:	:	:	:
#4+m+(r-1)	controlMask#r]	C_2	00-FF

a. M1: Mandatory: ControlState#1 can be used as either an InputOutputControlParameter or an additional controlState. If it is used as an InputOutputControlParameter, then it shall be implemented as defined in Annex E.1 of ISO 14229-1.
M1：强制的：控制状态#1可以作为输入输出控制参数或者一个额外的控制状态来使用。如果作为输入输出控制参数来使用，那么它应（shall）按照ISO 14229-1附录E的定义来执行。

b. The presence of the C1 parameter depends on the dataIdentifier#1 and the inputOutputControlParameter of controlOptionRecord#1 (if controlState#1 of controlOptionRecord#1 is used as an inputOutputControlParameter).
参数C1是否存在取决于DID#1参数与“控制可选记录#1”参数组中的输入输出控制参数（如果控制状态#1被用作输入输出控制参数的话）。

c. The presence of the C2 parameter depends on the dataIdentifier#1.
参数C2是否存在取决于DID#1。

13.1.2.2 Request message sub-function parameter definition/请求消息子功能参数定义

This service does not use a sub-function parameter.
本服务不使用子功能参数。

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13.1.2.3 Request message data parameter definition/请求消息数据参数定义

Table 13-2 Request message data parameter definition

Definition
dataIdentifier/数据标识符 This parameter identifies ECU local input signal(s), internal parameter(s) or output signal(s). The applicable range of values for this parameter can be found in the table of dataIdentifiers defined in Annex C, which is based on Annex C.1 of ISO 14229-1. The detailed data identifiers used by this service are defined in the individual diagnostic specification for each ECU. 该参数表示ECU的本地输入信号、内部参数或者输出信号。 该参数的可用范围在本规范附录C中定义，附录C是基于ISO 14229-1附录C.1制定的。该服务所使用的DID在ECU独立诊断规范中定义。
controlOptionRecord/控制可选记录 The controlOptionRecord of each dataIdentifier consists of one or multiple bytes (controlState#1/inputOutputControlParameter to controlState#m). ControlState#1 can be used as either an InputOutputControlParameter that describes how the ECU shall control its inputs or outputs, or as an additional controlState byte. 每个DID中的“控制可选记录”参数包含一个或多个字节（从控制状态#1/输入输出控制参数到控制状态#m），“控制状态#1”参数既可以用作输入输出控制参数或额外的控制状态，参数输入输出控制参数描述了ECU应（shall）如何控制本身输入或输出。 ● If it is used as an InputOutputControlParameter, then it shall be implemented as defined in Annex E, which is based on Annex E.1 of ISO 14229-1. 如果该参数作为输入输出控制参数来使用，那么它应（shall）按照本规范附录E来执行，附录E是基于ISO 14229-1附录E.1来制定的。 ● If it is used as a controlState#1, then the controlState shall consist of one or multiple byte with each byte or bit representing the respective adjustment value associated with one individual input signal, internal parameter or output signal of the group of parameters referenced by the particular dataIdentifier. The byte number (bit number respectively) complies with the order of parameters grouped together within one data record referenced by the dataIdentifier. The detailed controlStates used by this service are defined in the individual diagnostic specification for each ECU. 如果该参数作为控制状态#1来使用，那么“控制状态”参数应（shall）包含一个或多个字节，每个字节或者每位分别表示指定DID对应参数组内输入信号、内部参数或者输出信号的调整值。字节（位）的排布顺序遵从DID对应的数据记录参数的顺序。本服务所使用的控制状态的详细描述参见ECU独立诊断规范。
controlEnableMaskRecord/控制使能掩码记录 The ControlEnableMaskRecord of each dataIdentifier consists of one or multiple bytes (controlMask#1 to controlMask#r). The ControlEnableMask shall only be supported when the inputOutputControlParameter is used and the dataIdentifier to be controlled consists of more than one parameter (i.e. the dataIdentifier is bit-mapped or packeted by definition). There shall be one bit in the ControlEnableMask corresponding to each individual parameter defined within the dataIdentifier. The detailed controlEnableMaskRecords used by this service are defined in the individual diagnostic specification for each ECU. 每个DID中的“控制使能掩码记录”参数包含一个或多个字节（从控制掩码#1到控制掩码#r）。只有在使用输入输出控制参数情况下才支持“控制使能掩码记录”参数，并且所控制的DID包含多于一个参数（即：DID定义为位映射或者位封装格式）。在“控制使能掩码记录”参数中应（shall）有一位与DID中定义的某一独立参数位相对应。 本服务使用的“控制使能掩码记录”参数的详细描述参见ECU独立诊断规范。

13.1.3 Positive response message/肯定响应消息

13.1.3.1 Positive response message definition/肯定响应消息定义

Table 13-3 Positive response message definition

A_Data Byte	Parameter Name	Cvt	Hex value
#1	InputOutputControlByIdentifier Response Service ID	M	6F

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#2 #3	dataIdentifier#1[] = [byte #1(MSB) byte #2]	M M	00-FF 00-FF
#4 : #4+(m-1)	controlStatusRecord#1[] = [controlState#1/inputOutputControlParameter : controlState#m]	C_1[a] : C_2[b]	00-FF : 00-FF

a. The presence of the C_1 parameter depends on its usage in the request message. ControlState#1 is either used as an InputOutputControlParameter or as an additional controlState. If it is used as an InputOutputControlParameter then it shall be present in the response message and shall be the echo of the InputOutputControlParameter value given in the request message. In all other cases its presence is user-optional (depends on the usage of a controlStatusRecord).
参数C_1是否存在取决于对应的参数在请求消息中的应用。“控制状态#1”参数既可以用作输入输出控制参数也可以用作额外的控制状态。如果被用作输入输出控制参数，那么该参数应（shall）在响应消息中出现，并且是请求消息中输入输出控制参数值的映射。在其余所有情况下此参数是否出现取决于用户选择（取决于控制状态记录的使用）。

b. The presence of the C_2 parameter depends on the dataIdentifier and the inputOutputControlParameter (if controlState#1 is used as an inputOutputControlParameter).
参数C_2是否存在取决于DID与输入输出控制参数（如果“控制状态#1”参数被当做输入输出控制参数使用）。

13.1.3.2 Positive response message data parameter definition/肯定响应消息数据参数定义

Table 13-4 Positive response message data parameter definition

Definition
dataIdentifier/数据标识符 This parameter is an echo of the dataIdentifier(s) from the request message. 该参数为请求消息中DID的映射。
controlStatusRecord/控制状态记录 The controlState parameter of each dataIdentifier consists of one or multiple bytes (controlState#1/InputOutputControlParameter to controlState #m) which include e.g. feedback data. 每一个DID中的“控制状态”参数包含一个或多个字节（从控制状态#1/输入输出控制参数到控制状态#m），这些字节包含例如反馈数据等。 ● If controlState#1 was used as an InputOutputControlParameter in the request message, then the controlState#1 in the response is the echo of the InputOutputControlParameter value given in the request message (Annex E, which is based on Annex E.1 of ISO 14229-1.). 如果“控制状态#1”参数在请求信息中被用作输入输出控制参数，那么响应信息中的“控制状态#1”参数将是请求信息中输入输出控制参数值的映射。 ● If it is used as a controlState#1, then the controlState shall consist of one or multiple bytes which include e.g. the current parameter values of the controlled Inputs and/or Outputs. 如果该参数被用作““控制状态#1”参数”，那么控制状态应（shall）包含一个或多个字节，这些字节包含例如所控制的输入与/或输出的当前参数值。

13.1.4 Supported negative response codes/支持的否定响应码

Table 13-5 Supported negative response codes

Hex	Description	Cvt
7F	**serviceNotSupportedInActiveSession/激活会话不支持该服务**	M
33	**securityAccessDenied/安全访问拒绝** This code shall be sent if the ECU is not in an unlocked state.	M

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	如果处于锁闭状态，ECU应（shall）发送此否定响应码。	
31	**requestOutOfRange/请求超出范围** This code shall be returned if: 如果出现以下任一条件，ECU应（shall）发送此否定响应码： 1) the requested dataIdentifier value is not supported by the ECU; ECU不支持诊断仪请求的DID； 2) the dataIdentifier uses the controlState#1 parameter as an inputOutputControlParameter and the value contained in this parameter is invalid (see definition of inputOutputControlParameter in Annex E); DID使用控制状态#1参数作为“输入输出控制参数”，并且这个参数值是无效的（见附录E中输入输出控制参数定义）； 3) one or more of the applicable controlStates of the controlOptionRecord record are invalid. 一个或多个控制可选记录参数中可用的“控制状态”数据无效。	M
13	**incorrectMessageLengthOrInvalidFormat/不正确的消息长度或无效的格式**	M
22	**conditionsNotCorrect/条件不正确** This code shall be returned if the criteria for the requested InputOutputControl are not met. The correct condition for the ECU to implement the request message shall be defined in the individual ECU diagnostic specification. 当不满足请求的输入输出控制的操作条件时，ECU 应（shall）发送此否定响应码。ECU 执行该服务请求消息的正确条件应（shall）在 ECU 独立诊断规范内定义。	M

NOTE: The strategy of negative response codes can be found in Figure A.3 of Annex A.
注：否定响应码的策略见附录 A 图 A.3。

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14 Remote activation of routine functional unit/程序远程激活功能单元

14.1 RoutineControl (31 hex)/例程控制

14.1.1 Service description/服务描述

The RoutineControl service is used by the tester to
- start a routine,
- stop a routine, and
- request routine results.

A routine is referenced by a two-byte routineIdentifier.
诊断仪使用例程控制服务来完成以下操作:
- 启动一个程序，
- 停止一个程序，
- 请求程序运行结果。

该程序由一个2个字节的程序标识符来标识。

14.1.2 Request message/请求消息

14.1.2.1 Request message definition 请求消息定义

Table 14-1 Request message definition

A_Data Byte	Parameter Name	Cvt	Hex value
#1	RoutineControl Request Service Id	M	31
#2	sub-function = [routineControlType]	M	01/02/03
#3 #4	routineIdentifier [] = [byte#1 (MSB) byte#2]	M M	00-FF 00-FF
#5 : #n	routineControlOptionRecord[] = [routineControlOption#1 : routineControlOption#m]	C[a]/U : C/U	00-FF : 00-FF

a. The presence of the C parameter is user-optional for sub-function parameter startRoutine and stopRoutine.
对于子功能参数为启动程序（startRoutine）与停止程序（stopRoutine）的请求消息来说，参数C是否存在是用户可选的。

14.1.2.2 Request message sub-function parameter definition/请求消息子功能参数定义

Table 14-2 Request message sub-function parameter definition

Hex(bit 6-0)	Description	Cvt
01	startRoutine	M
02	stopRoutine	U
03	requestRoutineResults	U

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14.1.2.3 Request message data parameter definition/请求消息数据参数定义

Table 14-3 Request message data parameter definition

Definition
routineIdentifier/程序标识符 This parameter identifies an ECU local routine and is out of the range of defined routineIdentifiers (see Annex F.1). The applicable range of values for this parameter can be found in Annex F.1 of ISO 14229-1. The detailed routine identifiers used by this service are defined in the individual diagnostic specification for each ECU. 该参数标识一个ECU的本地程序，并且该参数值来自已定义的程序标识符范围（见附录F.1）。 该参数值的可用范围在ISO 14229-1的附录F.1有定义。本服务使用的“程序标识符”参数的详细描述参见ECU独立诊断规范。
routineControlOptionRecord/程序控制可选记录 This parameter record contains either: ● routine entry option parameters, which optionally specify start conditions of the routine (e.g. timeToRun, startUpVariables, etc.); or ● routine exit option parameters which optionally specify stop conditions of the routine (e.g. timeToExpireBeforeRoutineStops, variables, etc.). The detailed routine control option records used by this service are defined in the individual diagnostic specification for each ECU. 该参数记录可能包含： ● 程序入口操作参数，通过程序入口操作参数可以选择性地定义程序的启动条件（例如运行时间、启动变量等），或者 ● 程序退出操作参数，通过程序退出操作参数可以选择性地定义程序的停止条件（例如：程序停止前的总时间、变量等）。 本服务使用的“程序控制可选记录”参数的详细描述参见ECU独立诊断规范。

14.1.3 Positive response message/肯定响应消息

14.1.3.1 Positive response message definition/肯定响应消息定义

Table 14-4 Positive response message definition

A_Data Byte	Parameter Name	Cvt	Hex value
#1	RoutineControl Response Service Id	S	71
#2	routineControlType	M	00-7F
#3 #4	routineIdentifier [] = [byte#1 (MSB) byte#2]	M M	00-FF 00-FF
#5 : #n	routineStatusRecord[] = [routineStatus #1 : routineStatus #m]	U : U	00-FF : 00-FF

14.1.3.2 Positive response message data parameter definition/肯定响应消息数据参数定义

Table 14-5 Positive response message data parameter definition

Definition
routineControlType/程序控制类型 This parameter is an echo of bits 6 - 0 of the sub-function parameter from the request message. 该参数为请求消息中子功能参数第6-0位的映射。

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routineIdentifier/程序标识符 This parameter is an echo of the routineIdentifier from the request message. 该参数为请求消息中程序标识符的映射。
routineStatusRecord/程序状态记录 This parameter record is used to give to the tester either: ● additional information about the status of the ECU following the start of the routine; or ● additional information about the status of the ECU after the routine has been stopped (e.g. totalRunTime, results generated by the routine before stopped, etc.); or ● results (exit status information) of the routine which has been stopped previously in the ECU. The detailed routine status records used by this service are defined in the individual diagnostic specification for each ECU. 该参数记录用来发送以下信息给诊断仪： ● 启动程序后，有关ECU状态的附加信息， ● 停止程序后，有关ECU状态的附加信息（例如：总运行时间、程序停止之前产生的结果）， ● ECU内部之前已经停止的程序的结果（退出状态信息）。 本服务使用的“程序状态记录”参数的详细描述参见ECU独立诊断规范。

14.1.4 Supported negative response codes/支持的否定响应码

Table 14-6 Supported negative response codes

Hex	Description	Cvt
7F	**serviceNotSupportedInActiveSession/激活会话不支持该服务**	M
33	**securityAccessDenied/安全访问拒绝** This code shall be sent if the ECU is not in an unlocked state. 如果处于锁闭状态，ECU应（shall）发送此否定响应码。	M
12	**subFunctionNotSupported/不支持子功能**	M
31	**requestOutOfRange/请求超出范围** This code shall be returned if: 如果出现以下任一条件，ECU应（shall）发送此否定响应码： 1) the ECU does not support the requested routineIdentifier; ECU不支持请求的程序标识符； 2) the user optional routineControlOptionRecord contains invalid data for the requested routineIdentifier. 对请求的程序标识符来说，用户可选的“程序控制可选记录”参数包含无效数据。	M
13	**incorrectMessageLengthOrInvalidFormat/不正确的消息长度或无效的格式**	M
24	**requestSequenceError/请求序列错误** This code shall be returned if the “stopRoutine” or “requestRoutineResults” sub-function is received without first receiving a “startRoutine” for the requested routineIdentifier. 针对一个特定的程序标识符，如果在接收到一个子功能参数为“启动程序”的请求消息之前，首先接收到一个子功能参数为“停止程序”或者“请求程序结果”的请求消息，ECU应（shall）发送此否定响应码。	M
22	**conditionsNotCorrect/条件不正确** This code shall be returned if the criteria for the requested RoutineControl are not met. The correct condition for the ECU to implement the request message shall be defined in the individual ECU diagnostic specification. 当不满足请求的例程控制的操作条件时，ECU 应（shall）发送此否定响应码。ECU 执行该服务请求消息的正确条件应（shall）在 ECU 独立诊断规范内定义。	M

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NOTE: The strategy of negative response codes can be found in Figure A.2 of Annex A.
注：否定响应码的策略见附录 A 图 A.2。

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15 Upload download functional unit/上传下载功能单元

15.1 RequestDownload (34 hex)/请求下载

N/A

15.2 RequestUpload (35 hex)/请求上传

N/A

15.3 TransferData (36 hex)/传输数据

N/A

15.4 RequestTransferExit (37 hex)/请求退出传输

N/A

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16 ECU programming requirements and process/ECU 软件刷新要求及流程

16.1 ECU programming requirements/ECU 软件刷新要求

N/A

16.2 ECU programming process/ECU 软件刷新流程

N/A

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Annex A (normative) Global parameter definitions/附录 A（规范性附录）全局参数定义

A.1 Negative response codes/否定响应码

For this specification, the requirement of annex A of ISO 14229 is normative, only the negative response codes listed in annex A.1 of ISO 14229 shall be used. Table A.1 lists all negative response codes used within this specification.
就本规范而言，ISO 14229-1中附录A的要求是规范性的，只有ISO 14229-1附录A.1中的否定响应码可以使用，表A.1列出了本规范使用的所有否定响应码。

Table A.1 — Definition of Negative Response Code Values/否定响应码数值定义

Hex value	Negative Response Code/否定响应码
11	**ServiceNotSupported/服务不支持** This response code indicates that the requested action will not be taken because the ECU does not support the requested service. 该响应码表明请求的动作不能执行的原因是ECU不支持请求的服务。 The ECU shall send this response code in case the tester has sent a request message with a service identifier which is either unknown or not supported by the ECU. Therefore, this negative response code is not shown in the list of negative response codes to be supported for a diagnostic service because this negative response code is not applicable for supported services. 当诊断仪发送的请求消息中服务标识符无法识别或不支持时，ECU应发送该响应码。该否定响应码没有在诊断服务支持的负响应码列表中列出，因为对于支持的服务来说该负响应码不适用。
12	**SubFunctionNotSupported/不支持子功能** This response code indicates that the requested action will not be taken because the ECU does not support the service-specific parameters of the request message. The ECU shall send this response code if the tester has sent a request message with a known and supported service identifier but with “sub-function” which is either unknown or not supported. 该响应码表明请求的动作不能执行的原因是ECU不支持请求消息中的服务特定参数。如果诊断仪已经发送了一请求消息，并且该请求消息包含能识别且支持的服务标识符，但子功能要么无法识别要么不支持，此时ECU应（shall）发送此响应代码。
13	**IncorrectMessageLengthOrInvalidFormat/不正确的消息长度或无效的格式** This response code indicates that the requested action will not be taken because the length of the received request message does not match the prescribed length for the specified service or because the format of the parameters does not match the prescribed format for the specified service. 该响应码表明请求的动作不能执行的原因是ECU接收到的请求消息长度与特定服务规定的长度不匹配或者是参数格式与特定服务规定的格式不匹配。
22	**conditionsNotCorrect/条件不正确** This response code indicates that the requested action will not be taken because the ECU prerequisite conditions are not met. 该响应码表明请求的动作不能执行的原因是ECU的状态条件不允许。
24	**requestSequenceError/请求序列错误** This response code indicates that the requested action will not be taken because the ECU expects a different sequence of request messages or message to that sent by the tester. 该响应码表明请求的动作不能执行的原因是ECU收到一个非预期的请求消息序列或诊断仪发送的消息。
31	**requestOutOfRange/请求超出范围**

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	This response code indicates that the requested action will not be taken because the ECU has detected that the request message contains a parameter which attempts to substitute a value beyond its range of authority (e.g. attempting to substitute a data byte of 111 when the data is only defined to 100), or which attempts to access a dataIdentifier/routineIdentifier that is not supported or not supported in active session. 该响应码表明请求的动作不能执行的原因是ECU检测到请求消息包含一个超出允许范围的参数或者是不支持或者激活会话模式下不支持的数据标识符/例程标识符的访问。 This response code shall be implemented for all services which allow the tester to read data, write data or adjust functions by data in the ECU. 应（shall）允许诊断仪在ECU内部进行读数据、写数据或通过数据调整功能的服务使用该响应代码。
33	**securityAccessDenied/安全访问拒绝** This response code indicates that the requested action will not be taken because the ECU's security strategy has not been satisfied by the tester. 该响应码表明请求的动作不能执行的原因是诊断仪无法通过ECU的安全策略。 The ECU shall send this response code if one of the following cases occurs: 下列任一情况发生时，ECU应（shall）发送该代码： -the test conditions of the ECU are not met; -ECU的测试条件不允许； -the required message sequence, e.g. DiagnosticSessionControl, SecurityAccess, is not met; -请求消息的序列不对（例如：诊断会话控制，安全访问）； -the tester has sent a request message which requires an unlocked ECU. -诊断仪向一未解锁的ECU发送请求消息。 Besides the mandatory use of this negative response code as specified in the applicable services within this specification, this negative response code can also be used for any case where security is required and is not yet granted to perform the required service. 除了本规范中应用服务强制规定使用该否定响应码外，该否定响应码还能够用在需要安全访问但没通过安全访问的情况。
35	**invalidKey/密钥无效** This response code indicates that the ECU has not given security access because the key sent by the tester did not match with the key in the ECU's memory. This counts as an attempt to gain security. The ECU shall remain locked and increment its internal SAFC. 该响应码表明ECU不允许通过安全访问的原因是诊断仪发送的密钥与ECU内存中的密钥不匹配。该操作算作一次安全访问的尝试。此时ECU应（shall）保持锁定状态，并且增加内部SAFC的计数。
36	**exceedNumberOfAttempts/超出尝试次数** This response code indicates that the requested action will not be taken because the tester has unsuccessfully attempted to gain security access more times than the ECU's security strategy will allow. 该响应码表明请求的动作不能执行的原因是诊断仪尝试获得安全访问失败次数超过了ECU安全策略允许的值。
37	**requiredTimeDelayNotExpired/所需时间延迟未到** This response code indicates that the requested action will not be taken because the tester's latest attempt to gain security access was initiated before the ECU's required timeout period had elapsed. 该响应码表明请求的动作不能执行的原因是在ECU所需的请求延迟时间过去之前诊断仪又执行了一次请求。
70	**uploadDownloadNotAccepted/不允许上传下载** This response code indicates that an attempt to upload/download to a ECU's memory cannot be accomplished due to fault conditions. 该响应码表明试图向ECU内存上传/下载数据失败的原因是条件不允许。
71	**transferDataSuspended/数据传输暂停** This response code indicates that a data transfer operation was halted due to a fault. The active TransferData sequence shall be aborted.

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	该响应码表明由于错误导致数据传输操作的中止，当前激活的数据传输序列应（shall）被停止。
72	**generalProgrammingFailure/一般编程失败** This response code indicates that the ECU detected an error when erasing or programming a memory location in the permanent memory device (e.g. Flash Memory). 该响应码表明在不可擦除的内存设备中进行擦除或编程时ECU检测到错误发生。
73	**wrongBlockSequenceCounter/错误的数据块序列计数器** This return code shall be sent if the ECU detects an error in the sequence of the blockSequenceCounter. 如果ECU在数据块序列计数（blockSequenceCounter）序列中检测到错误发生，那么ECU应（shall）响应该代码。 The repetition of a TransferData request message with a blockSequenceCounter equal to the one included in the previous TransferData request message shall be accepted by the ECU. 如果当前接收到的发送数据请求消息中数据块序列计数器值与前一个发送数据请求消息中的一致，那么，ECU应（shall）能接收该发送数据请求消息而不能当做错误来处理。
78	**requestCorrectlyReceived-ResponsePending/正确接收请求消息-等待响应** This response code indicates that the request message was received correctly, and that all parameters in the request message were valid, but the action to be performed is not yet completed and the ECU is not yet ready to receive another request. As soon as the requested service has been completed, the ECU shall send a positive response message or negative response message with a response code different from this. 该响应码表明诊断仪请求的消息被ECU正确接收且请求消息中所有参数有效，但是将执行的动作未完成且ECU未准备好接收其它请求。一旦完成所请求的服务，ECU应（shall）发送一肯定响应消息或发送否定响应吗不为78的否定响应消息。
7F	**serviceNotSupportedInActiveSession/激活会话不支持该服务** This response code indicates that the requested action will not be taken because the ECU does not support the requested service in the session currently active. This response code shall only be used when the requested service is known to be supported in another session, otherwise response code SNS (serviceNotSupported) shall be used. 该响应码表明请求的动作不能执行的原因是当前会话模式下ECU不支持请求的服务。只有在已知另一个会话状态下支持该所请求的服务时，才能使用该否定响应码。否则，应（shall）使用否定响应码SNS（服务不支持）。 This response code is, in general, supported by each diagnostic service, unless otherwise stated in the data-link-specific implementation document; therefore, it is not listed in the list of applicable response codes of the diagnostic services. 一般情况下，每个诊断服务应支持该代码，除非在数据链路层的具体实施文件内有说明。因此，该代码没有在诊断服务支持的否定响应代码列表内列出。
92	**voltageTooHigh/电压过高** This response code indicates that the requested action will not be taken because the ECU prerequisite condition for voltage at the primary pin of the ECU is not met (current voltage is above a preprogrammed maximum threshold). 该响应码表明请求的动作不能执行的原因是ECU电压输入管脚的电压参数不正确（当前电压值超过了编程允许的最大门限值）。
93	**voltageTooLow/电压过低** This response code indicates that the requested action will not be taken because the ECU prerequisite condition for voltage at the primary pin of the ECU is not met (current voltage is below a preprogrammed minimum threshold). 该否定响应码表明请求的动作不能执行的原因是ECU电压输入管脚的电压参数不正确（当前电压值低于了编程允许的最小门限值）。

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Annex B (normative) Diagnostic and communication management functional unit data parameter definitions/附录 B（规范性附录）诊断与通信管理功能单元数据参数定义

B.1 baudrateIdentifier parameter definition/波特率标识符参数定义

Table B.1 Definition of baudrateIdentifier values/波特率标识符值定义

Hex	Description	Cvt
00	**ISOSAEReserved** This value is reserved by ISO 14229 for future definition. 该值作为ISO 14229将来定义的预留。	M
01	**PC9600Baud** This value specifies the standard PC baud rate of 9.6 KBaud. 该值定义了标准的PC波特率为9.6 KBaud。	U
02	**PC19200Baud** This value specifies the standard PC baud rate of 19.2 KBaud. 该值定义了标准的PC波特率为19.2 KBaud。	U
03	**PC38400Baud** This value specifies the standard PC baud rate of 38.4 KBaud. 该值定义了标准的PC波特率为38.4 KBaud。	U
04	**PC57600Baud** This value specifies the standard PC baud rate of 57.6 KBaud. 该值定义了标准的PC波特率为57.6 KBaud。	U
05	**PC115200Baud** This value specifies the standard PC baud rate of 115.2 KBaud. 该值定义了标准的PC波特率为115.2 KBaud。	U
06-0F	**ISOSAEReserved** This range of values is reserved by ISO 14229 for future definition. 该范围作为ISO 14229将来定义的预留。	M
10	**CAN125000Baud** This value specifies the standard CAN baud rate of 125 KBaud. 该值定义了标准的CAN波特率为125 KBaud。	U
11	**CAN250000Baud** This value specifies the standard CAN baud rate of 250 KBaud. 该值定义了标准的CAN波特率为250 KBaud。	U
12	**CAN500000Baud** This value specifies the standard CAN baud rate of 500 KBaud. 该值定义了标准的CAN波特率为500 KBaud。	U
13	**CAN1000000Baud** This value specifies the standard CAN baud rate of 1 MBaud. 该值定义了标准的CAN波特率为1 MBaud。	U
14-FF	**ISOSAEReserved** This range of values is reserved by this document for future definition. 该范围作为本规范将来定义的预留。	M

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Annex C (normative) Data transmission functional unit data parameter definitions/附录 C（规范性附录）数据传输功能单元数据参数定义

C.1 General DataIdentifier parameter definitions/统一数据标识符参数定义

N/A

C.2 Detailed DataIdentifier parameter definitions for ECU/vehicle identifiers options/ECU/车辆标识符选项的数据标识符参数详细定义

N/A

C.3 Diagnostic Addresses/诊断地址

N/A

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Annex D (normative) Stored data transmission functional unit data parameter definitions/附录 D（规范性附录）存储数据传输功能单元数据参数定义

D.1 groupOfDTC parameter definition/故障代码组参数定义

Table D.1 provides group of DTC definitions.
表D.1为故障代码组定义。

Table D.1 — Definition of groupOfDTC and range of DTC numbers/故障代码范围与故障代码组定义

Hex	Description/描述	Cvt
000000	Emission-related systems/排放相关系统	C[a]
FFFFFF	All Groups (all DTCs)/所有的组	M
a. This parameter is only supported by emission-related ECUs. 该范围参数仅用于排放相关的 ECU。		

D.2 DTCStatusMask and statusOfDTC bit definitions/故障码状态掩码与故障码状态位定义

N/A

D.3 Strategy of failure detection and fault handling/故障检测与处理策略

N/A

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Annex E (normative) Input output control functional unit data parameter definitions/附录 E（规范性附录）输入输出控制功能单元数据参数定义

E.1 InputOutputControlParameter definitions/输入输出控制参数定义

This section gives the requirements with that the ECUs and the tester shall comply, which is based on annex E.1 of ISO 14229-1. Other definitions and requirements can be found in Annex E of ISO 14229-1.
该部分给出了 ECU 与诊断仪应（shall）遵从的要求，该要求是基于 ISO 14229-1 附录 E.1 确定的，其它定义与要求见 ISO 14229-1 附录 E。

Table E.1 — inputOutputControlParameter definitions/输入输出控制参数定义

Hex	Description/描述	Cvt
00	**returnControlToECU/返回ECU控制** This value shall indicate to the ECU that the tester no longer has control over the input signal, internal parameter or output signal referenced by the inputOutputLocalIdentifier. Number of controlState bytes in request: 0 Number of controlState bytes in possible response: depends on the dataIdentifier, which is defined in the individual diagnostic specification for each ECU. 诊断仪通过该参数通知ECU它不再控制输入输出局部标识符（inputOutputLocalIdentifier）引用的输入信号、内部参数或者输出信号。 请求中的控制状态（controlState）字节数为：0 响应中的控制状态（controlState）字节数为：取决于在ECU独立诊断规范内定义的数据标识符。	U
01	**ResetToDefault/复位到默认状态** This value shall indicate to the ECU that it is requested to reset the input signal, internal parameter or output signal referenced by the inputOutputLocalIdentifier to its default state. Number of controlState bytes in request: 0 Number of controlState bytes in possible response: depends on the dataIdentifier, which is defined in the individual diagnostic specification for each ECU. 诊断仪通过该参数通知ECU把输入输出局部标识符（inputOutputLocalIdentifier）引用的输入信号、内部参数或者输出信号值复位到默认状态。 请求中的控制状态（controlState）字节数为：0 响应中的控制状态（controlState）字节数为：取决于在ECU独立诊断规范内定义的数据标识符。	U
02	**FreezeCurrentState/冻结当前状态** This value shall indicate to the ECU that it is requested to freeze the current state of the input signal, internal parameter or output signal referenced by the inputOutputLocalIdentifier. 诊断仪通过该参数通知ECU冻结输入输出局部标识符（inputOutputLocalIdentifier）引用的输入信号、内部参数或者输出信号值的当前状态。 Number of controlState bytes in request: 0 Number of controlState bytes in possible response: depends on the dataIdentifier, which is defined in the individual diagnostic specification for each ECU. 请求中的控制状态（controlState）字节数为：0 响应中的控制状态（controlState）字节数为：取决于在ECU独立诊断规范内定义的数据	U

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	标识符。	
03	**shortTermAdjustment/短时调整** This value shall indicate to the ECU that it is requested to adjust the input signal, internal parameter or output signal referenced by the inputOutputLocalIdentifier in RAM to the value(s) included in the controlOption parameter(s) (e.g. set Idle Air Control Valve to a specific step number, set pulse width of valve to a specific value/duty cycle). 诊断仪通过该参数通知ECU把RAM内输入输出局部标识符（inputOutputLocalIdentifier）引用的输入信号、内部参数或者输出信号值调整为控制选项（controlOption）参数内的值。 Number of controlState bytes in request: depends on the dataIdentifier, which is defined in the individual diagnostic specification for each ECU. 请求中的控制状态（controlState）字节数为：取决于在ECU独立诊断规范内定义的数据标识符。 Number of controlState bytes in possible response: depends on the dataIdentifier, which is defined in the individual diagnostic specification for each ECU. 响应中的控制状态（controlState）字节数为：取决于在ECU独立诊断规范内定义的数据标识符。	U

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Annex F (normative)Remote activation of routine functional unit data parameter definitions/附录 F（规范性附录）程序远程激活功能单元数据参数定义

F.1 RoutineIdentifier definition/程序标识符定义

N/A

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Annex G (informative) Examples for addressAndLengthFormatIdentifier parameter values/附录 G（参考性附录）地址与长度格式标识符参数值例子

G.1 addressAndLengthFormatIdentifier example values/地址与长度格式标识符示例值

Table G.1 contains examples of combinations of values for the high and low nibble of the addressAndLengthFormatIdentifier. The following shall be considered.
表G.1包含了地址与长度格式标识符（addressAndLengthFormatIdentifier）的高4位与低4位的示例值，应（shall）考虑以下两点：

● Values, which are either marked as “not applicable” for the “manageable memorySize” or the “memoryAddress range”, shall not be used and shall be rejected by the ECU via a negative response message.
那些在管理内存大小或内存地址范围列标明“不适用”的项目，表示ECU不应（shall not）使用地址与长度格式标识符并且应（shall）通过否定响应消息拒绝。

● Values with an applicable “manageable memorySize” and “memoryAddress range” are allowed for this parameter.
那些在管理内存大小或内存地址范围列标明“适用”的项目，表明允许使用地址与长度格式标识符（addressAndLengthFormatIdentifier）。

Table G.1 — addressAndLengthFormatIdentifier example/地址与长度格式标识符示例

Hex	Description/描述			
	Bit 7-4 (high nibble) number of memorySize bytes Bit 7-4（高4位）内存大小字节数		Bit 3-0 (low nibble) number of memoryAddress bytes Bit 3-0（低4位）内存地址字节数	
	Bytes used for memorySize parameter 内存大小参数使用字节数	Manageable size 可管理的大小	Bytes used for memoryAddress parameter 内存地址参数使用的字节数	Addressable Memory 可寻址的内存
00	not applicable	not applicable	not applicable	not applicable
01	not applicable	not applicable	1	256 byte – 1
02	not applicable	not applicable	2	64 KB – 1
03	not applicable	not applicable	3	16 MB – 1
04	not applicable	not applicable	4	4 GB – 1
05	not applicable	not applicable	5	1.024 GB – 1
06-0F	:	:	:	:
10	1	256 byte	not applicable	not applicable
11	1	256 byte	1	256 byte – 1
12	1	256 byte	2	64 KB – 1
13	1	256 byte	3	16 MB – 1
14	1	256 byte	4	4 GB – 1
15	1	256 byte	5	1.024 GB – 1

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16-1F	:	:	:	:
20	2	64 KB	not applicable	not applicable
21	2	64 KB	1	256 byte – 1
22	2	64 KB	2	64 KB – 1
23	2	64 KB	3	16 MB – 1
24	2	64 KB	4	4 GB – 1
25	2	64 KB	5	1.024 GB – 1
26-2F	:	:	:	:
30	3	16 MB	not applicable	not applicable
31	3	16 MB	1	256 byte – 1
32	3	16 MB	2	64 KB – 1
33	3	16 MB	3	16 MB – 1
34	3	16 MB	4	4 GB – 1
35	3	16 MB	5	1.024 GB – 1
36-3F	:	:	:	:
40	4	4 GB	not applicable	not applicable
41	4	4 GB	1	256 byte – 1
42	4	4 GB	2	64 KB – 1
43	4	4 GB	3	16 MB – 1
44	4	4 GB	4	4 GB – 1
45	4	4 GB	5	1.024 GB – 1
46-FF	:	:	:	:

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